

Heads in the Sand: Theory and Experiment on the Interdependence of Information Avoidance

Silvia Sonderegger

Fabio Tufano

Marcelo Woo*

**Job Market Paper*

This version: June 12, 2026

[Click here for the most recent version](#)

Abstract

Information avoidance in individual decision-making is well-documented: individuals often forgo useful, freely available information, even when doing so leads them to worse outcomes. Yet in many settings — such as global warming, infectious disease, organizational performance — outcomes are interdependent across individuals. In this paper, we develop a theoretical framework and design a novel experiment to study information avoidance in environments where information is freely available and ignorance is individually and collectively harmful. In such settings, when individuals are more likely to avoid information if others do so, groups risk becoming trapped in damaging collective ignorance equilibria. Our theoretical model shows, however, that the opposite pattern may also arise: individuals may become more likely to acquire information precisely when others avoid it. Consistent with both patterns, we find substantial heterogeneity in the data. Around 20% of subjects exhibit strategic complementarity, avoiding information when others do so, while a similar share exhibit strategic substitutability, acquiring information when others avoid it. The coexistence of these tendencies implies that group composition shapes equilibrium outcomes and can determine whether groups converge to widespread awareness or collective ignorance.

JEL Codes: C92, D70, D83, D84, D91

Keywords: Information Avoidance, Groupthink, Experiment, Anticipatory Utility

Silvia Sonderegger: University of Nottingham, CeDEx, silvia.sonderegger@nottingham.ac.uk. Fabio Tufano: University of Leicester, fabio.tufano@leicester.ac.uk. Marcelo Woo: University of Nottingham, CeDEx, marcelo.woo@nottingham.ac.uk.

We thank participants at the Nottingham CeDEx brown bag seminar for helpful comments and suggestions. We are especially grateful to Simon Gächter and Florian Zimmermann for insightful discussions. We also thank Thomas Barnes, Adrian Brown, Niramay Chugh, Tong Fang, Pierce Gately, Matias Golman, Yifan Li, Jesús Rodríguez, Kieran Stockley, Lara Suraci and Xue Wang for assistance with invigilation. We gratefully acknowledge funding from the University of Leicester and the University of Nottingham. The usual disclaimers apply.

1 Introduction

Information avoidance — the deliberate choice to forgo readily available and valuable information — is well documented in individual decision-making (Thornton, 2008; Ho et al., 2021).¹ People often choose not to learn whether they carry a potentially fatal disease, despite the availability of accurate tests (Oster et al., 2013), and investors have been found to behave like “ostriches,” avoiding potentially negative financial news (Galai & Sade, 2006; Karlsson et al., 2009; Fong & Hunter, 2022; Olafsson & Pagel, 2025) — even though doing so prevents them from making informed decisions and achieving better outcomes. A common thread underlying such behavior is the desire to avoid anxiety and stress: bad news that signals future adversity is painful to receive because it generates negative anticipatory feelings. By remaining ignorant, individuals preserve more favorable beliefs about the future, safeguarding their anticipatory utility at the cost of poorer decisions.

In many situations, however, individual outcomes are not independent but are instead shaped by the actions of others — giving rise to negative externalities. Examples include global warming, infectious diseases, and corporate settings.

Consider an investment bank holding assets that may be either high quality or junk. The bank’s future depends on its employees’ decisions — whether they maintain the current strategy (optimal if the assets are high quality) or liquidate their holdings (optimal if they are junk). Making the right decision, however, requires assessing the assets’ quality and potentially acknowledging that the bank may be risking bankruptcy — a deeply stressful prospect, as bankruptcy would leave all employees facing unemployment. Crucially, if the assets are junk, the severity of the outcome for each employee depends on what others do: if colleagues continue as usual rather than acting, the damage to the bank is far greater. Each employee’s decision to bury their head in the sand therefore worsens everyone’s payoff and shapes the incentives faced by others when deciding whether to remain ignorant or acquire information. The collapse of Lehman Brothers offers a stark real-world illustration. As warning signs mounted, a prevailing culture of willful blindness discouraged employees from investigating the true quality of the bank’s holdings, and this collective avoidance made things progressively worse, ultimately contributing to the institution’s downfall.

This paper studies information avoidance in groups where information is freely available and remaining ignorant is individually and collectively harmful. As the Lehman Brothers case suggests, contagious ignorance — what Bénabou (2013) calls “groupthink” — is an important force in such settings: if everybody else buries their heads in the sand, the bad state worsens to the point where each individual would rather remain ignorant than risk bearing the dread and anxiety associated with that reality. This raises a number of important questions. Is contagious ignorance the

¹For a comprehensive review on information avoidance, see Golman et al. (2017). Hertwig & Engel (2016) define the related concept of *deliberate ignorance* as “the conscious individual or collective choice not to seek or use information.” They focus on settings where the marginal acquisition costs are negligible and the potential benefits are large, so that acquiring information would appear to be rational from an economics of information perspective.

rule, or is the opposite tendency — strategic substitutability in information avoidance, whereby individuals become more inclined to acquire information when others avoid it — also possible? What determines equilibrium information acquisition within a group? Understanding which force dominates is key to predicting whether information avoidance necessarily self-sustains or unravels within groups and, ultimately, to designing policies that reduce the risk of organizations and societies sleepwalking into collective disaster. To answer these questions, we combine theory, experiment, and simulations. At its core, the problem is one of strategic interaction: each individual’s decision to acquire or avoid information depends on what others do. A controlled laboratory environment allows us to strip away confounding factors and isolate the key forces shaping these strategic interactions. We design a novel experiment for this purpose, zooming in on the strategic nature of information avoidance in groups.

Our investigation begins with a theoretical model that builds on the seminal contribution of [Bénabou \(2013\)](#). In addition to complementarity in information avoidance —which generates contagious ignorance and arises when anticipatory utility is concave — we show that individuals may also exhibit strategic substitutability: when others bury their heads in the sand, they become *less* likely to avoid information. This emerges when anticipatory utility is reference-dependent. Intuitively, when others’ avoidance makes the bad state outcome sufficiently severe, learning that the state is bad decreases anticipatory utility only marginally relative to remaining ignorant, whereas learning that it is good raises anticipatory utility substantially — making information acquisition optimal. The opposite holds when others acquire information and the bad state outcome is relatively mild: learning that the state is good improves anticipatory utility only slightly relative to ignorance, while learning that it is bad decreases it substantially — making information avoidance optimal. Individuals may also exhibit unconditional behavior, acquiring or avoiding information regardless of what others do. In addition, the model allows for social preferences, whose effects are theoretically ambiguous: while altruism may increase the propensity to acquire information by inducing individuals to internalize the externality of their choices, it may also amplify the emotional cost of bad news and thereby strengthen the incentive to avoid information.

Overall, therefore, our theoretical findings indicate that the dynamics of information avoidance in groups may vary considerably. Informed by the theory, we design a novel lab experiment in which participants are matched in groups and informed that there are two possible states of the world: *Good* and *Bad*. In the *Bad* state, everyone in the group is exposed to an aversive future event: at the end of the experiment, they hear a series of unpleasant screams, a consumption event known to produce anticipatory anxiety.² In the *Good* state, no noises are heard. At the beginning of the experiment, each participant chooses whether to discover the state immediately (“Now” option) or to wait until later (“Later” option). Importantly, in the *Bad* state the volume of the screams heard

²This is shown in [Beaurenaut et al. \(2020\)](#). We further discuss the rationales for the use of screams as a consumption event in Section 4.2.

by all group members decreases with the number of members who acquire information about the state (that is, who choose Now), reflecting the idea that acquiring information allows participants to improve outcomes in the bad state by taking remedial action.

To characterize the interdependence in participants' decisions, we elicit each participant's best response to the information avoidance decisions of other group members using the strategy method (Selten, 1967; Fischbacher et al., 2001). In line with the theory, we find substantial heterogeneity. Around 20% exhibit strategic complementarity, becoming *more* likely to avoid information when others do—a pattern consistent with contagious ignorance. A roughly equal share exhibit strategic substitutability, becoming *less* likely to avoid information when others do, suggesting that others' avoidance can serve as a wake-up call rather than a license to look away. The remaining participants exhibit unconditional behavior, with most choosing to acquire information independently of others' choices and around 10% unconditionally avoiding information.

The precise shares of different strategy types are less important than the pattern itself, as context is likely to matter. For example, individuals may choose to acquire information unconditionally in benign environments—where even the worst possible outcome is relatively mild—yet shift toward strategic complementarity or substitutability when the stakes are higher and the potential consequences more severe. The key finding here is the heterogeneity of strategy types, which implies that the aggregate dynamics of information avoidance within groups cannot be reduced to a single dominant force. These types correspond to distinct best-response patterns to others' behavior. Whether contagious ignorance takes hold or unravels therefore depends critically on group composition.

To study more closely how individuals move *along* their best-response functions, we exogenously shift the participants' expectations about their peers by presenting them with information from a pilot study—*Many Avoided* versus *Few Avoided*. This intervention strongly shifts expectations about peers' information decisions, while importantly leaving the distribution of strategy types unaffected. In the *Many Avoided* treatment, the share of participants avoiding information increases from 24% to 33%, in line with contagious ignorance. This is at first glance surprising, since strategic complements and substitutes are present in roughly equal shares and might therefore be expected to offset each other. The resolution of this apparent puzzle lies in the heterogeneity of thresholds: the share of information avoiders above which an individual begins avoiding information (if a complement) or acquiring it (if a substitute) varies considerably. In particular, strategic substitutes tend to cross their threshold at relatively low levels of avoidance in their group, while strategic complements begin avoiding information only once avoidance among peers is quite widespread. As a result, the shift in expectations has little bite for strategic substitutes but meaningfully affects the behavior of strategic complements, thereby driving the aggregate response.

The heterogeneity in best-response schedules implies that group composition shapes the equilibrium prevalence of information avoidance. To study this, we employ simulations that examine

how the interaction of strategy types determines equilibrium outcomes within a group. Groups composed primarily of strategic complements generate multiple equilibria and risk becoming trapped in what we term a *Mutually Assured Ignorance* (MAI) equilibrium — one in which everybody prefers to remain ignorant and the collective outcome in the bad state is as severe as possible. The presence of unconditional avoiders precipitates this outcome: as their share in the group grows, the probability of converging to a MAI equilibrium rises sharply, increasing steeply once the unconditional avoiders approach one-third of the group. Conversely, unconditional information acquirers can steer the group toward a *Mutually Assured Awareness* (MAA) equilibrium, in which everybody acquires information and the collective outcome in the bad state is as high as possible. Groups that include strategic substitutes, by contrast, never reach a MAI equilibrium but are equally unable to sustain universal information acquisition. Strategic substitutes act as buffers, preventing the group from sliding into collective blindness but also limiting its ability to converge on full awareness. Understanding these dynamics is of first-order importance in practice: whether a group converges to collective awareness or ignorance may hinge on the presence of just a small number of unconditional avoiders or strategic substitutes, with consequences that can cascade across entire organizations, markets, and societies.

The remainder of the paper is organized as follows. In Section 2, we discuss related literature. In Section 3, we present the theoretical framework. In Section 4, we describe the experimental design. In Section 5, we report the empirical findings. In Section 6, we use our data to inform a simulation exercise that explores how group composition shapes equilibrium levels of avoidance. In Section 7, we conclude.

2 Related Literature

Our paper relates to several strands of literature. [Golman et al. \(2017\)](#) survey information avoidance in individual decision-making, documenting its prevalence across a wide range of contexts. A large empirical literature, both in the lab and in the field, confirms that individuals often prefer to remain ignorant even when information is costless and useful for subsequent decisions (e.g., [Dana et al., 2007](#); [Karlsson et al., 2009](#); [Pagel, 2018](#); [Ho et al., 2021](#); [Masatlioglu et al., 2023](#); [Momsen & Ohndorf, 2023](#)). Avoidance rates vary widely across studies and contexts, ranging from 5% ([Ganguly & Tasoff, 2017](#)) to over 90% ([Oster et al., 2013](#)). These studies focus primarily on individual decision-making.

Most relevant for our paper, [Falk & Zimmermann \(2024\)](#) use a clean experimental design to show that anticipatory utility — the dread of future negative events — generates intrinsic preferences for avoiding or delaying information. Relatedly, [Engelmann et al. \(2024\)](#) show across five experiments that anticipatory anxiety drives wishful thinking, with participants systematically less likely to correctly identify patterns associated with negative future events. We build on the design

of [Falk & Zimmermann \(2024\)](#) but extend it to group settings where information has instrumental value and generates externalities: becoming informed benefits the decision-maker and the other group members. We make several contributions. First, we document that even when avoidance harms both the individual and others, a substantial share of subjects still prefer to remain ignorant. Second, we provide the first test, to our knowledge, of whether information avoidance decisions are interdependent. Third, we document substantial heterogeneity in strategy types and study its implications for equilibrium information avoidance.

A growing literature examines how group settings can amplify rather than attenuate individual biases. [Oprea & Yuksel \(2022\)](#) show that when subjects observe the beliefs of a similarly performing counterpart about their own IQ test performance, biases amplify as subjects overweight their counterpart’s beliefs when these reinforce their own motivated reasoning. [Enke et al. \(2023\)](#) show that biased agents self-select into group interactions, further amplifying distortions, and [Suvorov et al. \(2024\)](#) show that selective sharing of ability feedback within groups generates collective overconfidence and suboptimal investment decisions. A separate literature on strategic information acquisition in committees finds that costly search generates free-riding and under-acquisition through standard incentive problems ([Gerardi & Yariv, 2008](#)). Our paper focuses instead on information avoidance driven by anticipatory utility and shows that amplification of individual biases is not inevitable in this environment.

Against this background, the paper most closely related to ours is [Bénabou \(2013\)](#), who studies collective motivated beliefs in organizations and markets and shows that when payoffs are interdependent, individual cognitive strategies can be strategic complements, generating groupthink and mutually reinforced denial. We build on his framework and show that strategic substitutability is also possible: ignorance in the group can push an individual toward seeking information rather than avoiding it. Our substitutability result is driven by reference dependence in anticipatory utility. We model anticipatory utility as convex below a reference threshold and concave above it — a structure with a long tradition in prospect theory ([Kahneman & Tversky, 1979](#)), axiomatized within a standard expected utility framework by the success-failure model of [Alaoui & Penta \(2026\)](#), which is here applied to anticipatory utility. When others remain ignorant, payoffs in the bad state worsen, pushing expected anticipatory utility below the threshold. In this region, convexity generates risk-loving behavior: the agent becomes more willing to acquire information, exhibiting strategic substitutability rather than complementarity. The resulting heterogeneity in strategy types — encompassing both complements and substitutes — is our main new result. It implies that whether ignorance spreads or is contained depends critically on group composition, with substitutes and unconditional acquirers potentially counteracting the pull toward collective ignorance.

3 Theoretical Framework

We now describe our theoretical framework and key results. Consider the following setup, inspired by B  nabou (2013).³ There are two states of the world, $\sigma \in \{G, B\}$: a good state G , realized with probability $q \in (0, 1)$, and a bad state B , realized with probability $1 - q$. The game has three periods:

- At Date 2, material payoffs are realized and generate utility $u_i(\sigma)$, with $u_i(G) = u_G$ and $u_i(B) = u_B$, where $u_G > u_B$.
- At Date 1, agents derive utility from anticipating future payoffs. Let $U_i \equiv \mathbb{E}[u_i(\sigma)]$ denote expected material utility. Anticipatory utility is given by $v_i(U_i)$, with $v_i(\cdot)$ strictly increasing. This captures *anticipatory feelings* – such as hope or anxiety about future outcomes – or, more generally, preferences over the timing of uncertainty resolution, in the spirit of Kreps & Porteus (1978).
- At Date 0, before any uncertainty is resolved, agents choose whether to avoid ($y_i = 1$) or acquire ($y_i = 0$) information that perfectly reveals the state of the world. Agents evaluate the choice to acquire or avoid information according to Date-1 anticipatory utility, $\mathbb{E}[v_i(U_i)]$. If an agent avoids, she remains uncertain at Date 1 and experiences anticipatory feelings over expected Date-2 utility. If she acquires information, the uncertainty is resolved immediately, and the anticipatory utility reflects the revealed state. Figure 1 summarizes the model’s timeline.

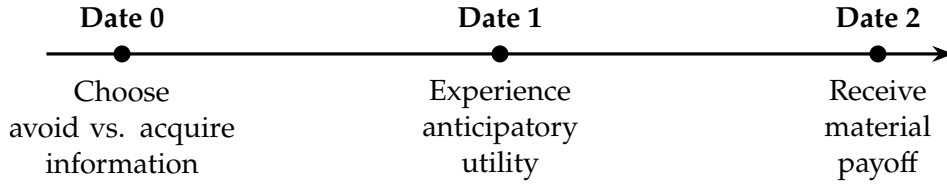


Figure 1: Timeline of the model

While the good-state payoff is constant, a key feature of the model is that each agent’s bad-state payoff depends on the whole group’s information acquisition decisions and deteriorates as more group members avoid information — generating negative payoff externalities within the group. This captures the intuition that an agent cannot respond appropriately to an adverse condition without first being aware of it.⁴ Formally, the Date-2 consumption utility in the bad state is

$$u_B = \bar{u}_B - \Delta \left(y_i + \sum_{j \neq i} y_j \right), \quad (1)$$

³The correspondence between the notation used in our model and B  nabou (2013) is provided in Appendix A, along with all the formal proofs of our results.

⁴This is equivalent to exogenously imposing that uninformed agents take no corrective action in the bad state, worsening the outcome — a feature that could readily be endogenized but is retained here for simplicity. Removing this choice lets participants in our experiment focus on the core decision to acquire or avoid information and eliminates potential confounds from introducing additional uncertainty about how many agents act on their information.

where $\Delta > 0$ and $\bar{u}_B < u_G$ is the utility obtained in the bad state when all group members acquire information. This makes it clear that each agent who avoids information worsens her own as well as everyone else's Date-2 utility by an amount Δ , representing the *instrumental value* of information.⁵

Since our experimental design elicits each agent's best response as a function of the share of fellow group members who avoid information, we work directly with this share, λ_{-i} , defined as

$$\lambda_{-i} \equiv \frac{\sum_{j \neq i} y_j}{N - 1},$$

where N is the number of agents in the group. Accordingly, we express i 's material payoff when she avoids information as $b(\lambda_{-i}) \equiv \bar{u}_B - \Delta[1 + (N - 1)\lambda_{-i}]$, so that $u_B = b(\lambda_{-i}) + \Delta(1 - y_i)$.

We focus on Nash equilibrium. An individual's *net* expected utility from avoiding information is

$$\begin{aligned} \varphi_i(\lambda_{-i}) &\equiv \mathbb{E}[v_i(U_i)|y_i = 1] - \mathbb{E}[v_i(U_i)|y_i = 0] \\ &= \underbrace{v_i((1 - q)b(\lambda_{-i}) + q u_G)}_{\text{Anticipatory utility when avoiding information}} - \underbrace{\left[(1 - q) \cdot v_i(b(\lambda_{-i}) + \Delta) + q \cdot v_i(u_G)\right]}_{\text{Expected anticipatory utility when acquiring information}}. \end{aligned} \quad (2)$$

Agent i 's *best response* is therefore

$$y_i^*(\lambda_{-i}) = \begin{cases} 1 & \text{if } \varphi_i(\lambda_{-i}) > 0, \\ 0 & \text{if } \varphi_i(\lambda_{-i}) < 0. \end{cases} \quad (3)$$

3.1 Best Response and Strategy Types

We now study how the shape of v_i affects an individual's best response. We begin by considering v_i that is either everywhere convex or everywhere concave.

Lemma 1 *If v_i is everywhere convex, individual i is an "Always Getter": she acquires information at Date 0 independently of the information choices of other group members.*

If v_i is convex, the utility gain from learning that the state is good outweighs the utility loss from learning that the state is bad. Consequently, the agent always prefers to acquire information. We next turn to the case in which v_i is concave, as in [Bénabou \(2013\)](#).

Lemma 2 (Bénabou, 2013) *If v_i is everywhere concave, individual i is either (i) an Always Getter, (ii) is an "Always Avoider", or (iii) exhibits "Strategic Complementarity": there exists a threshold $\lambda_i^* \in (0, 1)$ such that i strictly prefers to avoid information if $\lambda_{-i} > \lambda_i^*$ and strictly prefers to acquire information if $\lambda_{-i} < \lambda_i^*$.*

⁵The linearity of utility in y mirrors the model by [Bénabou \(2013\)](#). It highlights how complementarity and substitutability may arise from strategic, rather than mechanical considerations. Throughout the analysis, we assume that the expected Date-2 utility when uninformed exceeds the utility from acquiring information when the state is bad, i.e., $(1 - q)b(\lambda_{-i}) + q u_G > b(\lambda_{-i}) + \Delta$ for all λ_{-i} . This simplifies exposition but is not necessary for our results.

Appendix A.2 characterizes the necessary and sufficient conditions for each case. Always Getters and Always Avoiders make the same information acquisition choice independently of the choices of others in their group. Alternatively, the agent may condition her decision on the choices of others in her group, exhibiting strategic complementarity. Intuitively, the concavity of the anticipatory utility means that the distress from bad news outweighs the relief from good news. The instrumental value of information may nonetheless make information worth acquiring — but as more agents avoid information, the bad-state outcome worsens, amplifying the distress of discovering bad news. This raises the bar for information acquisition, generating strategic complementarity: an agent acquires information only if enough peers do the same.

We now show that strategic substitutability may also arise in our setting. This is a novel result relative to [Bénabou \(2013\)](#): in that paper, the presence of negative externalities from information avoidance necessarily generates strategic complementarity. Consider a reference-dependent v_i with reference utility level $\hat{u}_i$ such that $v_i''(u) < 0$ for $u > \hat{u}_i$ and $v_i''(u) > 0$ for $u < \hat{u}_i$.

Lemma 3 (Strategic substitutability) *If v_i is reference dependent, then under suitable conditions individual i exhibits “Strategic Substitutability”: there exists a threshold $\lambda_i^{**} \in (0, 1)$ such that i strictly prefers to avoid information if $\lambda_{-i} < \lambda_i^{**}$ and strictly prefers to acquire information if $\lambda_{-i} > \lambda_i^{**}$.*

Appendix A.2 provides sufficient conditions for strategic substitutability. Intuitively, consider v_i that is very sensitive to expected Date-2 utility near the reference point but is relatively flat away from it. Suppose Date-2 utility in the bad state is always below the reference level, while in the good state it is above it. When many group members avoid information, expected utility under ignorance falls below the reference level. Relative to remaining ignorant, bad news then carries only a modest loss in anticipatory utility while good news yields a substantial gain — so agent i prefers to acquire information. When instead many group members become informed, expected utility under ignorance rises above the reference level. Now bad news carries a large loss in anticipatory utility relative to good news, so agent i prefers avoidance. Note that reference dependence here operates over *anticipatory* rather than *consumption* utility — a meaningful departure from standard models à la [Kahneman & Tversky \(1979\)](#) and [Kőszegi & Rabin \(2006\)](#), in which utility depends on the gap between realized and reference consumption. In our setting, reference dependence operates one stage earlier: when contemplating future outcomes, agents experience anticipatory feelings that depend on how expected Date-2 utility compares to a reference utility level.

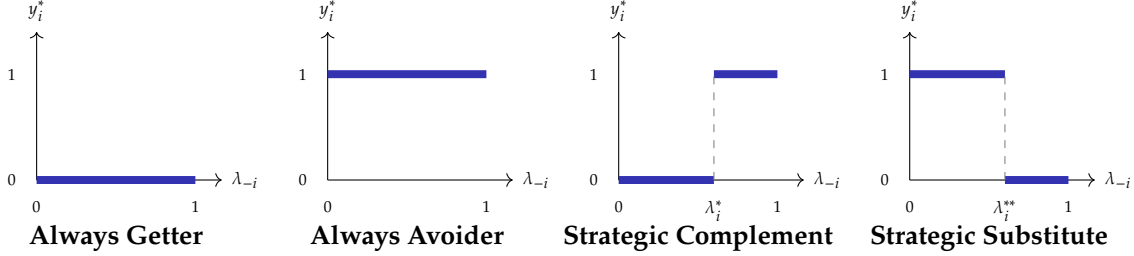
Another observation is that anticipatory utility may also be shaped by social preferences such as altruism or reciprocity. For example, agents may experience anticipatory feelings over their own Date-2 consumption utility and also, altruistically, over that of others. In Appendix A.3, we show that stronger altruistic concerns have an ambiguous effect on information avoidance, while stronger reciprocity increases the propensity for strategic complementarity.

The following proposition summarizes our main theoretical results.

Proposition 1 (Heterogeneity of interdependence) *Depending on the nature of v_i , individual i 's propensity to avoid information may be (i) independent of, (ii) increasing in (strategic complementarity), or (iii) decreasing in (strategic substitutability) the share of fellow group members who avoid information.*

We use the term “strategy type” to denote whether an individual is an Always Getter (AG), Always Avoider (AA), Strategic Complement (C), or Strategic Substitute (S).

Figure 2: Strategy types' best response functions



Notes: Each panel plots agent i 's best response $y_i^* \in \{0, 1\}$ against the share of other group members who avoid information, $\lambda_{-i} \in [0, 1]$, where $y_i^* = 1$ denotes avoiding and $y_i^* = 0$ denotes acquiring information.

Information-avoidance equilibria depend on the distribution of strategy types within the group. For example, [Bénabou \(2013\)](#) studies groups populated by Strategic Complements, where a “Mutually Assured Ignorance” (MAI) equilibrium — in which no one acquires information and no corrective action is taken — may arise, potentially with severe consequences.⁶ A “Mutually Assured Awareness” (MAA) equilibrium, in which all agents acquire information, is also possible, with the prevailing equilibrium depending on agents' expectations about others' behavior. Groups of Strategic Substitutes, by contrast, can only sustain equilibria with a mixture of information acquisition and avoidance. Let $\mu_k \in [0, 1]$ denote the share of strategy type $k \in \{C, S, AA, AG\}$ in a group, with $\mu_C + \mu_S + \mu_{AA} + \mu_{AG} = 1$. Among Strategic Complements (resp. Substitutes), suppose that λ_i^* (resp. λ_i^{**}) is distributed according to F_C (resp. F_S), and let r_C (resp. r_S) denote the share of Strategic Complements (resp. Substitutes) who avoid information in equilibrium.⁷

Proposition 2 *In equilibrium, the shares r_C and r_S of individuals who avoid information among Strategic Complements and Strategic Substitutes satisfy the following fixed-point conditions:*

$$(i) F_C(r_C \mu_C + r_S \mu_S + \mu_{AA}) = r_C \text{ and } (ii) 1 - F_S(r_C \mu_C + r_S \mu_S + \mu_{AA}) = r_S.$$

Corollary 1 (i) *In a group of Strategic Complements, both MAI and MAA equilibria may arise. When introducing Always Getters (resp. Always Avoiders) into such a group, there exists a threshold μ_{AG}^* (resp. μ_{AA}^*) such that, if the share of unconditional agents exceeds this threshold, MAA (resp. MAI) becomes the unique equilibrium. (ii) A positive share of Strategic Substitutes rules out both MAI and MAA equilibria.*

⁶[Bénabou \(2013\)](#) refers to this as a MAD (Mutually Assured Delusion) equilibrium. In the context of information avoidance, MAI provides a more natural label.

⁷For simplicity, we consider groups composed of a continuum of individuals.

4 Experimental Design and Procedures

4.1 Design

Guided by the theoretical model in Section 3, we design an experiment to study information avoidance in groups when remaining ignorant imposes negative externalities on others. The experiment is designed to establish whether information-avoidance decisions are causally interdependent and – crucially – to characterize heterogeneity in how individuals’ information choices respond to others’ behavior. To this end, we combine an informational intervention that exogenously shifts beliefs about peers’ information avoidance with a strategy-method elicitation of individual best-response schedules, complemented by additional measures that allow us to examine the mechanisms sustaining strategic complementarity.

4.1.1 Decision Environment

The experimental environment is designed to capture three core ingredients of the model in Section 3: (i) a payoff-relevant future consumption event, determined by an underlying state of the world and inducing anticipatory utility, (ii) a decision to acquire or avoid information about the state, and (iii) payoff externalities arising when group members acquire or avoid information. We build on [Falk & Zimmermann \(2024\)](#) by extending the experimental setting to groups with payoff externalities and by introducing methodological innovations, including a negative consumption event new to economics, the application of the strategy method to elicit complete best-response schedules and a novel measure of anticipatory utility. We describe each of these ingredients in turn.

Future consumption event. Participants are assigned to groups and informed that one of two payoff-relevant states of the world will be randomly drawn. The state is realized during a four-minute period at the end of the session (the “Risk Period”). If the group’s state is Bad (“Screams”), participants experience a negative consumption event in the form of hearing a series of aversive screams at random times during this period; if the state is Good (“Quiet”), no such event occurs. Crucially, the prospect of this event has been shown to generate sustained anticipatory anxiety, making information about the state relevant well before its realization ([Beaurenaut et al., 2020](#)).

Information decision. Before the realization of the future consumption event, each participant chooses whether to acquire information (option “Now”; $y_i = 0$) that perfectly reveals the group’s state of the world or avoid information (option “Later”; $y_i = 1$). Acquiring information resolves the uncertainty immediately by revealing the group’s state, while avoiding information postpones the resolution of uncertainty until just before the Risk Period. Importantly, “information decisions” are costless and simultaneously made, precluding any social learning. The value of the information

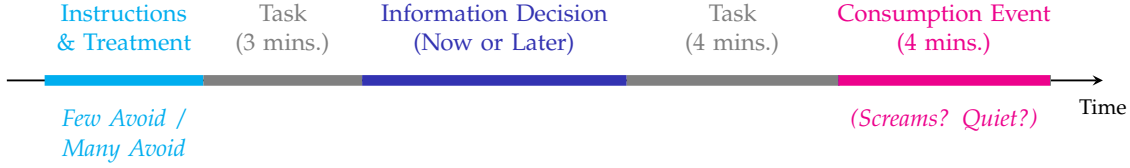


Figure 3: Timeline of experiment

decisions is therefore not monetary but psychological and instrumental: learning the state shapes anticipatory utility prior to the event and, in the bad state, enables corrective behavior.

Externality. Unlike studies of information avoidance in individual settings, our design features a payoff externality from information decisions. Acquiring information generates a positive return for both the decision-maker (through its standard instrumental value) and other group members (externality). In particular, in the Bad state, the severity of the negative consumption event depends on how many group members acquire information. Concretely, the *volume* of the screams decreases with the number of members who choose Now: each member who acquires information lowers the volume by two points. As a result, avoiding information is socially costly.⁸

Incentivized Task. Following [Falk & Zimmermann \(2024\)](#), participants complete an incentivized general-knowledge quiz at two points in time: once before the information decision and once between the information decision and the consumption event. Earnings increase with the number of correct answers and depend solely on individual performance. Apart from the show-up fee, this task constitutes the only source of monetary earnings in the experiment. The experimental timeline is summarized in Figure 3.

4.1.2 Elicitation of Best-Response Schedules

Proposition 1 characterizes heterogeneity in how individuals respond to others' information decisions. To identify this heterogeneity empirically, we use the strategy method ([Selten, 1967](#); [Fischbacher et al., 2001](#)) to elicit each participant's best-response schedule $y_i^*(\lambda_{-i})$ over a grid of possible shares of avoiders among other members (0%, 20%, 40%, 60%, 80%, and 100%).

The elicitation is incentive-compatible: with strictly positive probability, a participant's conditional choice is implemented. Specifically, after all group members make their unconditional information decisions, we randomly select all but one group member and compute the realized share of avoiders among them; the remaining participant's implemented choice is then her stated best response to that share.

⁸Each member's contribution to volume reduction is independent of others' choices (additive separability). We therefore study whether strategic interdependence in information decisions can arise even in the absence of any in-built interdependencies in the technology. See Section 4.2 for details.

Following the theoretical classification illustrated in Figure 2, we classify participants as Strategic Complements (Substitutes) if they choose Later only when the share of avoiders is sufficiently large (small). Participants who choose the same option regardless of others' choices are classified as Always Getters (always acquire) or Always Avoiders (always avoid). Participants switching multiple times are classified as "Others".

4.1.3 Information Provision Treatment

The strategy method identifies individual-level heterogeneity in best responses. Complementing this, we implement a between-subjects information provision treatment (e.g., [Haaland et al., 2023](#)) that exogenously shifts participants' expectations about the prevalence of avoidance among other members in their group (λ_{-i}). This belief variation serves as a gateway for assessing average interdependence and for probing the mechanisms underlying the heterogeneous best-response patterns identified above. Participants are randomized at the group level into two conditions: *Many Avoided* and *Few Avoided*. In *Many Avoided* (*Few Avoided*), participants are informed that 80% (20%) of participants in a group from a similar past pilot study chose Later.⁹

To verify that the manipulation works as intended, we elicit the participants' posterior beliefs about the share of avoiders among other group members after the information provision.

4.1.4 Additional Measurements

Anticipatory utility. To shed light on the mechanism, we introduce a novel elicitation designed to measure anticipatory utility over future consumption events. Crucially, we measure anticipatory utility *conditional on each information set*. Specifically, we ask participants:

- i) "How happy would you be *thinking about the outcome ahead* if you chose Now and then discovered that the lottery outcome is Screams?";
- ii) "How happy would you be *thinking about the outcome ahead* if you chose Now and then discovered that the lottery outcome is Quiet?"; and
- iii) "How happy would you be *thinking about the outcome ahead* if you chose Later?"

These questions allow us to measure anticipatory utilities $v_i(b(\lambda_{-i} + \Delta))$, $v_i(u_G)$, and $v_i((1-q)b(\lambda_{-i}) + q u_G)$, respectively. To infer the net utility of avoiding information, φ_i , we also elicit $(1 - q_i)$, the participant's subjective belief that the state is Screams. Together, these four questions measure the components of Equation (2), which defines the net utility of avoiding versus acquiring information. We refer to this elicitation as the *Kreps-Porteus* (KP) questions.

⁹We also inform each participant that, if other group members behave like pilot participants, the volume in the Screams state would be 89 or 91 in *Many Avoided* (60 or 62 in *Few Avoided*), depending on whether she herself avoids or acquires information. The 2-point difference within each condition reflects the focal participant's choice; the difference between conditions reflects others' choices.

Our theory predicts an asymmetric effect: as the share of avoiders λ_{-i} increases, the bad-state payoff deteriorates, lowering the anticipatory utility, $v_i(b(\lambda_{-i} + \Delta))$, while leaving anticipatory utility upon learning good news, $v_i(g)$, unchanged (Section 3). We test this prediction by examining whether the treatment affects only the former.

Baseline attitudes towards information. We administer the “Information Preferences Scale” (IPS; [Ho et al., 2021](#)), which measures an individual’s willingness to obtain potentially undesirable but instrumentally valuable information across multiple domains.¹⁰ Higher IPS scores indicate a stronger general preference for obtaining information. We use the IPS to (i) assess balance across experimental conditions, (ii) control for baseline information preferences in the analysis, and (iii) provide a benchmark for our KP elicitation.

Social preferences. Theoretically, social preferences interact with the interdependence of information avoidance (Appendix A.3). To assess this relationship empirically, we elicit individual measures by administering selected items from the Global Preferences Survey (GPS; [Falk et al., 2018](#)), capturing altruism, reciprocity, trust, risk attitudes, and time preferences. Each dimension is elicited through responses to two questions. Following [Falk et al. \(2018\)](#), we standardize responses and transform them into z-scores, with higher values indicating greater risk seeking, patience, altruism, reciprocity, and trust.

4.2 Procedures

4.2.1 Session Structure

At the start of each session, participants were matched into groups, and each group was randomly assigned to a treatment condition.¹¹ Participants then received detailed instructions on computer screens.¹² The information provision treatment was implemented during the instructions phase.

A computerized lottery determined the group outcome (Screams or Quiet). If the outcome was Screams, during the Risk Period all group members listened to a sequence of unpleasant screams through their headphones, drawn from the validated stimulus set in [Beaurenaut et al. \(2020\)](#).¹³ To ensure hearing safety, the maximum possible volume was calibrated at 80dB in accordance with

¹⁰The IPS consists of 18 questions eliciting subjects’ willingness to obtain information in hypothetical scenarios across five domains: health, finance, social relations, ego-relevant characteristics, and occupation. Responses range from “Definitely don’t want to know” (1) to “Definitely want to know” (4). Following [Ho et al. \(2021\)](#), we construct an overall IPS score by averaging responses across all 18 items, as well as domain-specific subscales.

¹¹Treatment assignment was blocked by week of year and time of session (morning and afternoon).

¹²Full instructions are available in Appendix D. The experiment was programmed in Lioness ([Giamattei et al., 2020](#)).

¹³Participants did not hear any sample screams before making information decisions, but were informed that “[p]revious research has shown that people consider similar screams as *more aversive (disturbing)* than the sound of *nails sliding on a chalkboard*” (see [Kumar et al., 2009](#)).

UK regulations. Headphone use was mandatory throughout the session, and compliance was monitored by experimenters.

We use aversive auditory stimuli (screams) as the negative future consumption event. Prior work shows that such stimuli generate sustained anticipatory anxiety, which is particularly suitable for group settings that unfold over longer experimental periods (Beaurenaut et al., 2020). The severity of the event can be manipulated in a fine-grained and intuitive way through volume, a feature that becomes especially relevant as group size increases and the marginal impact of any individual’s information acquisition diminishes. The use of auditory stimuli also facilitates implementation in larger groups and alleviates ethical concerns associated with more invasive procedures. In our setting, the relevant theoretical requirement is that the future event is perceived as undesirable. In Section 5.1, we provide evidence that this condition is satisfied in our sample.

Conditional on a *Screams* outcome, the volume experienced by participant i depended on group members’ information decisions. Let N denote group size and let $y_i = 1$ if participant i chose Later (and 0 otherwise). The volume was given by

$$vol_i = 50 + \frac{50}{N} \left(y_i + \sum_{j \neq i} y_j \right). \quad (4)$$

Each group correspond to a full experimental session, with approximately $N = 30$ participants. Avoiding information imposed a cost (in terms of volume increase) on all group members, including the decision-maker, equal to $50/N \approx 1.67$ volume points.¹⁴

While the volume function in Equation (4) may resemble a Public Good Game (PGG), our decision environment differs in key respects. In a standard PGG, contributing entails a known and homogeneous monetary cost that is independent of others’ actions. Here, there is no monetary cost of contributing: instead, the cost is psychological-stemming from the possibility of learning bad news – and is heterogeneous across individuals and endogenous to others’ information choices, which affect the severity of the bad state. As a result, a purely material-payoff-maximizing agent would always prefer to acquire information in our setting, whereas free-riding is optimal in a standard PGG. Accordingly, the central question here is not why individuals contribute, but why some choose to avoid information and how this decision depends on others’ choices.

The binary information decision (Now vs. Later) is our main outcome of interest. As a complementary, continuous measure, we ask participants – after choosing Now or Later – to report the strength of preference in relation to their choice using a Likert scale ranging from 0 (“indifferent”) to 10 (“very strong preference”). We construct the Information Avoidance (IA) scale by assigning this value a positive sign for participants who choose Later and a negative sign for those who choose Now. The IA scale ranges from -10 (very strong preference for Now) to +10 (very strong

¹⁴We described this to participants a change of “around 2 out of 100 volume level points.”

preference for Later), with 0 indicating indifference, and provides more variation than the binary outcome. Participants also report their reasons for the chosen option in an open text box.

4.2.2 Recruitment, Sample and Earnings

The experiment was conducted at the Centre for Decision Research and Experimental Economics (CeDEx) Lab, University of Nottingham (UK).¹⁵ We ran 17 experimental sessions, each forming an independent group, across two waves of data collection (December 2023 and May-June 2024), recruiting participants from the CeDEx subject pool. The final sample comprised 465 participants, with an average of roughly 27 subjects per session.¹⁶ Each session lasted about one hour, and participants earned on average £12.52 (± 2.00), including a £5 show-up fee. Full descriptive statistics of baseline characteristics are reported in Table B.1 (Appendix B.1).

5 Results

This section presents the analysis of our experimental data. We begin by establishing two necessary preconditions for the study. In Section 5.1, we document the existence of information avoidance in our decision environment and verify that, consistent with our assumptions, participants perceive the prospect of listening to screams as a bad. We then turn to our primary results. Our central objective is to characterize the structure of strategic interdependence in information avoidance decisions at the individual level. In Section 5.2, we use the strategy method to recover participants' best-response schedules and document substantial heterogeneity in strategic behavior, including both complementarity and substitutability. In Section 5.3, we investigate the mechanisms underlying strategic complementarity. Finally, in Section 5.4, we use experimental variation in beliefs to establish that information decisions are causally interdependent on average.

As we show below, individuals differ systematically in how they respond to others' choices, exhibiting both strategic complementarity and strategic substitutability. These heterogeneous best-response types generate offsetting aggregate effects, naturally attenuating the average treatment effect, yet imply large and highly non-linear differences in equilibrium information avoidance depending on group composition. We therefore treat the average treatment effect as a gateway result — establishing the existence of causal interdependence — and place primary emphasis on individual heterogeneity and its implications for group-level outcomes. Formally, aggregate responses—including the average treatment effect— can be interpreted as composition-weighted averages of

¹⁵The experiment was preregistered, AsPredicted #153871 and #175588, and was approved by the University of Nottingham's School of Economics Research Ethics Committee.

¹⁶Originally, we recruited 493 participants in 18 sessions. Due to technical issues, data were not recorded for 22 subjects in one session of 28, who are therefore excluded from the analysis; results are robust to the inclusion of the 6 participants from that session for whom data are available. In a separate incident, one participant left early reducing observations for final-stage variables by one.

individual best-response functions. From this perspective, average effects provide only a reduced-form summary of an underlying heterogeneous strategic environment.

5.1 The Prevalence of Information Avoidance

We first conduct a validation check that participants actually perceive the prospect of listening to screams as a bad. Our evidence suggests this is the case. First, we asked subjects how “(un)happy” they would be if screams were played at volume 50 and, separately, at volume 100. A large majority of subjects (82%) report strictly lower utility levels for screams at higher volumes.¹⁷ Second, we asked subjects to rate their “(un)happiness” upon discovering that the state was Screams (resp. Quiet) after choosing Now. As expected, 81 % report that discovering that the Screams state is strictly worse than discovering that the Quiet state.¹⁸

We find a positive and statistically significant share of information avoiders. Panel A of Figure 4 shows the prevalence of avoidance by data collection wave and for the pooled sample. In the pooled data, the share of information avoiders is approximately 28%. This share is stable across data collection waves.¹⁹

Panel B of Figure 4 plots the distribution of preference strength among avoiders (people who chose Later), elicited on a 0-10 scale (0=“indifferent”; 10=“very strong preference”). The average strength of preference is 6.6, well above indifference. Moreover, the cumulative distribution (orange line) shows that only 6% of avoiders report indifference, while more than 80% report a strength of at least 5. Thus, evidence suggests that most information avoiders strongly prefer to remain uninformed.

5.2 Strategy Types

This section presents our main results. As Proposition 1 shows, theory allows substantial heterogeneity in best-response behavior: some individuals may be more likely to avoid information when others do so (strategic complementarity), while others may react in the opposite direction (strategic substitutability), or not condition their decisions on others at all.

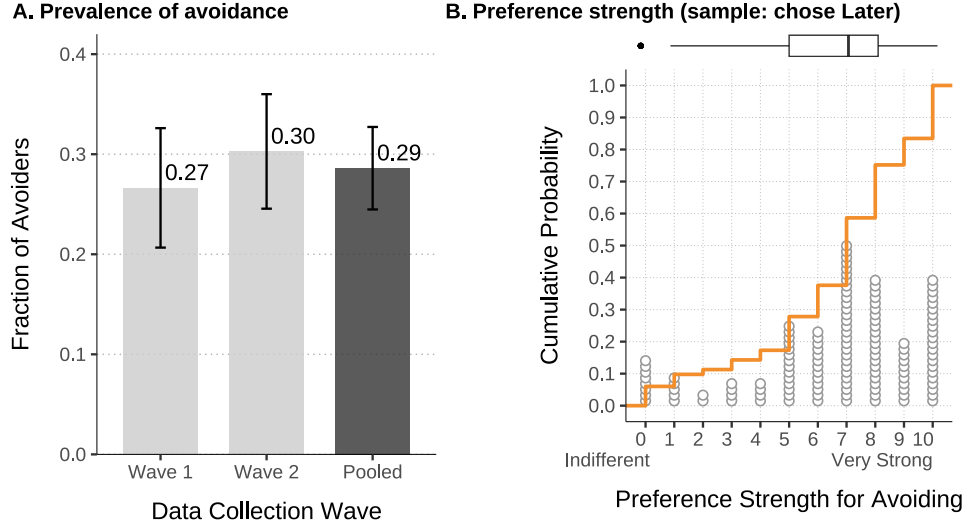
We use the strategy method to recover each participant’s full best-response schedule – namely, how their information decision varies with the expected shares of avoiders in their group. This

¹⁷Among the remaining subjects, 3% are indifferent between volume levels, and 9% would prefer volume 100. We call the latter type of subjects “volume-lovers”.

¹⁸Among the remaining subjects, 6% are indifferent between the two states and 14% would prefer discovering the Screams state. We call the latter type of subjects “scream-lovers”. The distribution of types of preference screams and volume is shown in Figure B.2 and Figure B.1 (Appendix B.2). All our main results are robust to excluding “scream lovers” and “volume lovers”, as shown in Appendix B.6.

¹⁹In an individual decision making context, [Falk & Zimmermann \(2024\)](#) find that 52% of subjects avoid information, i.e. a higher share than in our study. Note however that in their setup information acquisition provides no material benefit (i.e., it does not affect the severity of the future negative consumption event), eliminating instrumental and prosocial motives for information acquisition.

Figure 4: Information avoidance: prevalence and preference strength



Notes: Panel A shows the fraction of subjects who chose Later (avoided information), by data collection wave (light-gray bars) and in the full sample (dark-gray bar). Whiskers indicate 95% confidence intervals. Pairwise comparisons using two-sample proportion tests yield p-values greater than 0.4. Panel B restricts attention to the subset of subjects who chose Later (avoided information) and shows the distribution of the reported strength of preference for this choice (0 = “I am indifferent” between Now and Later; 10 = “I have a very strong preference” for the chosen option). Gray circles represent the absolute frequency of each preference strength (one dot per subject). The orange line shows the empirical cumulative distribution function (CDF). The boxplot shows the median and the interquartile range, with whiskers extending to the most distant point within 1.5 times the interquartile range.

allows us to directly characterize the structure of interdependence at the individual level and to decompose aggregate behavior into underlying strategy types. To complement this characterization, the experiment also includes an information-provision treatment that shifts participants’ beliefs about others’ behavior, which we exploit below to study how individuals move along their best-response schedules.

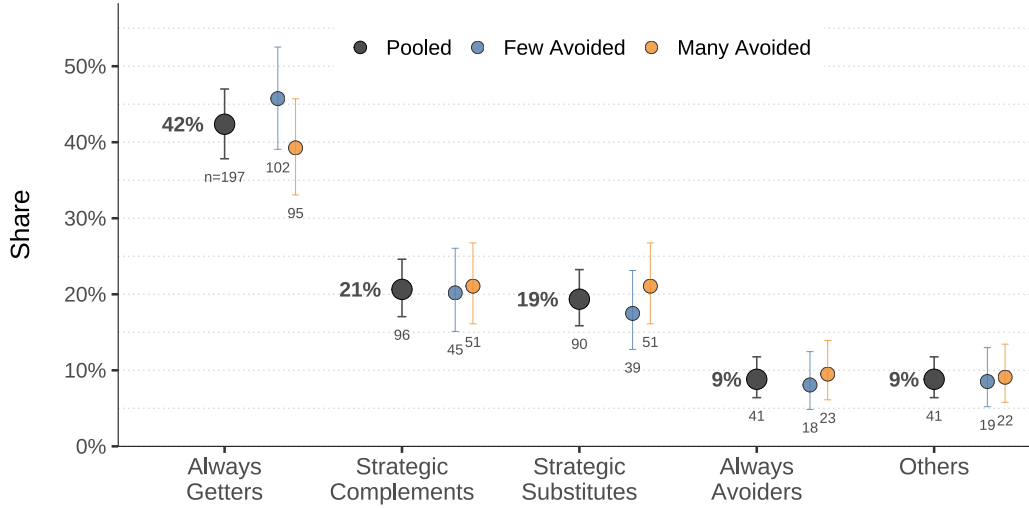
The data reveal substantial heterogeneity in how individuals respond to the information avoidance decisions of others. Figure 5 reports the empirical distribution of individual strategy types in our sample (black dots). The most and least prevalent types are Always Getter (42.4%) and Always Avoider (8.8%), whose information acquisition choices are independent of others’ choices.

On the other hand, approximately 40% of participants exhibit interdependent strategies. Among them, 21% are Strategic Complements: they avoid information only if a sufficiently high share of their group members also avoids it. A similar proportion (19.4%) are Strategic Substitutes, who avoid information only if a sufficiently *low* share of their group members avoids it.²⁰ Therefore, we find substantial heterogeneity in strategy types, with a significant share of individuals exhibiting dependence on the choices of others.

We now analyze how individuals of different strategy types respond to changes in beliefs in-

²⁰The remaining minority (8.8%) are subjects who switch their information decisions multiple times and, hence, whose best responses oscillate.

Figure 5: Distribution of strategy types



Notes: Empirical distribution of strategy types, by treatment and pooled. Dots indicate the share of each strategy type in the full sample (black), as well as within the *Few Avoided* (blue) and *Many Avoided* (orange) treatment conditions. Error bars represent 95% confidence intervals. Numbers below the error bars indicate the sample size in each cell.

duced by the treatment conditions, along both the extensive margin (whether the treatment conditions affect the distribution of types) and the intensive margin (how the treatment changes the probability of avoidance conditional on the type).

On the extensive margin, we find no effect of the treatment conditions on the distribution of strategy types. In Figure 5, blue and orange dots show the distribution of types in the *Few Avoided* and *Many Avoided* treatment conditions, respectively. The distributions appear remarkably similar across treatments. A regression analysis confirms this graphical intuition. Table B.11 (Appendix B.10) reports estimates from regressions where a dummy for each strategy type is regressed on the treatment indicator. In these regressions, the treatment coefficient captures the change in the proportion of each type between *Many Avoided* and *Few Avoided*. No statistically significant differences emerge for any type.

The distributional invariance across treatments is consistent with theory. Best-response *schedules* themselves should not change unless some (perceived) parameter of the environment—beyond the expected behavior of others—changes. Since the treatment was designed to change only the expectation about others' behavior, but no other feature of the environment, there is no apriori reason to expect the distribution of types to shift across treatments. Accordingly, we find that, empirically, the distribution of strategy types remains similar between treatments.

We now turn to the intensive margin. In Figure 6, Panel A compares the share of avoiders (who chose Later) between treatment conditions conditional on the strategy type. Table B.12 reports a linear regression of the avoidance dummy on the treatment for each type. In these regressions, the treatment coefficient captures a Conditional Average Treatment Effect (CATE): the difference in

the likelihood of avoidance between the *Many Avoided* and *Few Avoided* conditional on the strategy type. We find that the *Many Avoided* treatment significantly increases the likelihood of information avoidance for Strategic Complements. This is in line with theoretical intuitions. Strategic Complements are characterized by increasing best-response schedules, so higher beliefs about others' avoidance should lead them to avoid more often. As we document in Section 5.4, the experimental treatment indeed induces a substantial upward shift in such beliefs, providing the required variation to move individuals along their schedules. By contrast, however, we found no significant response among Strategic Substitutes.

The difference in responsiveness between Strategic Complements and Substitutes can be explained by the interaction between switching threshold distributions and the treatment-induced shift in beliefs, shown in Panel B of Figure 6. Average beliefs are shown in pink lines and average switching threshold in black lines. While the treatment leaves the average switching thresholds unchanged for both types, it substantially increases average beliefs.²¹ For Strategic Complements, the belief shift crosses the threshold, triggering increased avoidance. For Strategic Substitutes, however, the new beliefs remain on the same side of the threshold, resulting in no significant change in average avoidance rates. As expected, the treatment does not affect the avoidance rate for Always Getters and Always Avoiders, who, by definition, follow strategies that are irresponsive to the behavior of others (see Appendix B.11). Taken together, the pattern of intensive margins of treatment effects by types suggest that the positive average treatment effect documented in Section 5.4 is primarily driven by the responsive behavior of Strategic Complements.

Taken together, these results show that aggregate responses to belief changes arise from movement along heterogeneous best-response schedules, rather than from a uniform behavioral response. Strategic Complements respond strongly within the range of beliefs induced by the treatment, while Strategic Substitutes and unconditional types remain largely unresponsive. The average treatment effect therefore reflects a composition-weighted aggregation of these heterogeneous responses.

This perspective highlights that aggregate information avoidance is determined by the distribution of strategy types and their best-response functions. As a result, small average effects can mask large differences in equilibrium behavior across groups with different compositions.

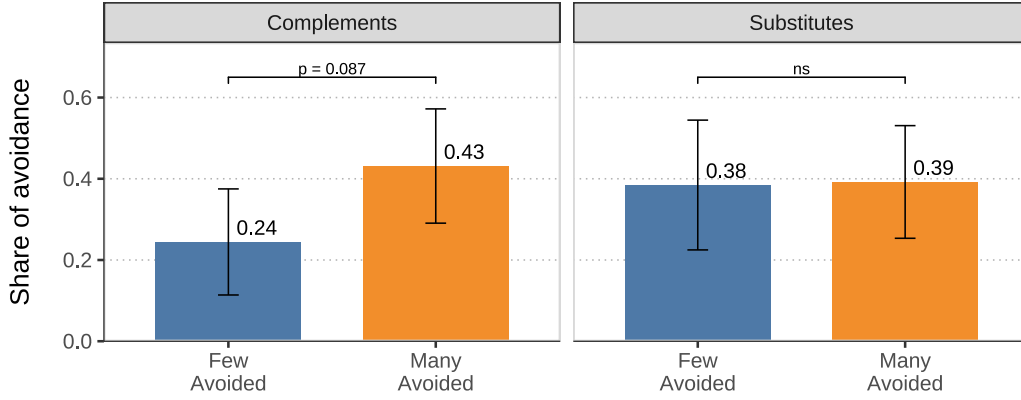
5.3 Strategic Complements and Mutually Assured Ignorance

Section 5.2 documented substantial heterogeneity in best-response behavior and showed that the positive average effect of higher expected avoidance by others is driven entirely by Strategic Complements. While the strategy method directly identifies how individuals respond to others' behav-

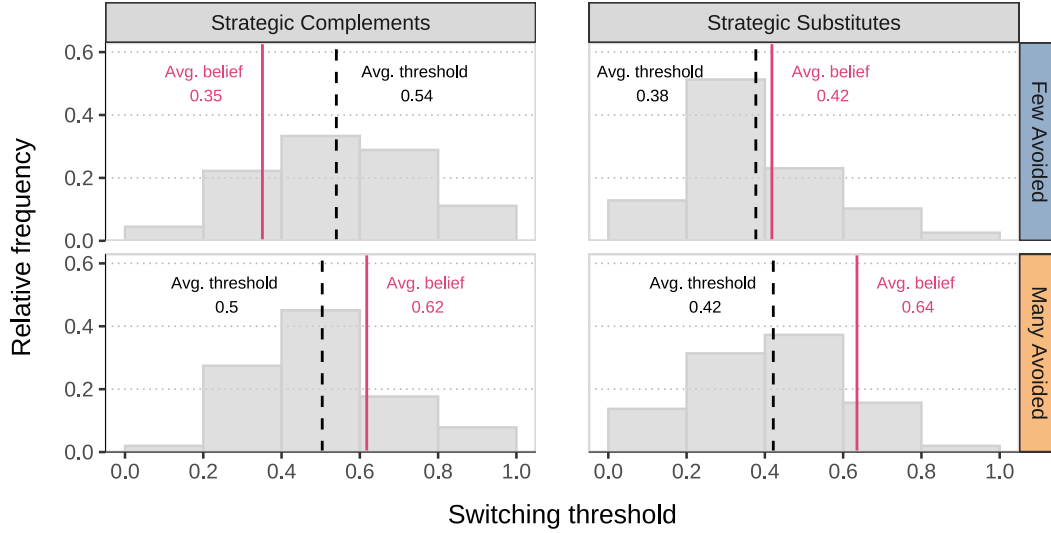
²¹As shown in Figure 6, average switching thresholds are approximately 0.5 for Strategic Complements and 0.4 for Strategic Substitutes, while average beliefs increase from 0.35 to 0.62 for Complements and from 0.42 to 0.64 for Substitutes.

Figure 6: Treatment effect heterogeneity by strategy type

A. Treatment effect on avoidance



B. Switching thresholds and beliefs



Notes: The figure shows the heterogeneity of treatment effects on information avoidance across interdependent types. Panel A shows the fraction of avoiders by treatment condition, separately for Strategic Complements and Strategic Substitutes. Error bars represent 95% confidence intervals. Panel B shows the switching thresholds—the fraction of others avoiding information at which a subject switches her own information decision—and beliefs about that fraction by strategy type (columns) and treatment condition (rows). The distribution of switching thresholds is shown by the histogram (gray bars); the average switching threshold by black, dashed lines; and the average belief about the fraction of others who avoid information by pink, solid lines.

ior, it is silent on *why* some individuals exhibit increasing best responses. We therefore examine a set of psychological and preference-based channels that may give rise to strategic complementarity in information avoidance, which could be driven by a variety of mechanisms. We first focus on the “Mutually Assured Ignorance” (MAI) channel, formalized by [Bénabou \(2013\)](#) and central in our model: the idea that when a subject expects others to remain ignorant about the state, her own payoffs in the bad state become worse, making the event of discovering bad news about the state more aversive. Below we provide evidence supporting this mechanism by using our KP mea-

asures of expected anticipatory utilities conditional on finding out good, bad, and no news. We then conclude this section by briefly discussing the plausibility of alternative mechanisms.

Anticipatory utility and information avoidance. To investigate the mechanisms underlying strategic complementarity in information avoidance, we use the KP questions, which elicit participants' expected anticipatory utility *conditional on each possible information set*: discovering good news ($\hat{v}_i(g)$), bad news ($\hat{v}_i(b + \Delta)$), and no news ($\hat{v}_i((1 - \hat{q}_i)b + \hat{q}_i g)$). In addition, we elicit their subjective belief that the state is bad ($1 - \hat{q}_i$). Because these questions are elicited prior the information decision, we interpret them as *expected* anticipatory utilities, evaluated before any information is acquired.

Based on the KP responses, we compute for each participant the expected anticipatory utility of acquiring information as $U_{0,i}^{Get} \equiv (1 - \hat{q}_i) \cdot \hat{v}_i(b + \Delta) + \hat{q}_i \cdot \hat{v}_i(g)$ and the expected anticipatory utility of avoiding information as $U_{0,i}^{Avoid} \equiv \hat{v}_i(1 - \hat{q}_i)b + \hat{q}_i g$. We then define the net expected anticipatory utility of avoiding information as

$$\begin{aligned}\hat{\varphi}_i &\equiv U_{0,i}^{Avoid} - U_{0,i}^{Get} \\ &= \hat{v}_i((1 - \hat{q}_i)b + \hat{q}_i g) - [(1 - \hat{q}_i) \cdot \hat{v}_i(b + \Delta) + \hat{q}_i \cdot \hat{v}_i(g)],\end{aligned}$$

which serves as our empirical estimate of the theoretical construct φ_i .

The model predicts that an individual i should avoid information whenever the net anticipatory utility is positive ($\varphi_i > 0$). We test this prediction by regressing the avoidance dummy y_{ig} on the empirical counterpart $\hat{\varphi}_i$, either as a continuous measure or as a dummy variable for $\hat{\varphi}_i > 0$, and benchmark these specifications against the IPS.²² All regressions control for field of study, strata fixed effects, demographic characteristics, and economic preferences.

First, consistent with the theoretical prediction, we find anticipatory utility is a strong predictor of information avoidance. A unit increase in the continuous measure $\hat{\varphi}_i$ is associated with a sizable increase in the probability of avoiding information ($\hat{\beta} = 0.023$, std.err. = 0.003, $p < 0.01$). Similarly, when focusing on the specification with the dummy variable for individuals with $\hat{\varphi}_i > 0$, individuals with a positive net anticipatory utility are substantially more likely to avoid information than those with $\hat{\varphi}_i < 0$ ($\hat{\beta} = 0.372$, std.err. = 0.046, $p < 0.01$). By comparison, the IPS is a statistically significant but much weaker predictor of avoidance ($\hat{\beta} = -0.129$, std.err. = 0.052, $p < 0.05$). Including the net utility measures increases the R^2 from 0.059 to 0.185 (continuous measure) and 0.211 (binary measure), compared to only 0.071 when including the IPS. Importantly, when both the IPS and KP-based measures are included in the same specification, the coefficients on the KP measures remain stable in magnitude and significance, indicating that anticipatory utility captures

²²The dummy is equal to 0 if $\hat{\varphi}_i < 0$, and encoded as a missing value for boundary cases where $\hat{\varphi}_i = 0$ (12 individuals in our sample). Therefore, the number of observations drops by 12 in regressions that include the dummy.

decision-relevant, environment-specific variation beyond general preferences for information.

Second, consistent with this interpretation, when the KP measures are included alongside the treatment indicator, we find the *Many Avoided* estimated coefficient attenuates markedly: starting from $\hat{\beta} = -0.122$ (std.err. = 0.049, $p < 0.05$), it shrinks toward zero and loses statistical significance ($\hat{\beta} = -0.081$, std.err. = 0.045, $p < 0.10$) once $\hat{\phi}_i$ is controlled for. By contrast, including the IPS alone leaves the *Many Avoided* coefficient largely unchanged ($\hat{\beta} = -0.123$, std.err. = 0.049, $p < 0.05$). This pattern suggests that the treatment operates primarily through changes in anticipatory utility. Complete regression results are reported in Appendix B.12.

To further illustrate this point, Figure 7 plots the relationship between the probability of avoiding information and the empirical net anticipatory utility $\hat{\phi}_i$, using a probit specification. The probability of avoidance increases with $\hat{\phi}_i$. Crucially, this relationship is statistically indistinguishable across the *Many Avoided* and *Few Avoided* conditions. We find the same pattern when estimating the model separately for Strategic Complements and Substitutes, where there is variation in behavior. This pattern of invariance is consistent with the theoretical prediction that, conditional on anticipatory utility, neither treatment conditions nor strategy type should have additional explanatory power for information avoidance.

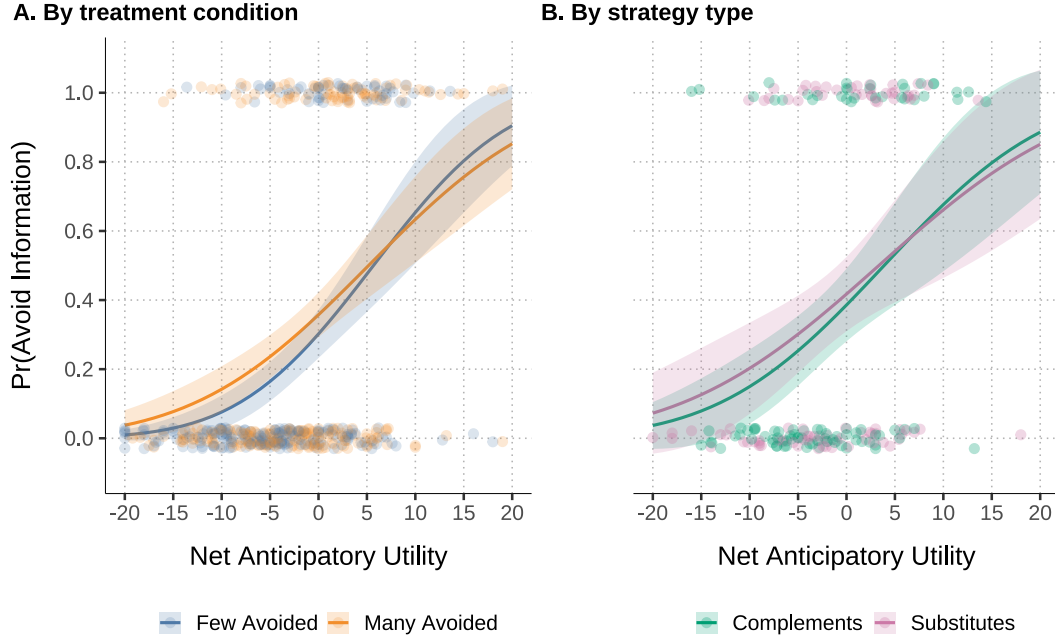
Overall, these results indicate that the KP measures proxy relevant determinants of information avoidance and provide support for their role as a key channel through which the treatment affects behavior.

Prevalence of avoidance and the MAI mechanism. We now decompose the effect of the *Many Avoided* condition on anticipatory utility to assess the plausibility of the MAI mechanism.

Figure 8 reports the effects of *Many Avoided* on the net anticipatory utility of avoiding information and on of its components, based on separate regressions with and without controls. The *Many Avoided* condition significantly increases the net utility of avoiding information. This effect is driven by a deterioration in the anticipatory utility of discovering *bad* news: in the *Many Avoided* condition, learning bad news becomes significantly more unpleasant relative to the *Few Avoided* condition. By contrast, the anticipatory utility of discovering *good* news is unaffected, consistent with the fact that the payoff in the good state (silence, g) is invariant with respect to the number of avoiders in the group. This pattern of asymmetric effects provides evidence for the theoretical MAI mechanism: when others avoid, the bad state deteriorates, making learning bad news more distressing and driving information avoidance.

The *Many Avoided* condition also increases subjects' belief that the state is bad. In principle, this raises the possibility that belief updating may interact with and amplify the MAI mechanism. However, existing experimental evidence suggests that exogenous shifts in prior beliefs about the bad state do not, by themselves, generate substantial changes in information avoidance (Falk & Zimmermann, 2024). Accordingly, the dominant channel operated by the *Many Avoided* condition

Figure 7: Anticipatory utility and information avoidance (Probit)



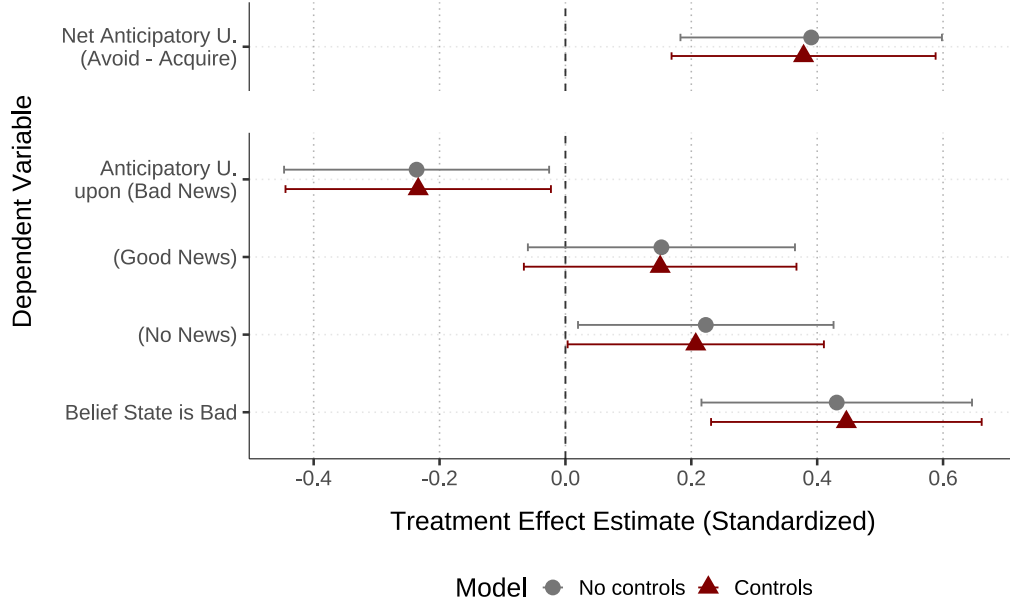
Notes: The figure illustrates the relationship between the measured net expected anticipatory utility of avoiding information and the propensity to avoid information. The net utility is calculated as the difference between the utility of avoiding information and the expected utility of acquiring information, elicited from the KP questions. Each point represents an individual participant's data, jittered vertically. The solid line shows the predicted probability of information avoidance from a probit regression model, estimated separately for each treatment condition (panel A) and for each strategy type (panel B). The shaded bands indicate the 95% confidence intervals.

in our setting appears to be the heightened disutility of discovering bad news when others avoid information.

Taken together, the decomposition exercise in Figure 8 shows stronger incentives to avoid information in groups arise primarily from the expected deterioration of anticipatory utility in the event bad news, consistent with the MAI mechanism of [Bénabou \(2013\)](#) and with our model.

While the evidence is consistent with the MAI channel, the interdependence of information avoidance could, in principle, arise from other behavioral mechanisms. We consider alternative explanations and find limited support for them. Conformity and information herding would predict that individuals are more likely to avoid information when they believe others have done so, *independently* of beliefs about the state; however, we find that beliefs about others' behavior are predictive of choices only when interacted with beliefs about the state (see Table B.14). In addition, the *Many Avoided* condition does not shift switching thresholds for strategic complements (Figure 6), nor the (anticipatory) utility of acquiring information and finding out good news, where both externalities and instrumental motives are shut down (Figure 8). Punishment also receives little support: although this mechanism predicts stronger effects of the *Many Avoided* condition among more negatively reciprocal subjects, the interaction between the condition and negative

Figure 8: Anticipatory utility and MAI mechanism



Notes: The figure reports the effects of the *Many Avoided* condition on net anticipatory utility and each of its components from separate regressions. Gray circles show estimates from models without further controls; red triangles correspond to models that include controls for demographics, preferences, and fixed effects (see notes in Figure 10). The top row reports effects on the net anticipatory utility (of avoiding information relative to acquiring it); lower rows report effects on its components: the anticipatory utilities upon learning bad, good, or no news, and the subjective belief that the state is bad. Coefficients are standardized; whiskers indicate 95% confidence intervals (robust standard errors).

reciprocity is insignificant (Table B.5). Finally, we consider non-linear preferences over volume reductions. For this channel to explain the results, perceptions of the instrumental value of information would have to differ substantially across conditions, despite the fact that the private return to information acquisition (a 2/100 volume change) is relatively small.

Overall, the evidence suggests that strategic complementarity in information avoidance is primarily driven by changes in anticipatory utility, consistent with the MAI mechanism; we next examine how heterogeneity in strategy types shapes equilibrium information avoidance at the group level.

5.4 The Interdependence of Information Avoidance

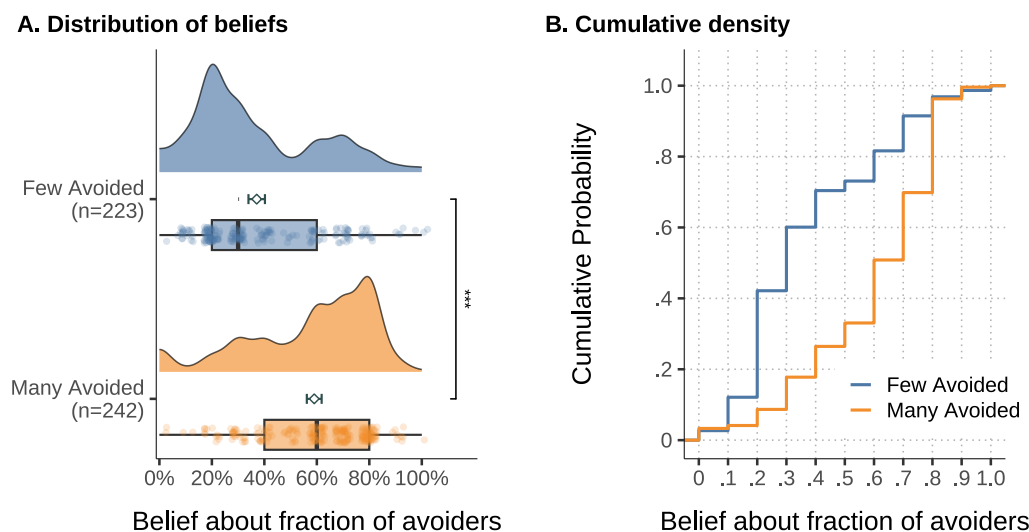
We now study the effect of exogenous variation in beliefs about the information acquisition choices of other individuals. Our aim here is not to document a large aggregate response, but to verify that individual information decisions react systematically to expectations about peers' behavior. As we show below, the resulting average effect is positive and statistically significant, though its magnitude reflects the aggregation of heterogeneous and offsetting individual responses — which we characterized in detail in the previous section.

Balance checks for each treatment are reported in the Table B.3. Importantly, IPS scores are

balanced between treatment groups, both for the overall scale and for each of the domain-specific subscale. Hence, differences in information avoidance between treatments cannot be attributed to baseline differences in general or domain-specific attitudes toward information.²³

Our treatment manipulation successfully shifts expectations about the share of avoiders among other members in the group. Figure 9 reports the effect of the information-provision treatment on beliefs about others' information decisions. Panel A displays the distribution and summary statistics of beliefs by treatment condition, while Panel B shows the corresponding empirical CDFs. Mean beliefs about the share of avoiders (indicated by hollow diamonds) were 37% in the *Few Avoided* condition versus 59% in the *Many Avoided* condition, a highly significant difference of 22 percentage points ($p < 0.001$). Table B.4 reports OLS estimates from a linear regression of beliefs on the treatment dummy and shows that the estimate of the average treatment effect is robust—to the inclusion of different sets of controls. Moreover, the distributional evidence in Panel B indicates a first-order stochastic dominance of beliefs in the *Many Avoided* condition relative to *Few Avoided*. In sum, the information-provision treatment performs as intended in shifting beliefs about others' behavior.

Figure 9: Manipulation check



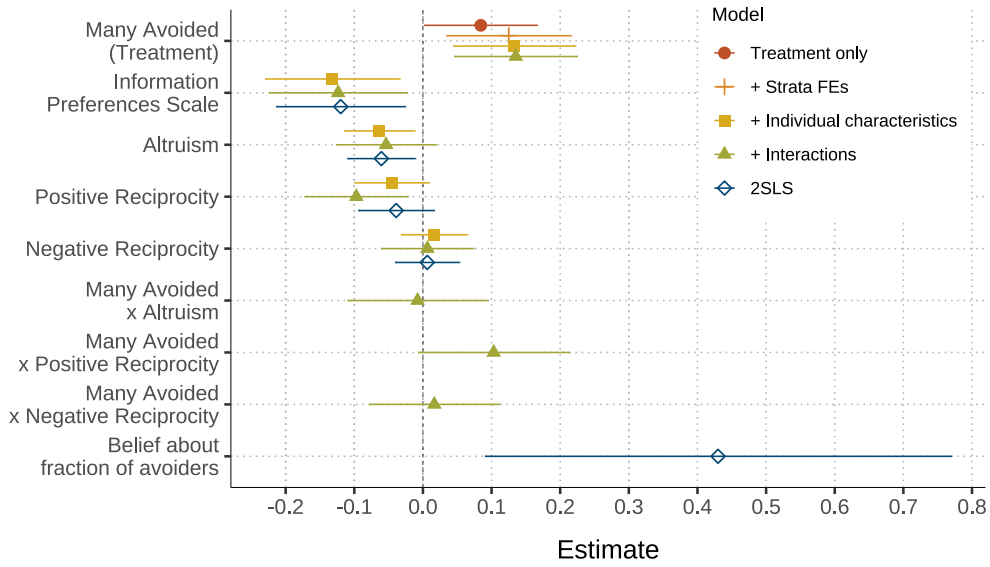
Notes: The figure shows the treatment effect on beliefs about the share of avoiders among other group members. Panel A shows the distribution of beliefs, by treatment condition. Dots (jittered) represent individual observations; violin plots show kernel density estimates; box plots show median (center line) and interquartile range (box); diamonds show mean beliefs with error bars representing 95% confidence intervals; p-value from a two-sample t-test. Panel B shows the empirical CDF of beliefs.

We now examine whether the treatment-induced shift in beliefs translates into differences in actual information decisions. Figure 10 reports regression-based estimates of the effect of the

²³Note that fields of study are slightly imbalanced: *Many Avoided* has fewer Business and of Economics students and more Humanities students than *Few Avoided*. To account for this, in subsequent analyses we report results both with and without field-of-study fixed effects in the regressions.

information-provision treatment on information avoidance across alternative specifications. In the baseline specification, exposure to the *Many Avoided* condition significantly increases the likelihood of information avoidance relative to the *Few Avoided* condition. While the estimated average effect is economically modest, it is precisely estimated and indicates that information avoidance decisions respond systematically to expectations about others' behavior. Raw treatment-control differences in avoidance rates are reported in Appendix B.5. As discussed above, we examine this heterogeneity directly using the strategy method in the next section.²⁴

Figure 10: Information avoidance: treatment effect and predictors



Notes: Regression estimates of the effect of the information-provision treatment on information avoidance. Coefficients are shown for alternative linear probability models and for a 2SLS specification instrumenting beliefs with treatment assignment. The dependent variable is an indicator equal to one if the participant chose to avoid information (Later = 1) and zero otherwise (Now = 0). Red circles report estimates from a specification including only the treatment indicator. Orange crosses additionally include strata and field-of-study fixed effects. Yellow squares further control for demographic characteristics, social and economic preferences (omitted from figure for conciseness), and the Information Preferences Scale (Ho et al., 2021, where higher values indicate a stronger preference for acquiring potentially undesirable information). Green triangles additionally include interactions between treatment assignment and social preferences. Blue hollow diamonds report 2SLS estimates instrumenting beliefs about the share of avoiders with treatment assignment. Error bars represent 95% confidence intervals based on robust standard errors.

To quantify the relationship between beliefs and behavior (the interdependence), we move beyond the reduced-form results and estimate a 2SLS model instrumenting beliefs with the treatment assignment (Table B.10).²⁵ The estimated coefficient ranges from $\hat{\beta}^{2SLS} \approx 0.4$ to 0.6, implying that

²⁴The positive average effect is robust across a wide range of alternative specifications, sample restrictions, and outcome measures. In particular, the effect remains statistically significant when excluding participants with non-standard preferences (Scream- and Volume-Lovers), under alternative sets of controls, and when using a continuous measure of information avoidance. Detailed robustness analyses are reported in Appendices B.6 and B.7.

²⁵The exogeneity assumption is satisfied by design. The monotonicity assumption—requiring that for every subject potential beliefs are weakly higher in *Many Avoided* relative to *Few Avoided*—is likely satisfied, as suggested by the first-order stochastic dominance of the belief distribution in *Many Avoided* relative to *Few Avoided* shown in Figure 9. The exclusion restriction requires that the effect of the information provision treatment operates only through posterior

a 10 percentage point increase in expected avoidance by others raises an individual’s probability of avoiding information by about 5 percentage points. This estimate should be interpreted as a reduced-form relationship among individuals whose beliefs respond to the treatment, rather than as a structural parameter of behavior. In light of the heterogeneity documented above, it reflects the composition of individuals who adjust their choices in response to belief changes. This indicates that, on average, higher expected avoidance by others increases the likelihood of avoiding information, consistent with strategic complementarity at the aggregate level. As with the reduced-form estimates, this average causal response should be interpreted as establishing the presence of interdependence rather than summarizing its underlying structure.

We also examine how the average treatment effect correlates with individual characteristics, including social preferences. General preferences for information (as captured by the IPS), reciprocity, and altruism are each associated with systematic differences in information decisions across treatment conditions.²⁶ These correlations suggest that individual characteristics help organize variation in information acquisition decisions, but do not by themselves capture the strategic structure of interdependence that we analyze below.

In light of the heterogeneity in best-response behavior documented above, this aggregate response should be interpreted as a composition-weighted summary of underlying strategy types rather than as a uniform behavioral effect, and therefore provides only a reduced-form view of a fundamentally heterogeneous strategic environment.

To quantify this point, we assess the relative importance of the heterogeneity of strategy types and of beliefs in explaining information avoidance. Following [Shorrocks \(2013\)](#), we implement a Shapley decomposition of the adjusted R^2 of regressions predicting information avoidance. The Shapley decomposition estimates each regressor’s marginal contribution to the adjusted R^2 (and hence, its importance in explaining variation in the outcome variable). The method has been applied to assess relative importance of contributing factors to economic growth ([Henderson et al., 2018](#)) and to income inequality ([Shorrocks, 1982, 2013](#)).

The results of the Shapley decomposition are reported in Table 1. Panel A shows that a model that includes both beliefs and dummies for strategy types, along with demographic characteristics (age and gender) and the Information Preferences Scale, explains 39% of the variation in information avoidance choices, as measured by the adjusted R^2 . Including only strategy types in the model drops the adjusted R^2 to 28.2% while including only beliefs reduces it to 7.2%, suggesting that strategy types contribute more explanatory power. The adjusted R^2 drops further to 3.8% and 0.7% when including only demographic characteristics and only the IPS, respectively. Panel

beliefs about others’ information decisions. While this assumption is not directly testable, we discuss mechanisms in Section 5.3.

²⁶Specifically, individuals with a stronger tendency to seek potentially undesirable information (higher IPS scores) are less likely to avoid information; more reciprocal individuals are on average less (more) likely to avoid in the *Few Avoided* (*Many Avoided*) condition; more altruistic individuals are less likely to avoid information ($\hat{\beta} = -0.068$).

B provides the exact Shapley decomposition of marginal contributions. Strategy types contribute 29 percentage points (72% of the joint contribution) while beliefs contribute 7 percentage points (18.5% of the joint contribution), demographics 7.0%, and the IPS 2.2%.

	(1) Estimates	(2) Relative importance
Panel A: Adjusted R^2		
Strategy Types only	0.330	—
Beliefs only	0.111	—
Demographics only	0.038	—
Information Preferences Scale only	0.007	—
<i>Joint</i>	<i>0.390</i>	—
Panel B: Shapley decomposition		
Strategy Types	0.282	72.2%
Beliefs	0.072	18.5%
Demographics	0.027	7.0%
Information Preferences Scale	0.009	2.2%
<i>Total</i>	<i>0.390</i>	<i>100.0%</i>

Notes: Each entry in Panel A represents an adjusted R^2 value from a separate regression of information avoidance on the right-hand side variables listed in the row headings. Panel B shows Shapley decomposition (column 1) for the set of variables shown in the row headings and their relative importance (column 2).

Table 1: R^2 and Shapley decomposition from regressions predicting information avoidance

The dominant relative importance of strategy types in explaining information decisions suggests that the group composition of types is a strong determinant of individual, and hence, aggregate avoidance. This motivates a study on how variations in the composition of strategy types map to aggregate decisions, which we turn to analyze next using simulations.

6 Group Composition and Equilibrium Avoidance

The presence of strategic complements, strategic substitutes, and unconditional types suggests that group-level avoidance depends on its composition, and that small compositional changes may generate large differences in aggregate information avoidance. In this section, we study how group composition maps into group-level outcomes. Using simulations based on the empirically estimated distribution of strategy types' best-response functions, we examine how composition determines *equilibrium* levels of information avoidance.

We refer to the equilibrium prevalence of information avoidance as an “information-avoidance equilibrium.” Formally, an information-avoidance equilibrium is a share λ^* of individuals who avoid information such that a proportion λ^* of group members best-responds by avoiding information, while the remaining proportion best-responds by acquiring it.

6.1 Equilibrium Avoidance across Strategy Types

6.1.1 Method

To derive the set of information-avoidance equilibria (“equilibrium set”) of a given group, we proceed in two steps: (1) from the individual best-response schedules $y_i^*(\lambda_{-i})$, we construct the group-level (“aggregate”) best-response schedule $Y^*(\lambda)$, and (2) we identify the equilibrium set as the set of fixed points of the schedule Y^* .

We derive the aggregate best-response schedule Y^* as follows. Each individual best-response schedule is a function $y_i^* : [0, 1] \rightarrow \{0, 1\}$ mapping each share of individuals avoiding information to a decision to avoid ($y_i = 1$) or acquire ($y_i = 0$) information. For each λ , we compute and sum individual best responses, $\sum_i y_i^*(\lambda)$, yielding the total number of information avoiders at λ . Dividing by the number of group members gives the proportion of individuals who avoid information $Y^*(\lambda)$. For example, consider the smallest possible group composed of one Always Getter (with $y_1^*(\lambda_{-1}) = 0$ for all λ_{-1}) and one Always Avoider (with $y_2^*(\lambda_{-2}) = 1$). Then the aggregate best-response function is $Y^*(\lambda) = 0.5$ for all λ : for any λ , half of the group members best-respond by avoiding information.

Since, in practice, the measurement of y_i^* is coarse (possible only at a finite set of values of λ), the derived aggregate schedule Y^* is also coarse, with the risk of missing equilibrium points. Estimating its full shape therefore requires an assumption about the missing intervals of λ . Our approach is to impute missing values using linear interpolation. This method is simple and intuitive, provides good linear approximations under continuity in Y^* , and ensures that the estimated schedule varies smoothly with λ . While discontinuities in Y^* are possible, their location is unknown, and the linear interpolation does not impose specific assumptions about where they occur.

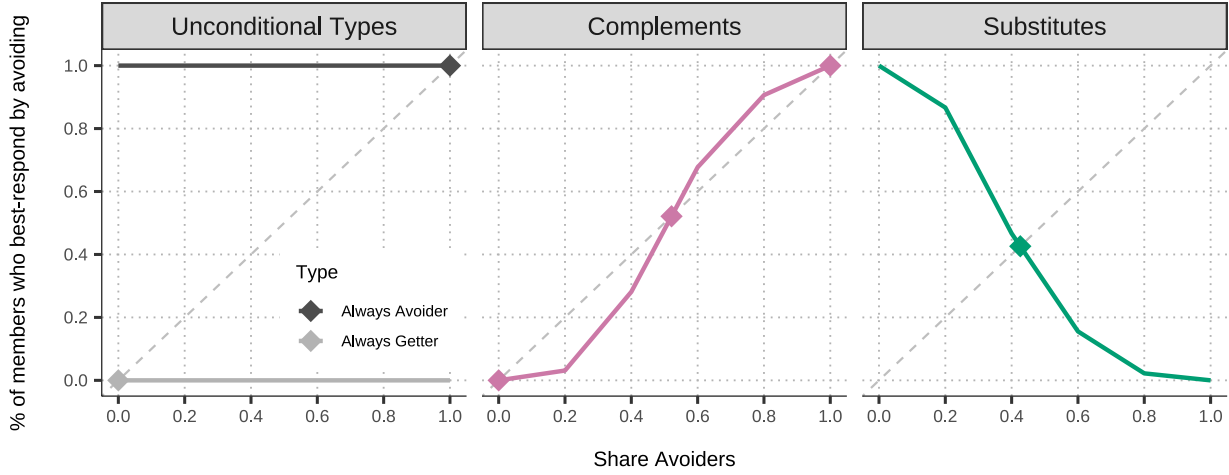
Once the shape of Y^* is estimated, equilibrium points are identified as the fixed points of Y^* , namely the set of values λ^* such that $Y^*(\lambda^*) = \lambda^*$. Graphically, these correspond to the points where the schedule Y^* intersects the identity line.

6.1.2 Results

To build intuition, we begin by considering homogeneous groups, that is, groups composed entirely of a single strategy type. We construct these groups using all participants of each type. Figure 11 plots the aggregate best-response function $Y^*(\lambda)$ for each type. Lines show the estimated schedule $Y^*(\lambda)$, while diamonds mark the equilibrium set, defined as the intersection(s) of $Y^*(\lambda)$ with the identity line (dotted).

Groups of unconditional types (Always Getters or Always Avoiders) display a flat aggregate schedule, reflecting the fact that all members’ choices are independent of others’ decisions. These groups have a unique information-avoidance equilibrium, in which everybody is either informed (Always Getters) or uninformed (Always Avoiders).

Figure 11: Aggregate best-responses and information-avoidance equilibria (homogeneous groups)



Notes: Aggregate (group-level) best response schedules $Y^*(\lambda)$ for homogeneous groups. Each panel shows the aggregate schedule constructed using all experimental participants of the indicated strategy type (not bootstrap samples). Lines show $Y^*(\lambda)$, the share of group members who best-respond by avoiding information at each share of avoiders λ . The dotted diagonal line is the identity line. Diamonds mark the information-avoidance equilibria. Always Getters and Always Avoiders have flat schedules with corner equilibria. Strategic Complements have an upward-sloping schedule with three equilibria ($\lambda^* = 0, 0.52, 1$). Strategic Substitutes have a downward-sloping schedule with one interior equilibrium ($\lambda^* = 0.43$).

In contrast, homogeneous groups of conditional types generate monotonic schedules: negatively sloped for Strategic Substitutes and positively sloped for Strategic Complements. Substitutes feature a unique interior equilibrium. Complements, however, exhibit multiple equilibria—including full awareness (MAA), full avoidance (MAI), and an interior equilibrium—providing empirical support for the theoretical prediction of B  nabou (2013). This multiplicity captures the risk of collective ignorance, even when full awareness is also sustainable. The figure therefore illustrates Corollary 1: Complements can sustain both MAA and MAI equilibria, while Substitutes can sustain neither.

6.2 Group Composition and Equilibrium Thresholds

The diversity of shapes in Y^* schedules suggests that equilibrium outcomes may vary substantially with group composition. To study this relationship systematically, we simulate groups with different compositions of strategy types based on empirical individual schedules and characterize the resulting equilibrium outcomes.

6.2.1 Method

To characterize how group composition affects equilibrium information acquisition, we conduct a bootstrap simulation exercise. For a given group size n and composition $\mu = (\mu_{AG}, \mu_{AA}, \mu_{SC}, \mu_{SS})$,

denoting the shares of Always Getters, Always Avoiders, Strategic Complements, and Strategic Substitutes, we repeatedly draw B groups of size n , with sampling probabilities given by μ . For each simulated group, we construct the aggregate best-response correspondence implied by its realized composition and compute all fixed points, classifying equilibria according to their stability. We then aggregate these equilibrium points across bootstrap draws to obtain the distribution of equilibrium avoidance associated with composition μ . Repeating this procedure over a sequence of compositions allows us to trace how changes in the distribution of strategy types affect the prevalence and stability of collective ignorance.

We simulate over a grid of compositions in which each type's proportion varies from 0 to 1 in increments of 0.1. Given the constraint that proportions sum to 1, the grid includes 286 distinct compositions. For each composition, we bootstrap $B = 100$ groups of $n = 13$.²⁷ We focus on small groups, as composition manipulation is most policy-relevant at this scale. The simulation thus involves 367,600 sampled individuals across 28,600 groups and 286 compositions.

Among the 286 compositions, we focus on four simple sequences that vary only two types: one conditional ($C \in \{SC, SS\}$) and one unconditional ($U \in \{AG, AA\}$), with $\mu_U + \mu_C = 1$ and μ_U varying from 0 to 1.²⁸ These restricted sequences isolate the effect of introducing unconditional types into groups initially composed entirely of conditional types. This design is motivated by the observation that groups of Strategic Complements can sustain multiple equilibria; introducing unconditional types—whose behavior is independent of others' choices—allows us to ask questions such as: “How many Always Avoiders does it take to tip the equilibrium toward full avoidance?”

6.2.2 Results

For any group composition, each of the B simulated groups generates each an aggregate best-response schedule with a corresponding set of information-avoidance equilibria. Aggregating across simulated groups yields a composition-specific distribution of best-response schedules and equilibria. We then study how key features of these distributions vary with group composition.

The effect of group composition on the aggregate best-response schedule Y^* is reported in Appendix C.1. Starting from homogeneous groups of conditional types (with upward- or downward-sloping schedules), increasing the share of unconditional progressively flattens the schedule and shifts it towards corner equilibria: full awareness for Always Getters ($Y^* = 0$) and full avoidance for Always Avoiders ($Y^* = 1$).

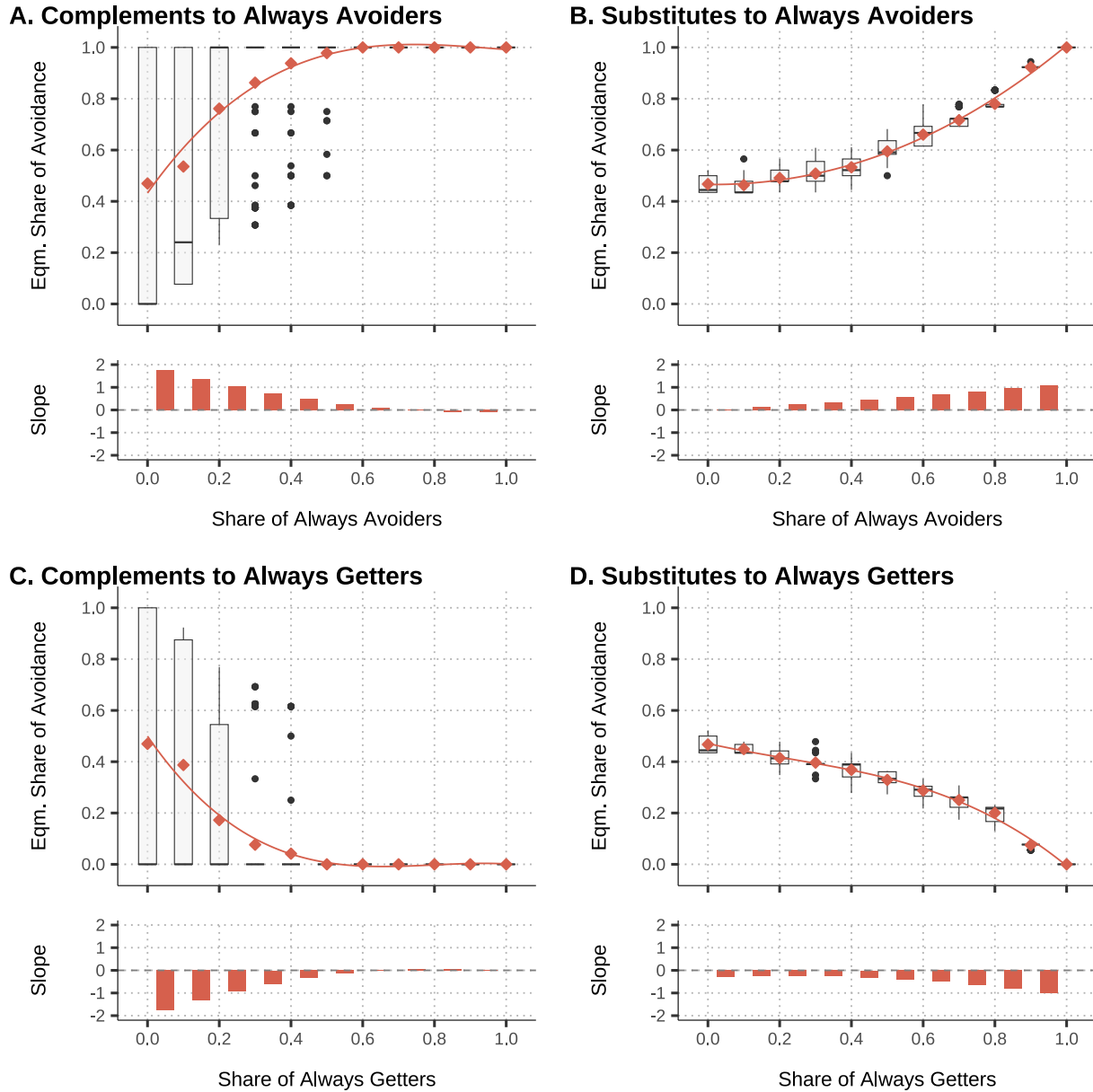
Figure 12 shows how the distribution of information-avoidance equilibria varies with group composition. Each panel focuses on one of the four sequences, starting from a conditional type

²⁷When n is an even number (e.g., $n = 10$), segments of Y_b^* can lie exactly on the identity line, generating infinitely many equilibrium points and apparent non-monotonicities in the distribution of equilibria across compositions. This creates technical issues when counting and calculating the relative frequency of equilibria. Choosing a non-round group size (e.g., $n = 13$) ensures that discrete steps in Y_b^* do not align with the identity line.

²⁸Higher-dimensional compositions are analyzed in Appendix C.

and gradually increasing the share of an unconditional type along the x -axis. The y -axis shows the equilibrium share of group members who avoid information, that is, the equilibrium prevalence of ignorance.

Figure 12: Effect of group composition on equilibrium distribution



Notes: Effect of group composition of strategy types on equilibrium share of avoidance. Each panel focuses on mixtures of two strategy types: one conditional (Complements or Substitutes) and one unconditional (Always Avoiders or Always Getters). The x -axis represents the group composition, indexed by the participation of the unconditional type. The y -axis measures the equilibrium share of avoidance. Boxplots summarize features of the distribution of equilibria (over bootstrapped groups) of the composition indicated on the x -axis. Diamonds indicate the average of the distribution; lines show a polynomial fit of the averages. The bottom panels show the slopes of the fitted line.

The boxplots summarize the distribution of information-avoidance equilibria associated with

each group composition and reveal clear differences between Complements and Substitutes.²⁹ When the share of unconditional types is sufficiently low (left side of x -axis), Complements exhibit a wide range of information-avoidance equilibria, including both full awareness ($y = 0$) and full avoidance ($y = 1$), while Substitutes display a more concentrated distribution of interior equilibria around $y = 0.5$. As the share of unconditional types increases (moving right along the x -axis), the distributions shift towards the characteristic behavior of the unconditional type, but the patterns differ markedly. Complements exhibit sharp tipping points: once the share of unconditional types crosses a threshold of around 30%, the distribution collapses toward either full avoidance (with Always Avoiders) or full awareness (with Always Getters). Accordingly, the dynamics of groups with Complements are highly non-linear. Substitutes, by contrast, exhibit only smooth transitions without strong tipping points, and the distribution collapses only when Substitutes are absent. These patterns reflect Complements' amplifying tendency and Substitutes' stabilizing responses: as the share of unconditional types increases, Complements reinforce the shift, whereas Substitutes counteract.

We illustrate this mechanism by focusing on the average equilibrium (red diamonds) for each composition. A polynomial fit (red line) shows that, for Complements, the average equilibrium responds most strongly when the share of unconditional types is low; while the opposite is true for Substitutes.³⁰ The bottom panels show the slopes of the fitted lines. These slopes can be interpreted as *composition elasticities*, measuring the change (in percentage points) in the average equilibrium induced by a one-percentage-point change in the share of the unconditional type.

When Complements are prevalent, the elasticity is close to 2, implying that a 1pp increase in the share of unconditional types changes the average equilibrium by 2pp. This elasticity declines rapidly (in absolute values) as Complements become less prevalent. In contrast, when Substitutes are prevalent, the elasticity is close to 0: increases in the share of unconditional types are largely offset by the responses of Substitutes. As Substitutes become less prevalent (moving right along the x -axis), the elasticity increases in absolute value, reaching 1 when Substitutes disappear.

Our findings have implications for organizational design. In groups of Strategic Complements, small interventions — such as introducing a few information seekers or, conversely, tolerating a few additional avoiders— can tip entire groups between widespread information acquisition and collective ignorance. In groups of Strategic Substitutes, similar interventions produce only incremental and contained effects.

Overall, our simulations show that the composition of strategy types strongly shapes equilibrium outcomes: different mixtures generate markedly different equilibrium sets, and critical thresholds exist where small compositional changes trigger large behavioral responses.

²⁹Histograms of the distributions of equilibria are shown in Appendix C.2.

³⁰We fit a third-order polynomial to allow non-monotonicities while minimizing overfitting.

7 Conclusion

We study the social interdependence of information avoidance in environments where an individual's information decision generates externalities for others. Theoretically, we extend [Bénabou \(2013\)](#) and show that individual best responses to information decisions are heterogeneous, giving rise to four broad strategy types. Empirically, we design a novel experiment to identify and measure this heterogeneity, documenting four strategy types and of the mediating role of anticipatory utility. Using simulations, we then examine the aggregate implications of this heterogeneity, showing that group composition generates nonlinear responses in the distribution of informational equilibria and can give rise to tipping points.

Organization designers and policy-makers are often interested in increasing the take-up of information. Our findings suggest that interventions aimed at this goal should account for the interdependence of information decisions. Existing recommendations from the literature on information preferences are typically derived from individual decision settings and therefore abstract from spillovers across individuals. By contrast, our findings imply that incentivizing a subgroup to acquire information may trigger positive or negative responses among others. Such “cognitive externalities” ([Bénabou, 2013](#)) imply that the aggregate effects of these interventions may differ from their direct effects on targeted individuals. Policy-makers and organization designers may therefore benefit from accounting for equilibrium effects when designing interventions to increase information acquisition in settings with interdependent outcomes.

When team composition is a choice variable, collective ignorance may be mitigated through group design. Our simulations show that group composition strongly influences the equilibrium prevalence of information avoidance. Aggregate responses can be highly nonlinear in the mixture of strategy types, with some compositions exhibiting particularly large responses to marginal changes. As a result, group design can be an effective lever for reducing collective ignorance.

Our investigation points to several avenues for future research. First, in our experimental setup—as well as in [Bénabou \(2013\)](#)—information choices are tightly linked to the severity of the future outcome. As a result, we cannot disentangle whether the observed interdependence is driven by changes in the expected outcome severity (a “severity effect”) or by changes in others’ expected information choices (a “herding effect”). We view our study as a proof of concept that establishes the empirical relevance of interdependence in information decisions. Further research is needed to disentangle these mechanisms more cleanly. Second, we focus on a static setting in which individuals make information decisions simultaneously. In many applications, however, such decisions unfold sequentially, creating scope for future work on the dynamics of willful ignorance in groups.

Our study highlights that understanding how individuals respond to others’ information decisions is crucial for designing environments that promote information acquisition and, ultimately, improve collective outcomes.

References

- Alaoui, L. & Penta, A. (2026). Attitudes toward Success and Failure. *American Economic Journal: Microeconomics*, 18(1), 105–145.
- Beaurenaut, M., Tokarski, E., Dezechache, G., & Grèzes, J. (2020). The ‘Threat of Scream’ paradigm: A tool for studying sustained physiological and subjective anxiety. *Scientific Reports*, 10(1), 12496.
- Belloni, A., Chernozhukov, V., & Hansen, C. (2014). Inference on Treatment Effects after Selection among High-Dimensional Controls. *The Review of Economic Studies*, 81(2), 608–650.
- Bénabou, R. (2013). Groupthink: Collective Delusions in Organizations and Markets. *The Review of Economic Studies*, 80(2), 429–462.
- Dana, J., Weber, R. A., & Kuang, J. X. (2007). Exploiting moral wiggle room: Experiments demonstrating an illusory preference for fairness. *Economic Theory*, 33(1), 67–80.
- Engelmann, J. B., Lebreton, M., Salem-Garcia, N. A., Schwardmann, P., & van der Weele, J. J. (2024). Anticipatory Anxiety and Wishful Thinking. *American Economic Review*, 114(4), 926–960.
- Enke, B., Graeber, T., & Oprea, R. (2023). Confidence, Self-Selection, and Bias in the Aggregate. *American Economic Review*, 113(7), 1933–1966.
- Falk, A., Becker, A., Dohmen, T., Enke, B., Huffman, D., & Sunde, U. (2018). Global Evidence on Economic Preferences. *The Quarterly Journal of Economics*, 133(4), 1645–1692.
- Falk, A. & Zimmermann, F. (2024). Attention and Dread: Experimental Evidence on Preferences for Information. *Management Science*, 70(10), 7090–7100.
- Fischbacher, U., Gächter, S., & Fehr, E. (2001). Are people conditionally cooperative? Evidence from a public goods experiment. *Economics Letters*, 71(3), 397–404.
- Fong, J. & Hunter, M. (2022). Can Facing the Truth Improve Outcomes? Effects of Information in Consumer Finance. *Marketing Science*, 41(1), 33–50.
- Galai, D. & Sade, O. (2006). The “Ostrich Effect” and the Relationship between the Liquidity and the Yields of Financial Assets. *The Journal of Business*, 79(5), 2741–2759.
- Ganguly, A. & Tasoff, J. (2017). Fantasy and Dread: The Demand for Information and the Consumption Utility of the Future. *Management Science*, 63(12), 4037–4060.
- Gerardi, D. & Yariv, L. (2008). Information acquisition in committees. *Games and Economic Behavior*, 62(2), 436–459.

- Giamattei, M., Yahosseini, K. S., Gächter, S., & Molleman, L. (2020). LIONESS Lab: A free web-based platform for conducting interactive experiments online. *Journal of the Economic Science Association*, 6(1), 95–111.
- Golman, R., Hagmann, D., & Loewenstein, G. (2017). Information Avoidance. *Journal of Economic Literature*, 55(1), 96–135.
- Haaland, I., Roth, C., & Wohlfart, J. (2023). Designing Information Provision Experiments. *Journal of Economic Literature*, 61(1), 3–40.
- Henderson, J. V., Squires, T., Storeygard, A., & Weil, D. (2018). The Global Distribution of Economic Activity: Nature, History, and the Role of Trade. *The Quarterly Journal of Economics*, 133(1), 357–406.
- Hertwig, R. & Engel, C. (2016). Homo Ignorans: Deliberately Choosing Not to Know. *Perspectives on Psychological Science*, 11(3), 359–372.
- Ho, E. H., Hagmann, D., & Loewenstein, G. (2021). Measuring Information Preferences. *Management Science*, 67(1), 126–145.
- Kahneman, D. & Tversky, A. (1979). Prospect Theory: An Analysis of Decision under Risk. *Econometrica*, 47(2), 263–291.
- Karlsson, N., Loewenstein, G., & Seppi, D. (2009). The ostrich effect: Selective attention to information. *Journal of Risk and Uncertainty*, 38(2), 95–115.
- Kőszegi, B. & Rabin, M. (2006). A Model of Reference-Dependent Preferences. *The Quarterly Journal of Economics*, 121(4), 1133–1165.
- Kreps, D. M. & Porteus, E. L. (1978). Temporal Resolution of Uncertainty and Dynamic Choice Theory. *Econometrica*, 46(1), 185.
- Kumar, S., Forster, H., Bailey, P., & Griffiths, T. (2009). Mapping unpleasantness of sounds to their auditory representation. *The Journal of the Acoustical Society of America*, 124, 3810–7.
- Masatlioglu, Y., Orhun, Y., & Raymond, C. (2023). Intrinsic Information Preferences and Skewness. *American Economic Review*, 113(10), 2615–2644.
- Momsen, K. & Ohndorf, M. (2023). Information avoidance: Self-image concerns, inattention, and ideology. *Journal of Economic Behavior & Organization*, 211, 386–400.
- Olafsson, A. & Pagel, M. (2025). The Ostrich in Us: Selective Attention to Personal Finances. *The Review of Economics and Statistics*, (pp. 1–26).

- Oprea, R. & Yuksel, S. (2022). Social Exchange of Motivated Beliefs. *Journal of the European Economic Association*, 20(2), 667–699.
- Oster, E., Shoulson, I., & Dorsey, E. R. (2013). Optimal Expectations and Limited Medical Testing: Evidence from Huntington Disease. *American Economic Review*, 103(2), 804–830.
- Pagel, M. (2018). A News-Utility Theory for Inattention and Delegation in Portfolio Choice. *Econometrica*, 86(2), 491–522.
- Selten, R. (1967). Die Strategiemethode zur Erforschung des eingeschränkt rationalen Verhaltens im Rahmen eines Oligopolexperimentes. *Beiträge zur experimentellen Wirtschaftsforschung*.
- Shorrocks, A. F. (1982). Inequality Decomposition by Factor Components. *Econometrica*, 50(1), 193–211.
- Shorrocks, A. F. (2013). Decomposition procedures for distributional analysis: A unified framework based on the Shapley value. *The Journal of Economic Inequality*, 11(1), 99–126.
- Simonsohn, U., Simmons, J. P., & Nelson, L. D. (2020). Specification curve analysis. *Nature Human Behaviour*, 4(11), 1208–1214.
- Suvorov, A., van de Ven, J., & Villeval, M. C. (2024). Selective Information Sharing and Group Delusion.
- Thornton, R. L. (2008). The Demand for, and Impact of, Learning HIV Status. *American Economic Review*, 98(5), 1829–1863.

A Theoretical Appendix

A.1 Mapping of Notation

Table A.1 shows the correspondence between the notation used in Section 3 and Bénabou (2013).

Concept	Bénabou (2013)	This Paper
Good state	$\sigma = H$	$\sigma = G$
Bad state	$\sigma = L$	$\sigma = B$
Date-2 payoff in good state	$U_2^i \equiv \sum_{j=1}^N a_H^{ji} e^j + b_H^{ji} (1 - e^j)$	u_G (constant)
Date-2 payoff in bad state	$U_2^i \equiv \sum_{j=1}^N a_L^{ji} e^j + b_L^{ji} (1 - e^j)$	$u_B = \bar{u}_B - \Delta \left(y_i + \sum_{j \neq i} y_j \right)$ $= b(\lambda_{-i}) + \Delta(1 - y_i)$, where $y_i = 1$ if i avoids information (0 otherwise), $b(\lambda_{-i}) \equiv \bar{u}_B - \Delta(1 + (N - 1)\lambda_{-i})$ and $\lambda_{-i} \equiv \frac{\sum_{j \neq i} y_j}{N-1}$
Effort	$e^i \in \{0, 1\}$; effort improves payoff in good state, worsens it in bad state; agents always choose $e = 1$ when uninformed.	No explicit effort choice; information avoidance automatically worsens everybody's payoff by Δ in bad state.
Return to i from j with $e^j = 1$ in good state	a_H^{ij}	0 for all i, j
Return to i from j with $e^j = 0$ in good state	b_H^{ij}	0 for all i, j
Return to i from j with $e^j = 1$ in bad state	a_L^{ij}	$-\Delta < 0$ for all i, j
Return to i from j with $e^j = 0$ in bad state	b_L^{ij}	0 for all i, j
Prior belief that state is Good	q	q
Damage to i if everybody else avoids information	d_L^i	$-\Delta(N - 1)$
Maximum material payoff in good state	A_H^i	0
Maximum material payoff in bad state	B_L^i	$\bar{u}_B$

Table A.1: Mapping of notation: Bénabou (2013) and this paper

A.2 Proofs

Proof of Lemma 1 Consider v_i everywhere convex. Agent i prefers to avoid information if $\varphi_i > 0$, i.e.

$$v_i((1 - q)b(\lambda_{-i}) + qu_G) > (1 - q)v_i(b(\lambda_{-i}) + \Delta) + qu_G \quad (\text{A.1})$$

Since $v_i(\cdot)$ is strictly increasing, $v_i(b(\lambda_{-i}) + \Delta) > v_i(b(\lambda_{-i}))$, and hence a necessary condition for (A.2) is that

$$v_i(pb + (1 - p)g) > pv_i(b) + (1 - p)v_i(g)$$

but this contradicts convexity. Therefore, the agent never avoids information, i.e., the agent is an Always Getter. ■

Proof of Lemma 2 Suppose v_i everywhere concave. First note that the net incentive to avoid information increases in the share of other group members who avoid information:

$$\begin{aligned}\frac{\partial \varphi_i}{\partial \lambda_{-i}} &= (1-q)v'_i((1-q)b(\lambda_{-i}) + qu_G) \frac{\partial b(\lambda_{-i})}{\partial \lambda_{-i}} - (1-q)v'_i(b(\lambda_{-i}) + \Delta) \frac{\partial b(\lambda_{-i})}{\partial \lambda_{-i}} \\ &= [v'_i((1-q)b(\lambda_{-i}) + qu_G) - v'_i(b(\lambda_{-i}) + \Delta)] (1-q) \frac{\partial b(\lambda_{-i})}{\partial \lambda_{-i}} > 0\end{aligned}$$

by concavity of v_i and our assumption that $(1-q)b(\lambda_{-i}) + qu_G > b(\lambda_{-i}) + \Delta$ for all λ_{-i} . It follows that: (i) If $\varphi_i(0) > 0$ then by monotonicity $\varphi_i(\lambda_{-i}) > 0$ for any λ_{-i} , that is, the agent is an Always Avoider; (ii) If $\varphi_i(1) < 0$ then by monotonicity $\varphi_i(\lambda_{-i}) < 0$ for any λ_{-i} , that is, the agent is an Always Getter; (iii) If $\varphi_i(0) < 0$ and $\varphi_i(1) > 0$, then by continuity there exists an interior λ_{-i}^* such that the agent acquires information whenever $\lambda_{-i} < \lambda_{-i}^*$ and avoids it when $\lambda_{-i} > \lambda_{-i}^*$ (strategic complementarity). ■

Proof of Lemma 3 We focus throughout on $u_G > \widehat{u}_i > b(0) + \Delta$. Since $b(\cdot)$ is a decreasing function, this ensures that $\widehat{u}_i > b(\lambda_{-i}) + \Delta$ for all λ_{-i} .

(i) Suppose that when everyone else avoids information, then the expected date-2 utility under ignorance falls below the reference level:

$$(1-q)b(1) + qu_G < \widehat{u}_i. \quad (\text{A.2})$$

Then v_i is convex at $(1-q)b(1) + qu_G$. This follows since $v_i(x)$ is convex for all $x < \widehat{u}_i$ and concave for all $x > \widehat{u}_i$. We now derive the sufficient condition for

$$(1-q)v_i(b(1)) + qv_i(u_G) > v_i((1-q)b(1) + qu_G). \quad (\text{A.3})$$

Step 1: Write

$$\begin{aligned}v_i((1-q)b(1) + q\widehat{u}_i) &= v_i((1-q)b(1) + qu_G) - d_1 \\ qv_i(\widehat{u}_i) &= qv_i(u_G) - d_2\end{aligned} \quad (\text{A.4})$$

for some $d_1 > 0$ and $d_2 > 0$.

Step 2: Note that, by convexity of $v_i(x)$ for all $x < \widehat{u}_i$, we have

$$(1-q)v_i(b(1)) + qv_i(\widehat{u}_i) > v_i((1-q)b(1) + q\widehat{u}_i) \quad (\text{A.5})$$

Step 3: Substitute for $qv_i(\widehat{u}_i)$ and $v_i((1-q)b(1) + q\widehat{u}_i)$ in (A.5) and rearrange to obtain

$$(1-q)v_i(b(1)) + qv_i(u_G) > v_i((1-q)b(1) + q\widehat{u}_i) + d_2 - d_1. \quad (\text{A.6})$$

If $d_2 \geq d_1$, then (A.3) follows directly from (A.6). In turn, (A.3) ensures that information avoidance is dominated by information acquisition when $\Delta \rightarrow 0$. Since $v_i(b(1) + \Delta) > v_i(b(1))$, a higher Δ makes avoidance even less desirable. We conclude that, when $d_2 \geq d_1$ or

$$v_i(u_G) - v_i(\widehat{u}_i) \geq \frac{v_i((1-q)b(1) + qu_G) - v_i((1-q)b(1) + q\widehat{u}_i)}{q} \quad (\text{A.7})$$

then $\varphi_i(1) < 0$: when all other agents acquire information, agent i strictly prefers to avoid.

(ii) Suppose that when everyone else acquires information, expected date-2 utility under ignorance is above the reference level:

$$(1-q)b(0) + qu_G > \widehat{u}_i. \quad (\text{A.8})$$

We now derive sufficient conditions for

$$v_i((1-q)b(0) + qu_G) > (1-q)v_i(b(0)) + qv_i(u_G). \quad (\text{A.9})$$

Step 1: Write

$$\begin{aligned} v_i((1-q)\widehat{u}_i + qu_G) &= v_i((1-q)b(0) + qu_G) + d_3 \\ (1-q)v_i(\widehat{u}_i) &= (1-q)v_i(b(0)) + d_4 \end{aligned} \quad (\text{A.10})$$

for some $d_3 > 0$ and $d_4 > 0$.

Step 2: Note that, since $v_i(x)$ is concave for all $x > \widehat{u}_i$, we have

$$v_i((1-q)\widehat{u}_i + qu_G) > (1-q)v_i(\widehat{u}_i) + qv_i(u_G). \quad (\text{A.11})$$

Step 3: Substitute for $v_i((1-q)\widehat{u}_i + qu_G)$ and $(1-q)v_i(\widehat{u}_i)$ in (A.11) and rearrange to obtain

$$v_i((1-q)b(0) + qu_G) > (1-q)v_i(b(0)) + qv_i(u_G) + d_4 - d_3. \quad (\text{A.12})$$

Condition (A.12) implies that if $d_4 \geq d_3$ or

$$v_i(\widehat{u}_i) - v_i(b(0)) \geq \frac{v_i((1-q)\widehat{u}_i + qu_G) - v_i((1-q)b(0) + qu_G)}{1-q} \quad (\text{A.13})$$

then (A.9) follows directly from (A.12). In turn, (A.9) ensures that, when $\Delta \rightarrow 0$, $\varphi_i(0) > 0$ holds: when all other agents acquire information, agent i prefers to avoid. By continuity, there exists a value $\Delta^* > 0$ such that when $\Delta < \Delta^*$ the agent strictly prefers to avoid information when all other

agents in the group acquire it.

To sum up, therefore, assuming that Δ is sufficiently small, conditions (A.2), (A.7), (A.8) and (A.13) are sufficient to ensure that $\varphi_i(0) > 0$ and $\varphi_i(1) < 0$. These conditions are satisfied when v_i rises very steeply in the neighborhood of $\widehat{u}_i$ but is approximately flat away from it, and when $b(0) - b(1)$ is large. By continuity of $\varphi_i(\lambda_{-i})$, there exists an interior threshold λ_i^{**} such that agent i strictly prefers to avoid information if $\lambda_{-i} < \lambda_i^{**}$ and strictly prefers to acquire information if $\lambda_{-i} > \lambda_i^{**}$ (strategic substitutability). ■

Proof of Proposition 1 Follows directly from lemmata 1-3. ■

A.3 Social Preferences and Information Avoidance

In our decision environment, acquiring information generates a positive externality on other agents. This creates a natural role for social preferences. An agent who internalizes others' payoffs faces stronger incentives to acquire information, but also experiences anticipatory utility from others' outcomes, which may introduce countervailing forces. We extend the baseline model to incorporate two forms of social preferences: altruism and reciprocity.

A.3.1 Altruism

First, we consider altruistic preferences. Suppose that if date-2 expected utility for the whole group is u , individual i experiences anticipatory utility $v_i(u) = w_i(u) + \alpha_i a(u)$ where $w_i(u)$ captures anticipatory utility over own date-2 consumption utility and $\alpha_i a(u)$ captures altruistic anticipatory utility over other people's date-2 consumption utility, where $\alpha_i \geq 0$ represents the weight placed by i on altruistic anticipation. Both w_i and a are continuous, increasing and differentiable functions. The net incentive to avoid information for agent i is:

$$\begin{aligned} \varphi_i^{Altr} = & \underbrace{w_i((1-q)b + qu_G) + \alpha_i a((1-q)b + qu_G)}_{\text{Anticipatory utility when avoiding information}} \\ & - \underbrace{(1-q)[w_i(b + \Delta) + \alpha_i a(b + \Delta)] + q[w_i(u_G) + \alpha_i a(u_G)]}_{\text{Expected anticipatory utility when acquiring information}} \end{aligned} \quad (\text{A.14})$$

where we remove the dependence on λ_{-i} from b for notational simplicity. Differentiating the net incentive to avoid with respect to α_i gives:

$$a((1-q)b + qu_G) - [(1-q)a(b + \Delta) + qa(u_G)] \quad (\text{A.15})$$

The sign of expression (A.15) is ambiguous and depends on the exact shape of $a(u)$. For instance,

if $a(u)$ is everywhere concave, (A.15) is positive, while if $a(u)$ is everywhere convex, it is negative. If $a(u)$ is linear, (A.15) is equal to 0. This shows that the effect of stronger altruism on the propensity to acquire or avoid information is generally ambiguous. Consider now the effect of stronger altruism on interdependence. This is captured by

$$\frac{\partial^2 \varphi_i^{Altr}}{\partial \lambda_{-i} \partial \alpha_i} = (1 - q) \{a'(b + \Delta) - a'((1 - q)b + qu_G)\} \quad (\text{A.16})$$

Again, the sign of (A.15) is ambiguous. For example, given $(1 - q)b + qu_G > b + \Delta$, the sign of (A.15) is positive if $a(u)$ is everywhere concave and it is negative if $a(u)$ is everywhere convex. If $a(u)$ is linear, (A.15) is equal to 0.

A.3.2 Reciprocity

We also extend the model to incorporate reciprocity concerns. An agent may derive utility from harming others if they avoid information and worsen her outcome, or from benefiting them if they become informed and help her. Formally, suppose that agent i 's reciprocity utility is given by $-\rho_i \cdot (y_i - \lambda_{-i})^2$, where $\rho_i \geq 0$ measures her sensitivity to reciprocity. Agent i 's net utility from avoiding information is now augmented by an additional component given by

$$-\rho_i[(1 - \lambda_{-i})^2 - \lambda_{-i}^2] = -\rho_i(1 - 2\lambda_{-i}) \quad (\text{A.17})$$

It is then easy to see that greater reciprocity concerns strengthen the propensity for strategic complementarity: $\partial^2 \varphi_i / \partial \lambda_{-i} \partial \rho_i > 0$. Intuitively, on top of the baseline incentives to avoid information, reciprocity concerns introduce an additional motive to match the actions of others, creating a tendency towards strategic complementarity.

B Additional analyses

All subsequent analyses are referenced in the document. Brief descriptions are provided for each analysis.

B.1 Descriptive statistics

Descriptive statistics of the experimental sample are reported below.

	N	Mean	SD	Median	Min	Max
<i>Demographic characteristics</i>						
Age	460	22.509	3.713	22.000	18.000	47.000
Gender (Female=1)	465	0.583	0.494	1.000	0.000	1.000
Has work experience	460	0.154	0.362	0.000	0.000	1.000
<i>General information preferences</i>						
IPS (overall scale)	464	3.116	0.414	3.111	1.000	4.000
<i>Social and economic preferences</i>						
Patience	464	-0.000	1.000	0.201	-3.493	1.124
Risk seeking	464	-0.000	1.000	0.296	-2.678	1.570
Positive reciprocity	464	-0.000	0.803	-0.017	-3.911	1.244
Negative reciprocity	464	0.000	0.836	0.117	-1.833	1.946
Altruism	464	0.000	0.836	0.096	-2.163	1.947
Trust	464	-0.000	1.000	-0.178	-1.662	2.048
<i>Field of study</i>						
Medicine	465	0.213	0.410	0.000	0.000	1.000
STEM	465	0.200	0.400	0.000	0.000	1.000
Humanities	465	0.228	0.420	0.000	0.000	1.000
Economics	465	0.108	0.310	0.000	0.000	1.000
Business	465	0.211	0.408	0.000	0.000	1.000
Other fields	465	0.056	0.230	0.000	0.000	1.000

Notes: The table shows descriptive statistics. The number of observations is lower for some variables elicited at the end of the session (e.g., age) because a few participants did not respond. For other variables (e.g., the IPS score and social preferences), one observation is missing due to a participant leaving the session early to attend a lecture.

Table B.1: Sample summary statistics

B.2 Are screams a bad? Preferences over screams and volume

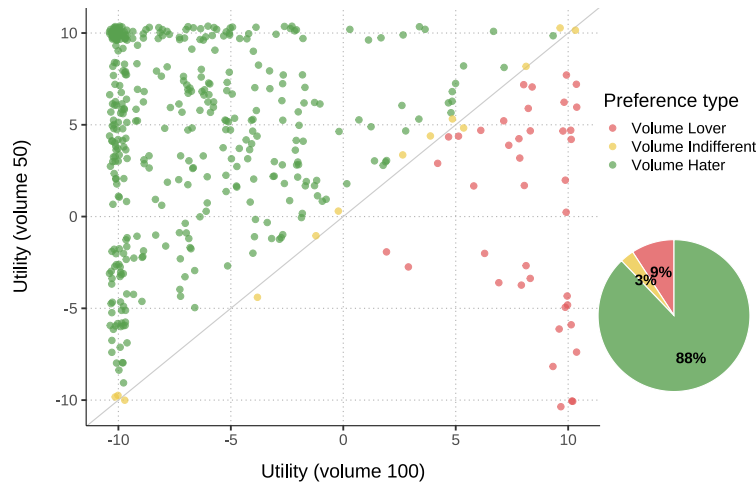
We classify “Volume Haters” (resp., “Lovers”) as subjects who report higher (lower) utility from hearing screams at volume 50 than at volume 100. Similarly, we classify “Scream Haters” (“Lovers”) as subjects who report a higher (lower) utility from learning that the state is Quiet than that it is

Screams. To capture the intensity of these preferences, we compute for each individual the difference between the utility of volume 50 and volume 100, and the difference between the utility of learning that the state is Quiet state relative to Screams.

Figure B.1 shows the joint distribution of subjects' reported utilities of hearing screams at different volumes, distinguishing each type of volume preference. In our sample, 88% of subjects are Volume Haters, 3% are Volume Indifferent, and 9% are Volume Lovers.

Figure B.2 presents the joint distribution of subjects' reported utilities of learning each state, distinguishing each type of scream preference. In our sample, 81% of subjects are Scream Haters, 6% are Scream Indifferent, and 14% are Scream Lovers.

Figure B.1: Preferences over volume of screams: reported utilities and preference types



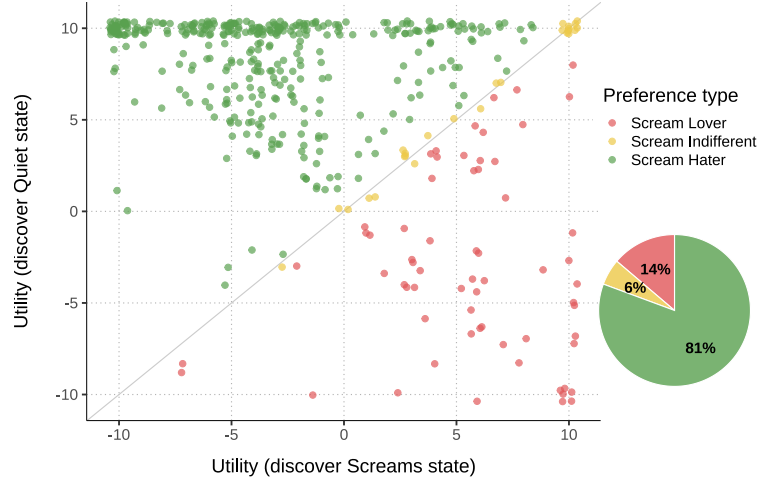
Notes: The figure displays reported utilities when screams were played at different volumes. Each point (jittered) corresponds to a participant's reported utilities for volume 50 (y -axis) and volume 100 (x -axis). The gray 45° line indicates indifference between volume levels. Observations are classified into three preference types: Volume Haters (green), who prefer quieter screams; Volume Indifferents (yellow), who report no difference; and Volume Lovers (red), who prefer louder screams. The pie chart summarizes the prevalence of each preference type in the sample.

The joint distribution of preference types over screams and volume is shown in Table B.2.

B.3 Balance checks

Table B.3 reports the treatment balance checks.

Figure B.2: Preferences over screams: reported utilities upon learning each state



Notes: The figure displays reported expected utilities upon learning the state of the world. Each point (jittered) corresponds to a participant's reported utilities for learning the Quiet state (y -axis) and the Screams state (x -axis). The gray 45° line indicates indifference across states. Observations are classified into three preference types: Scream Haters (green), who prefer to discover the Quiet state; Scream Indifferents (yellow), who report no difference; and Scream Lovers (red), who prefer to discover the Scream state. The pie chart summarizes the prevalence of each preference type in the sample.

Scream Preference Type	Volume Preference Type				Total
	Volume Lover	Volume Indifferent	Volume Hater		
Scream Lover	23	3	34	4	64
Scream Indifferent	1	5	19	1	26
Scream Hater	16	5	330	24	375
Total	40	13	383	29	465
Scream Lover	4.9%	0.6%	7.3%	0.9%	13.8%
Scream Indifferent	0.2%	1.1%	4.1%	0.2%	5.6%
Scream Hater	3.4%	1.1%	71.0%	5.2%	80.6%
Total	8.6%	2.8%	82.4%	6.2%	100.0%

Table B.2: Distribution of Scream- and Volume-preference types

	(1) Few Avoided	(2) Many Avoided	(3) Difference (2)-(1)
<i>Demographic characteristics</i>			
Age	22.53 (3.98)	22.49 (3.46)	-0.04 (0.35)
Gender (Female=1)	0.57 (0.50)	0.60 (0.49)	0.03 (0.05)
<i>Information preferences</i>			
IPS, overall scale	3.10 (0.42)	3.13 (0.41)	0.02 (0.04)
Health subscale	3.04 (0.72)	3.08 (0.68)	0.04 (0.06)
Finance	2.88 (0.67)	2.90 (0.63)	0.02 (0.06)
Personal	3.11 (0.55)	3.14 (0.55)	0.03 (0.05)
General	3.04 (0.70)	3.08 (0.70)	0.03 (0.06)
Occupational	3.29 (0.48)	3.30 (0.50)	0.01 (0.05)
<i>Field of study</i>			
Medicine	0.19 (0.40)	0.23 (0.42)	0.04 (0.04)
STEM	0.18 (0.39)	0.21 (0.41)	0.03 (0.04)
Humanities	0.19 (0.40)	0.26 (0.44)	0.07* (0.04)
Economics	0.14 (0.35)	0.08 (0.27)	-0.06** (0.03)
Business	0.25 (0.43)	0.18 (0.38)	-0.07* (0.04)
Other fields	0.06 (0.24)	0.05 (0.22)	-0.01 (0.02)
<i>Work experience</i>			
Has work experience	0.16 (0.37)	0.15 (0.36)	-0.01 (0.03)
Observations	223	242	465

* p < 0.1, ** p < 0.05, *** p < 0.01. Robust standard errors in parentheses.

Table B.3: Balance checks

B.4 Treatment effect on belief about prevalence of avoidance

Table B.4 reports estimates from linear regressions of beliefs on the treatment dummy and different sets of controls. Column (1) reproduces the results in Figure 9: on average, the expected share of avoiders among other group members is 21.9 percentage points higher in *Many Avoided* than in *Few Avoided*. This difference is highly significant and robust to the inclusion of the following controls: strata fixed effects (column 2), field of study fixed effects and individual characteristics (age and gender; column 3), and economic and social preferences (altruism, reciprocity, risk-seeking, patience, trust), as well as the IPS (column 4). Overall, the evidence confirms that the treatment successfully shifted beliefs about the prevalence of avoidance among other members.

	Dep. var: Belief about share of avoiders			
	(1)	(2)	(3)	(4)
Many Avoided	0.219*** (0.021)	0.215*** (0.025)	0.218*** (0.025)	0.221*** (0.025)
Control group mean	0.371*** (0.016)	0.471*** (0.072)	0.394*** (0.137)	0.494*** (0.163)
Strata FEs		✓	✓	✓
Field of study FEs			✓	✓
Individual characteristics			✓	✓
Preferences (economic, social, IPS)				✓
Observations	465	465	460	460

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

Notes: The table shows the estimated coefficients of a regression of beliefs about the share of information avoiders among other members on the treatment dummy. The dependent variable is a continuous measure of beliefs, ranging from 0 (0% of the subject's group will avoid information") to 1 (100%). The treatment dummy is equal to one if the subject was in the *Many Avoided* condition and zero if in the *Few Avoided* condition. Individual characteristics include age and gender. "Preferences" include economic and social preferences (risk-seeking, time preferences, altruism, reciprocity, and trust) and the Information Preferences Scale (Ho et al., 2021), measuring an individual's general preference for obtaining information. Field of study fixed effects include dummies for fields of study. Strata-fixed effects include dummies for week of year and time of day. The number of observations drops in columns (3) and (4) because five participants did not provide information about their age.

Table B.4: Manipulation check (regressions)

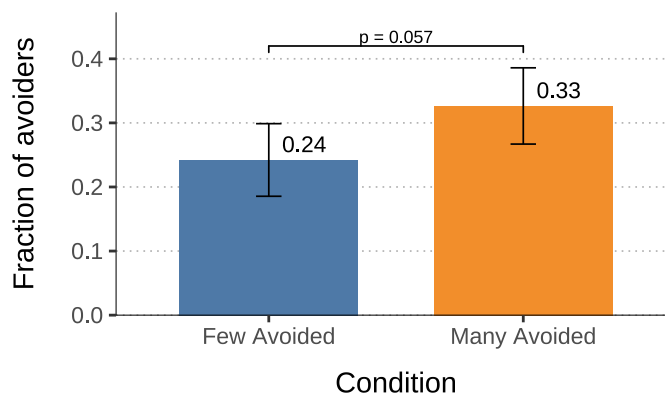
B.5 Treatment effect on information avoidance

This section presents further details on the treatment effect on information avoidance. Figure B.3 reports the raw treatment-control differences in information avoidance rates, showing a that the probability of avoiding information is higher in *Many Avoided* than in *Few Avoided*. This difference is statistically significant at 5%.

Table B.5 reports regression-based estimates of the treatment effect. Column 1 reports the unconditional treatment effect. As columns 2 to 5 show, the treatment effect is robust to the inclusion of individual characteristics, as well as fixed effects by strata and field of study (column 2; $\hat{\beta} = 0.109$, significant at 5%), and increases to 0.123 when all controls are included, significant at the 1% level (column 4). In this specification, the treatment effect corresponds to roughly half of the baseline prevalence of avoiders.

The estimate of the treatment effect is robust to a range of alternative specifications. Excluding Scream- and Volume-Lovers from the sample makes the effect even stronger (Tables B.6 and B.7). In addition, the estimate remains robust and significant across alternative specifications, regardless of whether the set of controls is selected by the researcher (Simonsohn et al., 2020) or via the data-driven post-double selection procedure (Belloni et al., 2014), as shown in Appendix B.7. Finally, we obtain qualitatively similar results when using as the outcome variable the IA scale (the continuous preference strength for information avoidance) (Appendix B.8).

Figure B.3: Treatment effect on information avoidance



Notes: The figure shows raw treatment-control differences on information avoidance (chose Later). Bars represent the fraction of avoiders under the *Few Avoided* (blue) and *Many Avoided* (orange) treatments. Error bars represent 95% confidence intervals.

The regressions in Table B.5 reveal that individual characteristics predict information avoidance. First, the Information Preferences Scale significantly predicts choices in our experiment: individuals with a stronger tendency to seek potentially undesirable information (higher IPS scores) are less likely to avoid information in our experiment.

Second, social preferences are also associated with information avoidance. More altruistic in-

	Dependent variable: Avoided information				
	(1)	(2)	(3)	(4)	(5)
Many Avoided	0.084** (0.042)	0.109** (0.049)	0.114** (0.049)	0.123** (0.049)	0.124** (0.049)
Information Preferences Scale			-0.125** (0.051)	-0.130** (0.052)	-0.118** (0.054)
<i>Social preferences</i>					
Altruism				-0.065** (0.028)	-0.052 (0.041)
× Many Avoided					-0.006 (0.057)
Positive reciprocity				-0.046 (0.029)	-0.101** (0.041)
× Many Avoided					0.109* (0.060)
Negative reciprocity				0.013 (0.026)	-0.005 (0.036)
× Many Avoided					0.030 (0.051)
Constant	0.242*** (0.029)	0.126 (0.225)	0.326 (0.301)	0.280 (0.311)	0.273 (0.321)
Strata FEs		✓	✓	✓	✓
Field of study FEs		✓	✓	✓	✓
Individual characteristics			✓	✓	✓
Observations	465	465	460	460	460

* p < 0.1, ** p < 0.05, *** p < 0.01. Robust standard errors in parentheses.

Notes: The table reports estimates from a linear probability model regressing information avoidance on the treatment dummy. The dependent variable is a binary indicator equal to one if the subject avoid information (chose Later) and zero otherwise. The treatment dummy equals one if the subject was in the *Many Avoided* condition and zero in the *Few Avoided* condition. The regressions include strata fixed effects, demographic controls, field of study, and measures of economic and information preferences (see notes in Table B.4). The number of observations drops in column (3) to (5) since five participants did not provide information about their age.

Table B.5: Treatment effect on information avoidance: regressions

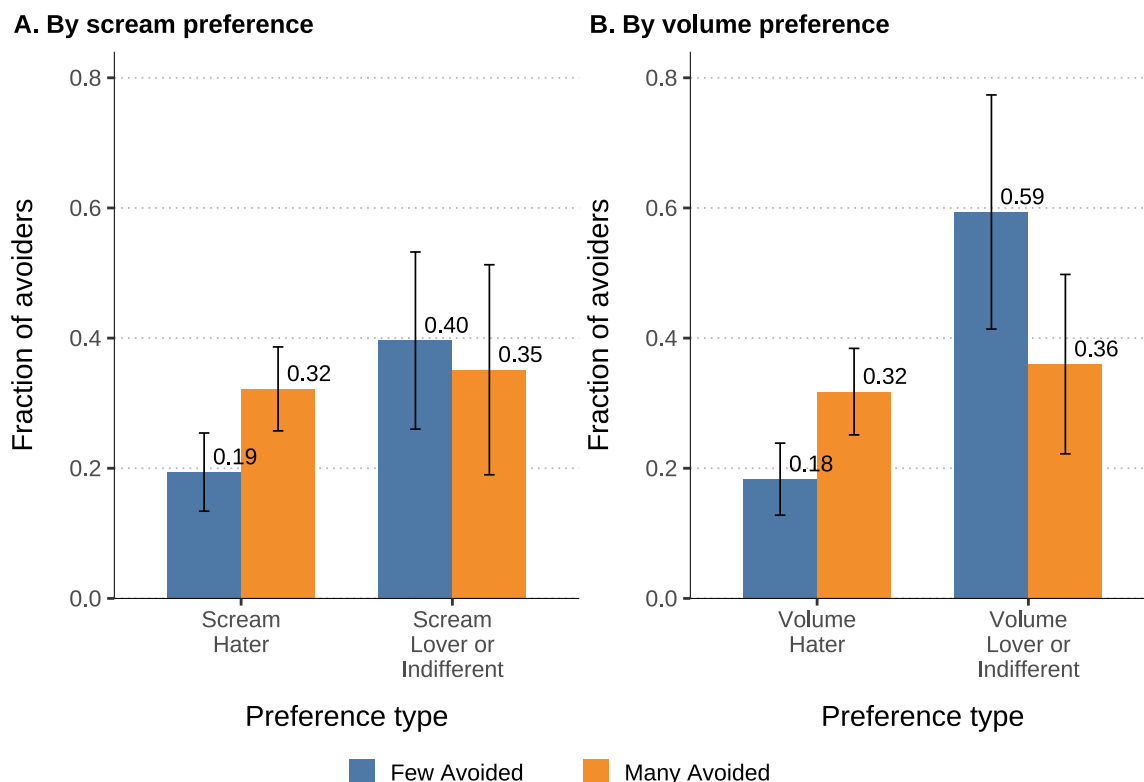
dividuals are less likely to avoid information ($\hat{\beta} = -0.068$), consistent with the idea that prosocial motives encourage information acquisition to benefit others. While the estimated coefficients cannot be interpreted causally, the observed patterns align with the theoretical predictions (Appendix A.3). For reciprocity, we observe a significant and positive interaction with the treatment: more reciprocal individuals are on average less likely to avoid in the *Few Avoided*, and more likely in the *Many Avoided* condition, again as predicted in Appendix A.3. However, as Table B.6 and Table B.7 show, the estimated coefficients shrink when Scream- and Volume-Lovers are excluded, becoming

statistically insignificant, although the signs of the point estimates remain unchanged.

B.6 Robustness to exclusion of Scream- and Volume-Lovers

Here, we show that the main regression results are robust to the exclusion of Scream- and Volume-Lovers. Figure B.4 shows the treatment effect on information avoidance by type of preference for scream (panel A) and for volume (panel B). The treatment effect is positive for scream- and volume-haters, but the direction of the effect is is, if anything, the opposite for scream- and volume-lovers. This suggests that, by not excluding scream- and volume-lovers from our main regressions, the treatment effect is rather attenuated.

Figure B.4: Treatment effect on information avoidance by preference over screams and volume



Notes: The figure shows treatment effects on information avoidance (chose Later) conditional on individual preference types. Panel A splits participants by scream preference: Scream-Haters versus Scream-Lovers or -Indifferents. Panel B splits by volume preference: Volume-Haters versus Volume-Lovers or -Indifferents. For each subgroup, the bars report the fraction of avoiders under the *Few Avoided* (blue) and *Many Avoided* (orange) treatments. Error bars represent 95% confidence intervals.

Table B.6 shows our main regression results if we exclude Scream-Lovers. Table B.7 shows our main regression results if we exclude Volume-Lovers. In both subsamples, the magnitude of the estimated treatment effect increases, compared to the full sample (reported in Table B.5), for all the reported specifications. The treatment coefficient becomes significant at 1% for all specifications.

	Dependent variable: Avoided information				
	(1)	(2)	(3)	(4)	(5)
Many Avoided	0.128*** (0.045)	0.162*** (0.052)	0.157*** (0.052)	0.169*** (0.052)	0.170*** (0.053)
Information Preferences Scale			-0.126** (0.056)	-0.124** (0.059)	-0.114* (0.060)
<i>Social preferences</i>					
Altruism				-0.049 (0.031)	-0.072 (0.047)
× Many Avoided					0.050 (0.064)
Positive reciprocity				-0.030 (0.032)	-0.061 (0.046)
× Many Avoided					0.064 (0.066)
Negative reciprocity				0.031 (0.028)	0.037 (0.037)
× Many Avoided					-0.008 (0.054)
Constant	0.194*** (0.031)	-0.041 (0.240)	0.188 (0.324)	0.123 (0.340)	0.112 (0.352)
Strata FEs		✓	✓	✓	✓
Field of study FEs		✓	✓	✓	✓
Individual characteristics			✓	✓	✓
Observations	375	375	371	371	371

* p < 0.1, ** p < 0.05, *** p < 0.01. Robust standard errors in parentheses.

Notes: The table reports estimates from a linear probability model regressing information avoidance on the treatment dummy, restricted to the subsample of Scream-Haters. See notes in Table B.5.

Table B.6: Treatment effect on information avoidance (subsample of Scream-Haters)

	Dependent variable: Avoided information				
	(1)	(2)	(3)	(4)	(5)
Many Avoided	0.134*** (0.044)	0.159*** (0.050)	0.161*** (0.049)	0.166*** (0.050)	0.164*** (0.051)
Information Preferences Scale			-0.133** (0.054)	-0.132** (0.057)	-0.127** (0.058)
<i>Social preferences</i>					
Altruism				-0.062** (0.030)	-0.076* (0.045)
× Many Avoided					0.033 (0.061)
Positive reciprocity				-0.023 (0.030)	-0.037 (0.041)
× Many Avoided					0.032 (0.063)
Negative reciprocity				0.021 (0.027)	0.007 (0.035)
× Many Avoided					0.024 (0.054)
Constant	0.183*** (0.028)	-0.136 (0.162)	0.015 (0.270)	-0.088 (0.284)	-0.094 (0.297)
Strata FEs		✓	✓	✓	✓
Field of study FEs		✓	✓	✓	✓
Individual characteristics			✓	✓	✓
Observations	383	383	380	380	380

* p < 0.1, ** p < 0.05, *** p < 0.01. Robust standard errors in parentheses.

Notes: The table reports estimates from a linear probability model regressing information avoidance on the treatment dummy, restricted to the subsample of Volume Haters. See notes in Table B.5

Table B.7: Treatment effect on information avoidance (subsample of Volume-Haters)

B.7 Robustness to alternative specifications

Specification curve analyses To assess the robustness of our main treatment effect, we conducted a specification curve analysis (Simonsohn et al., 2020) by systematically varying sample restrictions, control sets, and model specifications. Specifically, we estimated the treatment effect across all combinations of four subsamples (full sample, excluding Scream-Lovers, excluding Volume-Lovers, excluding either) and twelve covariate sets (ranging from no controls to different combinations of demographics, field-of-study fixed effects, GPS and IPS measures). The results are presented in Figure B.5. Each point in the top panel represents the estimated effect size under an alternative specification, sorted by magnitude. Black dots indicate that the estimate is statistically significant at 5% level. The lower panels display the corresponding sample, control, and model choices. The main specification is highlighted with a red square. Figure B.6 show a similar plot but at level $\alpha = 0.01$.

Across specifications, the estimated average treatment effect remains consistently positive and is statistically significant in the majority of cases. Excluding subgroups (e.g., Volume-Lovers, Scream-Lovers) or adding or removing controls (demographics, IPS, GPS scales) does not qualitatively change the result. These findings indicate that the estimated treatment effect remains robust to a broad set of alternative specifications.

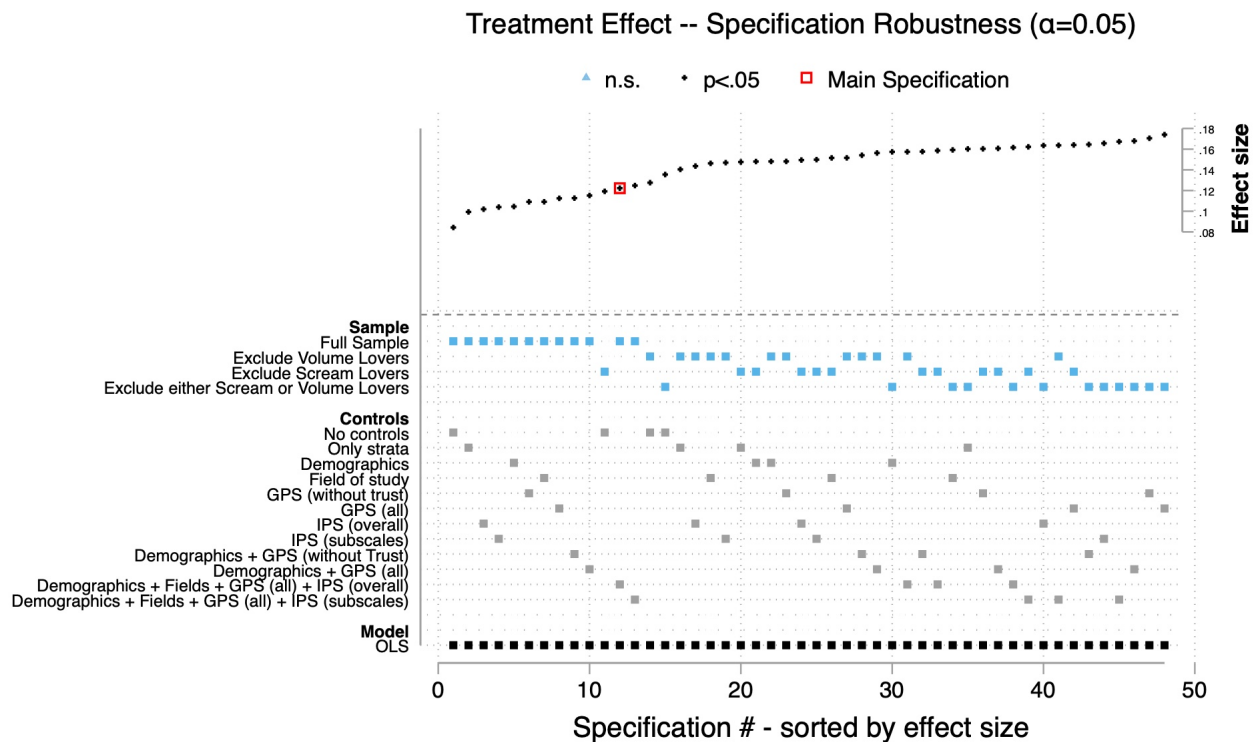
Post-double selection The specification curve analyses above rely on pre-defined subsets of covariates that are either included or excluded jointly as blocks. This design tests robustness across broad groups of control variables but lacks some flexibility: for instance, social preferences may be either included or excluded as a block, without considering combinations where altruism and reciprocity enter separately. To allow for such flexibility, we employ the post-double selection procedure (Belloni et al., 2014), a machine-learning-based, data-driven method for selecting controls from a potentially large set of covariates with the goal of causal estimation of parameters.

Results are reported in Table B.8. We estimate the treatment effect on information avoidance using both the plug-in method (columns 1–4) and cross-validation (columns 5–8). All specifications include strata fixed effects, and we progressively add four sets of controls: (i) baseline controls (economic and social preferences; IPS scales, subscales, and individual questions; field of study dummies; age; gender; and work experience), (ii) experimental context and comprehension controls (group size, perceived ease of instructions, and quiz difficulty), (iii) quiz performance until the information decision (questions answered and number correct), and (iv) quiz performance at the end of the experimental session (questions answered and number correct, performance level, and earnings).

Across all specifications, the estimated treatment effect remains positive and statistically significant, with magnitudes similar to those in the main regressions. These findings confirm that the

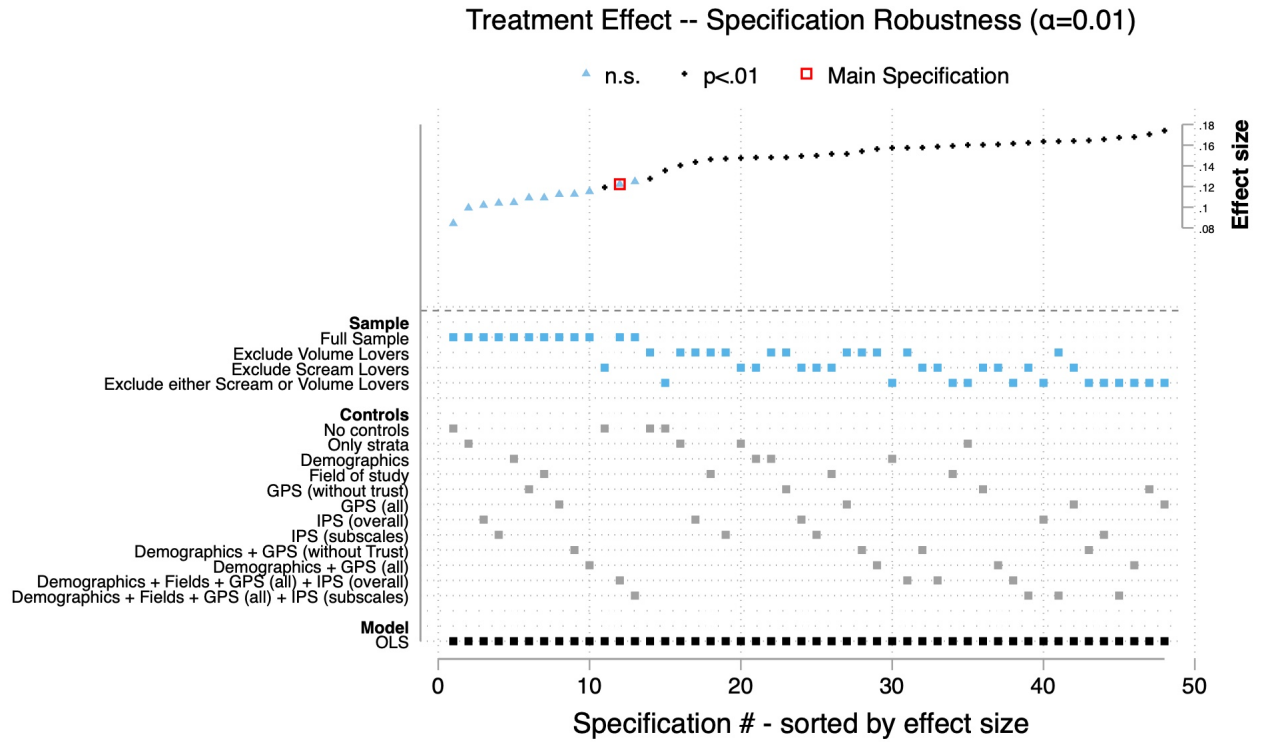
main result (higher information avoidance in the *Many Avoided* condition) is robust to alternative specifications where selection of controls is data-driven.

Figure B.5: Specification Curve Analysis ($\alpha = 0.05$)



Notes: The figure reports a specification curve analysis of the treatment effect. Each point represents the estimated effect under an alternative model specification, sorted by effect size. The top panel plots estimated effect sizes, with black dots indicating specifications significant at the 5% level, blue triangles indicating non-significant estimates, and the red square denoting the main specification reported in Section 5.4. The lower panels display the sample restrictions, control sets, and model choices corresponding to each specification. Effect sizes remain consistently positive across specifications, indicating robustness of the main result.

Figure B.6: Specification Curve Analysis ($\alpha = 0.01$)



Notes: The figure reports a specification curve analysis of the treatment effect. Each point represents the estimated effect under an alternative model specification, sorted by effect size. The top panel plots estimated effect sizes, with black dots indicating specifications significant at the 1% level, blue triangles indicating non-significant estimates, and the red square denoting the main specification reported in Section 5.4. The lower panels display the sample restrictions, control sets, and model choices corresponding to each specification. Effect sizes remain consistently positive across specifications, indicating robustness of the main result.

	Method: Plug-in				Cross-Validation			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Many Avoided	0.095* (0.050)	0.101* (0.056)	0.101* (0.056)	0.101* (0.056)	0.123** (0.049)	0.109* (0.056)	0.110* (0.056)	0.110* (0.057)
Strata FEs	✓	✓	✓	✓	✓	✓	✓	✓
Set 1: Baseline controls	✓	✓	✓	✓	✓	✓	✓	✓
Set 2: Context & comprehension		✓	✓	✓		✓	✓	✓
Set 3: Quiz performance A			✓	✓			✓	✓
Set 3: Quiz performance B				✓				✓

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

Notes: The table reports post-double selection estimates of the effect of being in the Many Avoided condition on information avoidance. Columns (1)–(4) present results using the plug-in method, while columns (5)–(8) use cross-validation. All specifications include strata fixed effects and progressively add four sets of controls to the covariate selection algorithm: (i) baseline controls (age, gender, field of study dummies, social and economic preferences, Information Preferences Scale (IPS), and work experience), (ii) context and comprehension controls (group size, perceived ease of instructions, and quiz difficulty), (iii) quiz performance before making information decision (number of questions answered and number correct), and (iv) quiz performance at the end of experiment (number of questions answered and number correct, performance level, and earnings).

Table B.8: Post-double selection estimates

B.8 Alternative outcome: continuous measure of avoidance (IA Scale)

As a robustness check, we replicate our main analysis using the continuous measure of information avoidance (the “IA Scale”) rather than the binary indicator. Recall that the IA Scale captures the strength of preference for the chosen information decision, ranging from -10 (strong preference for acquiring information) to $+10$ (strong preference for avoiding).

Table B.9 reports the results. The overall pattern closely mirrors that of the main regressions with the binary outcome (Table B.5): signs of coefficients and statistical significance remain unchanged. Across specifications, the treatment effect of *Many Avoided* remains positive and significant. The coefficient is stable with the inclusion of controls and fixed effects, indicating that our main finding—that information avoidance increases under *Many Avoided*—remains robust to the choice of outcome measure.

Moreover, the results with IA Scale also echo the findings on social preferences documented earlier: altruism significantly predicts lower avoidance, while reciprocity remains directionally similar as before, but without though reaching statistical significance. The IPS remains negatively and significantly associated with strength of preference for avoiding information.

	Dependent variable: IA Scale				
	(1)	(2)	(3)	(4)	(5)
Many Avoided	1.271** (0.624)	1.610** (0.746)	1.677** (0.740)	1.846** (0.743)	1.838** (0.753)
Information Preferences Scale			-1.441* (0.772)	-1.487* (0.807)	-1.389* (0.834)
<i>Social preferences</i>					
Altruism				-1.094** (0.428)	-1.168* (0.638)
× Many Avoided					0.291 (0.873)
Positive reciprocity				-0.326 (0.444)	-0.855 (0.620)
× Many Avoided					1.098 (0.937)
Negative reciprocity				0.224 (0.405)	0.120 (0.586)
× Many Avoided					0.181 (0.820)
Constant	-3.821*** (0.431)	-4.525 (3.463)	-2.052 (4.706)	-2.815 (4.889)	-2.761 (5.065)
Strata FEs		✓	✓	✓	✓
Field of study FEs		✓	✓	✓	✓
Individual characteristics			✓	✓	✓
Observations	465	465	460	460	460

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

Notes: The table reports estimates from a linear regression of the IA Scale on the treatment dummy. The IA Scale captures the strength of preference for the chosen information decision, ranging from -10 (strong preference for acquiring information) to +10 (strong preference for avoiding). The treatment dummy equals one if the subject was in the *Many Avoided* condition and zero in the *Few Avoided* condition. Regressions include strata fixed effects, demographic controls, field of study, and measures of economic and information preferences (see notes in Table B.4). The number of observations drops in column (3) to (5) since five participants did not provide information about their age.

Table B.9: Treatment effect on IA Scale

B.9 Instrumental variable regressions (2SLS)

Table B.10 reports instrumental variable regressions (2SLS) of information avoidance on beliefs about the share of avoiders. Beliefs are instrumented with the random treatment assignment. The model specifications (columns 1-5) match those of the main regressions (Table B.5), except that the treatment dummy is replaced by the instrumented belief.

Once again, the overall pattern is very similar to that of the main regressions (Table B.5). Across specifications (columns 1–5), a higher belief that others avoid information significantly increases the likelihood of avoiding information. The estimated coefficient on beliefs is stable, positive, and statistically significant across specifications (ranging from $\hat{\beta}^{2SLS} = 0.384$ to 0.604).

The coefficients associated to social preference measures remain directionally similar. Altruism is associated with lower avoidance, while positive reciprocity reduces avoidance in the Few Avoided condition but reverses under the Many Avoided condition, although coefficients lose statistical significance relative to the main regressions.

Finally, the IPS is negatively associated with the decision to avoid information.

	Dependent variable: Avoided information				
	(1)	(2)	(3)	(4)	(5)
Belief about share of avoiders	0.384** (0.181)	0.504** (0.213)	0.523** (0.206)	0.556*** (0.201)	0.604** (0.239)
Information Preferences Scale			-0.123*** (0.045)	-0.122*** (0.046)	-0.065 (0.060)
<i>Social preferences</i>					
Altruism				-0.072*** (0.025)	-0.076 (0.144)
× Belief					0.030 (0.314)
Pos. reciprocity				-0.032 (0.028)	-0.296* (0.165)
× Belief					0.635 (0.386)
Neg. reciprocity				-0.000 (0.024)	-0.080 (0.124)
× Belief					0.091 (0.275)
Constant	0.100 (0.088)	-0.130 (0.232)	0.117 (0.282)	0.006 (0.288)	-0.209 (0.359)
Strata FEs		✓	✓	✓	✓
Field of study FEs		✓	✓	✓	✓
Individual characteristics			✓	✓	✓
Observations	465	465	460	460	460
First Stage F-stat	104.71	75.93	76.49	81.34	12.77

* p < 0.1, ** p < 0.05, *** p < 0.01. Robust standard errors in parentheses.

Notes: The table reports the second stage 2SLS estimates from regressing the information avoidance indicator (equal to one if the subject chose Later and zero otherwise) on the subject's reported belief about the share of information avoiders in his group, where the belief is instrumented by the treatment assignment dummy (equal to one if the subject was in the *Many Avoided* condition and zero otherwise). Depending on the specification, listed in columns 1 to 5, regressions include strata fixed effects, demographic controls, field of study, and measures of economic and information preferences (see notes in Table B.4). The number of observations drops in column (3) to (5) since five participants did not provide information about their age.

Table B.10: 2SLS estimates: effect of beliefs on information avoidance, using treatment assignment as instrument

B.10 Treatment effect on the distribution of strategy types (extensive margin)

	Dependent variable: Subject belongs to strategy type			
	(1) Always Getter	(2) Always Avoider	(3) Complement	(4) Substitute
Many Avoided	-0.065 (0.046)	0.014 (0.026)	0.009 (0.038)	0.036 (0.037)
Control group mean	0.457*** (0.034)	0.081*** (0.018)	0.202*** (0.027)	0.175*** (0.026)
Observations	465	465	465	465

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

Notes: The table examines whether the treatment impacts the distribution of strategy types. It reports estimated coefficients from linear regressions where each column corresponds to a different strategy type. The dependent variable is a dummy equal to 1 if the subject's best-response schedule matches the given type and 0 otherwise. The independent variable is the treatment dummy (equal to 1 for the *Many Avoided* condition and 0 for the *Few Avoided* condition).

Table B.11: Treatment effect on distribution of strategy types (extensive margin)

B.11 Treatment effect on probability of avoiding for each strategy type (intensive margin)

Table B.12 shows the intensive margin of the treatment: the effect on the probability of avoiding information for each type. Each column shows the estimates of separate linear probability models, fitted in the subsample of each strategy type.

	Dependent variable: Avoided information			
	(1) Always Getters	(2) Always Avoiders	(3) Complements	(4) Substitutes
Many Avoided	0.014 (0.033)	0.056 (0.057)	0.187* (0.096)	0.008 (0.106)
Control group mean	0.049** (0.022)	0.944*** (0.057)	0.244*** (0.066)	0.385*** (0.080)
Observations	197	41	96	90

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

Notes: The table presents estimates from linear probability models where the dependent variable is an indicator equal to 1 if the subject avoided information (chose Later). Each column reports a separate regression estimated on the subsample of the strategy type indicated by the column label. The independent variable is the treatment dummy (equal to 1 for the *Many Avoided* condition and 0 for the *Few Avoided* condition).

Table B.12: Treatment effect on information avoidance, by strategy type (intensive margin)

B.12 Mechanisms: anticipatory utility (KP questions)

Table B.13 assesses the predictive validity and mediation properties of our Kreps-Porteus (KP) measures of anticipatory utility. Panel A examines predictive power; Panel B examines whether the treatment effect operates through anticipatory utility.

Panel A: Predictive power. Column (1) shows that controls alone explain only 6% of variation in information avoidance (adjusted $R^2 = 0.059$). Column (2) adds the Information Preferences Scale (IPS), a validated measure of general preferences for information. The IPS significantly predicts avoidance (individuals with higher willingness to know are less likely to avoid), but only modestly improves explanatory power ($R^2 = 0.071$).

Columns (3)-(4) replace the IPS with our KP measures of net anticipatory utility—continuous and dichotomous versions, respectively. Both measures are highly significant and positively associated with avoidance: higher net utility from avoiding predicts higher avoidance rates. Including the KP measures increases the R^2 to 0.185 (continuous) and 0.211 (dichotomous), nearly three times the explanatory power of the IPS. This suggests the KP measures capture decision-relevant aspects specific to our environment.

Columns (5)-(6) include both the IPS and KP measures. The coefficients on both remain stable, and the KP measures explain variance beyond the IPS. This pattern supports both the predictive and discriminant validity of our KP measures: they predict behavior independently of general information preferences and are not mere proxies for the IPS.

Panel B: Treatment and mediation. Panel B adds the treatment indicator. Column (1) shows the baseline treatment effect ($\beta = 0.122$, $p < 0.01$). Adding the IPS (column 2) leaves the treatment effect nearly unchanged, suggesting the IPS does not likely mediate the treatment effect. In contrast, adding the KP measures (columns 3-4) reduces the treatment coefficient by approximately 50% and renders it marginally significant or insignificant, suggesting the treatment operates through shifting anticipatory utility. Columns (5)-(6) confirm this pattern holds when controlling for both the IPS and KP measures.

	Dependent variable: Avoided information					
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Predictive power						
Information Preferences Scale (IPS)		-0.129** (0.052)			-0.136*** (0.047)	-0.148*** (0.049)
Net utility of avoidance (continuous)			0.023*** (0.003)		0.023*** (0.003)	
Net utility of avoidance (dichotomous)				0.372*** (0.046)		0.379*** (0.045)
Controls and fixed effects	✓	✓	✓	✓	✓	✓
Observations	460	460	460	448	460	448
Adjusted R^2	0.059	0.071	0.185	0.211	0.199	0.226
Panel B: Treatment and utility						
Many Avoided	0.122** (0.049)	0.123** (0.049)	0.061 (0.047)	0.081* (0.045)	0.062 (0.046)	0.080* (0.045)
Information Preferences Scale (IPS)		-0.130** (0.052)			-0.136*** (0.047)	-0.147*** (0.048)
Net utility of avoidance (continuous)			0.022*** (0.003)		0.022*** (0.003)	
Net utility of avoidance (dichotomous)				0.365*** (0.047)		0.372*** (0.046)
Controls and fixed effects	✓	✓	✓	✓	✓	✓
Observations	460	460	460	448	460	448
Adjusted R^2	0.070	0.082	0.187	0.214	0.200	0.230

Notes: The table presents OLS estimates analyzing the predictive power and potential mediating role of the KP measures on actual information decisions. The dependent variable is an indicator equal to 1 if the subject chose to avoid information about the state (option Later) and 0 otherwise. Panel A assesses the predictive power of a number of measures. “Information Preferences Scale (IPS)” (Ho et al., 2021) measures overall preference to know potentially undesirable information; higher values indicate stronger preference to know. “Net utility of avoidance (continuous)”, $\hat{\phi}^i$, is the empirical measure of the net expected utility of avoiding information (relative to acquiring it), derived from the KP questions. “Net utility of avoidance (dichotomous)” is an indicator equal to 1 if the empirical net utility of is higher than zero, $\hat{\phi}^i > 0$. In Panel A, coefficients are standardized for comparability across measures. Panel B focuses on how the treatment dummy (“Many Avoided”) changes upon inclusion of the predictive measures; coefficients are not standardized for comparability with Table B.5. All regressions control for demographic characteristics (age and gender), economic and social preferences (patience, risk-seeking, trust, altruism, and positive and negative reciprocity), and fixed effects by strata and field of study of the participant. Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B.13: Anticipatory Utility (KP Measures) and Information Avoidance

B.13 Mechanisms: supplementary tables

	Dependent variable: Avoided information				
	(1)	(2)	(3)	(4)	(5)
Belief about share of avoiders	0.090 (0.173)	0.122 (0.175)	0.098 (0.176)	0.085 (0.173)	0.063 (0.169)
Pr(screams)	-0.194 (0.150)	-0.176 (0.148)	-0.151 (0.148)	-0.156 (0.144)	-0.208 (0.145)
Belief about share of avoiders $\times$ Pr(screams)	0.852*** (0.294)	0.800*** (0.294)	0.787*** (0.297)	0.826*** (0.293)	0.909*** (0.292)
Constant	0.111 (0.076)	0.136 (0.166)	-0.116 (0.215)	0.125 (0.279)	0.029 (0.285)
Strata FEs		✓	✓	✓	✓
Field of study FEs			✓	✓	✓
Individual characteristics				✓	✓
Information preferences				✓	✓
Social and economic preferences					✓
Observations	465	465	465	460	460

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

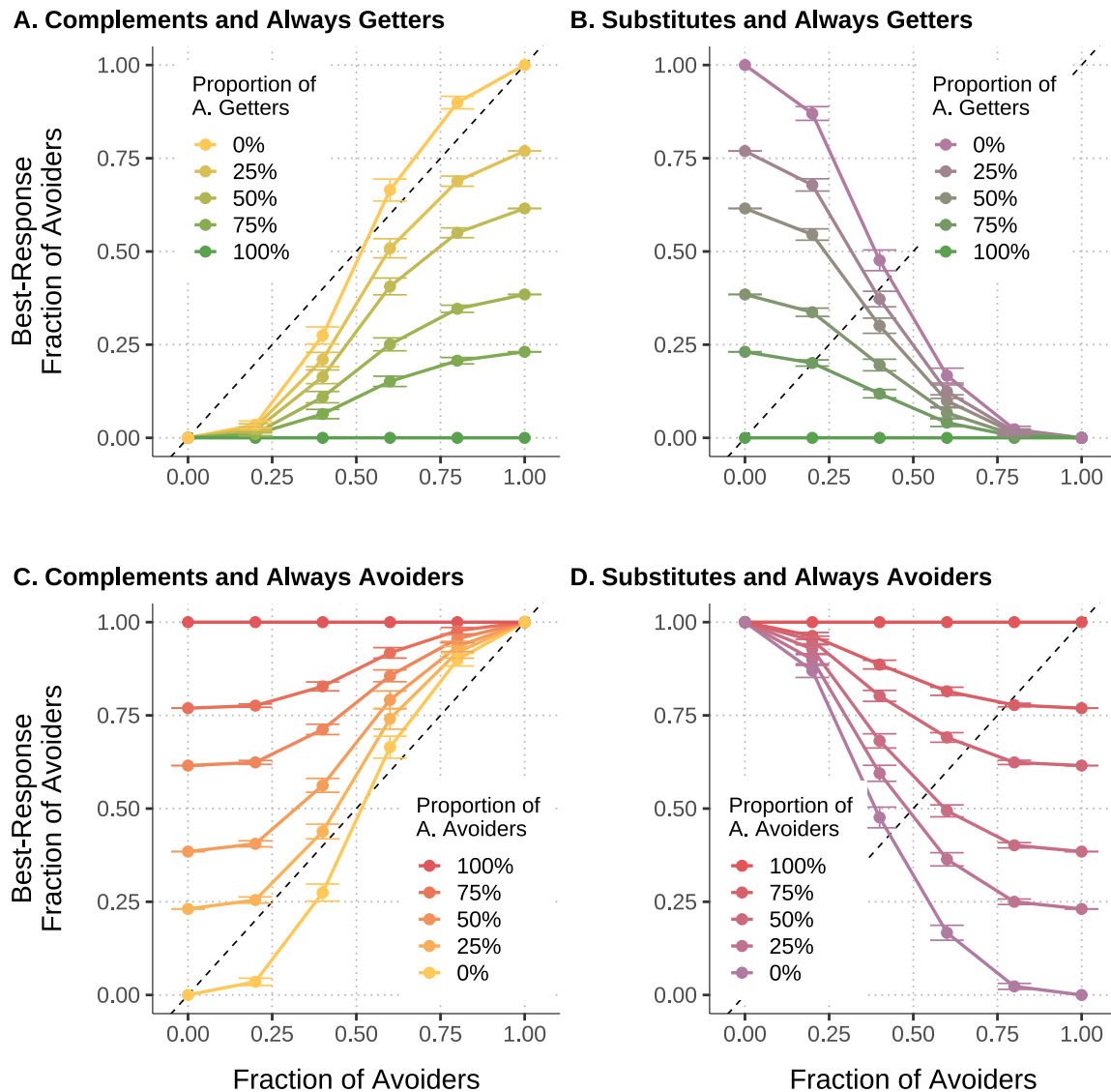
Notes: The table reports estimates from a linear probability model regressing information avoidance on the subject's belief about the share of information avoiders among others (i.e., who would choose Later), the subjective belief that the state is Screams ("Pr(screams)"), and their interaction. The regressions include a variety of controls; see notes in Table B.5.

Table B.14: Interaction of beliefs and information avoidance

C Group composition and equilibrium — additional analyses

C.1 Best-response schedules

Figure C.1: Effect of group composition on aggregate best-response schedules

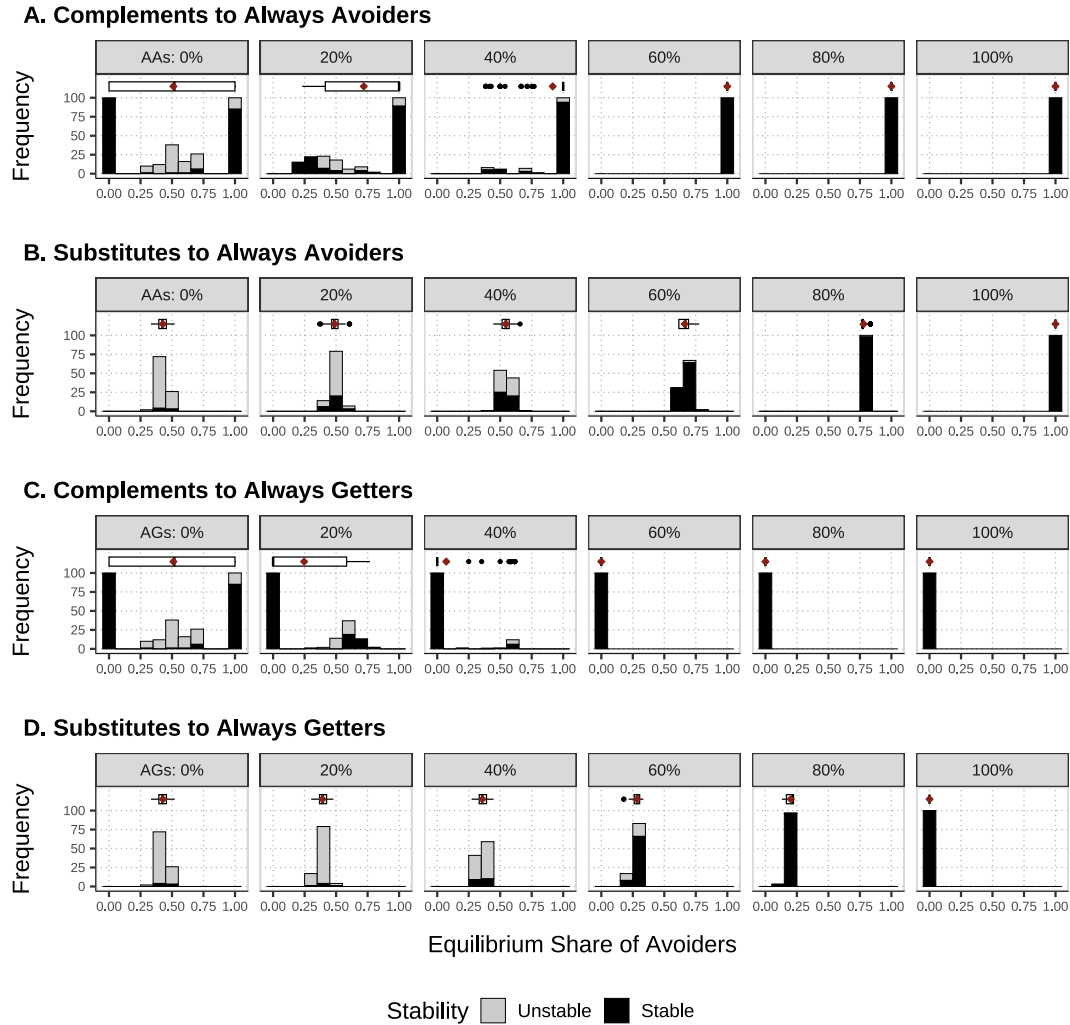


Notes: The figure shows aggregate best-response schedules by group composition. Each panel focuses on mixtures of two strategy types. Lines show the average (over bootstrapped groups) aggregate best-response schedule of a given composition. Groups composed entirely of conditional types are shown in yellow (Strategic Complements) and purple (Strategic Substitutes). Within each panel, additional lines represent increasing shares of unconditional types—Always Getters (green) or Always Avoiders (red). Error bars denote 95% confidence intervals.

C.2 Distribution of equilibria

The composition-specific distribution of information-avoidance equilibria is shown in Figure C.2. Each panel starts with groups composed entirely of one conditional type (Strategic Complements in Panels A and C; Strategic Substitutes in Panels B and D). Moving rightward, each facet gradually increases the participation of one unconditional type (Always Avoiders in Panels A and B; Always Getters in Panels C and D).

Figure C.2: Distribution of information-avoidance equilibria across group compositions



Notes: Distribution of information-avoidance equilibria (i.e., equilibrium prevalence of avoidance) across group compositions. Each facet corresponds to group composition, indexed by the share of an unconditional type—Always Avoiders (“AA”, panels A and B) or Always Getters (“AG”, panels C and D)—and a remainder share of a conditional type (Strategic Complements or Substitutes). Bars report the absolute frequency (across bootstrapped groups) of the equilibrium share of avoiders indicated on the x-axis. Stable equilibria are shown in dark bars, and unstable equilibria in light bars. Boxplots summarize the distribution. Red diamonds show the average of the distribution.

C.3 Higher-dimensional group compositions

In the main analysis, we focused on “restricted” sequences of group compositions, where only two strategy types were mixed. We now turn to richer cases involving more than two types. To visualize these, it is possible to map the richer composition space onto a unit probability triangle. Since Substitutes are highly likely to produce unstable equilibria, we omit them in this analysis. This allows redefining compositions as 3-dimensional probability vectors, which can be fully represented in the triangle.

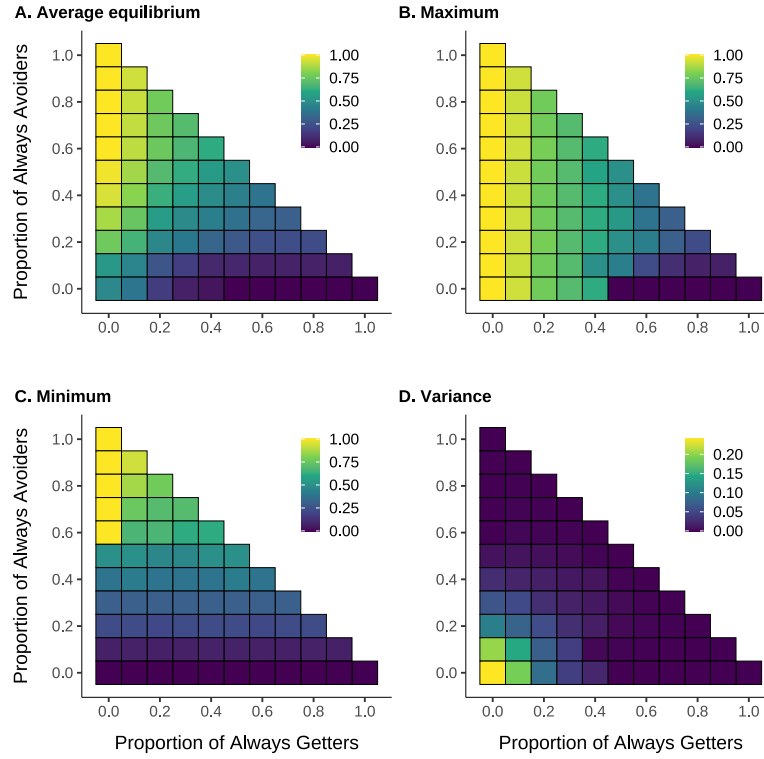
Figure C.3 illustrates this representation. The proportion of Always Getters is represented on the x -axis; and Always avoiders on the y -axis.³¹ Each cell corresponds to a unique group composition, associated with a distribution of fixed points (with examples visualized in the histograms above). Panel A presents, for each cell, the average of the set of stable equilibria. A group of pure Complements (the cell at $(0, 0)$) has an average close to 0.5. As we increase the share of Always Getters (moving horizontally along the x -axis) the average equilibrium share of avoiders decreases smoothly towards zero. Conversely, increasing the share of Always Avoiders (moving upward from $(0, 0)$ along the y -axis) pushes the average smoothly toward one. Overall, across the triangle, average equilibrium outcomes respond smoothly—although nonlinearly—to group composition.

Panel B shows the maximum equilibrium in the distribution, for each cell. Intuitively, if a policy-maker regards avoidance of information as an undesirable behavior, panel B shows the “worst” equilibrium in the distribution, for each cell. For groups composed entirely of Complements, the worst equilibrium corresponds to full avoidance, and the addition of Always Avoiders does not change this. By contrast, introducing Always Getters improves the worse-case outcome: the maximum equilibrium share of avoiders declines monotonically. Notably, a strong discontinuity emerges around $p_{AG} = 0.5$. More generally, such discontinuities—where a fixed amount of change in composition produces large, abrupt shifts in the worst-case equilibrium—appear across the composition space, and these are stronger the higher the proportion of Complements. These discontinuities cluster around $p_{AG} = 0.5$: when the proportion of Always Getters is sufficiently high, the aggregate best-response function shifts downward enough to eliminate any stable interior equilibria, generating the jump in the maximum. The discontinuities weaken as the share of Always Avoiders increases, as their presence helps sustain interior equilibria.

Panel C shows the minimum among the equilibria (the “best” equilibrium in the distribution), with patterns analogous to those for the maximum. Here, the relevant probability is p_{AA} . Finally, panel D exhibits the variance of the equilibrium set, which may matter for a risk-averse policy-maker. Naturally, the variance is highest in groups consisting of pure Complements, where multiple equilibria exist. Adding unconditional types flattens the aggregate schedule, progressively collapsing the distribution of equilibria (as shown in Figure C.2) and hence reducing variance.

³¹The proportion of Complements is implied by the complementary probability and visible as the horizontal (or ver-

Figure C.3: Group composition and summary features of equilibrium distributions



Notes: The figure illustrates how group composition affects summary features of the distribution of information-avoidance equilibria. Each cell in the unit probability triangle represents a distinct group composition, defined by the proportion of Always Getters (x -axis), Always Avoiders (y -axis), and Strategic Complements (implied by the vertical, or horizontal, distance to the hypotenuse). Each panel reports a different feature of the equilibrium distribution—the average (A), maximum (B), minimum (C), or variance (D)—with values represented by the color gradient. Only stable equilibria are considered; Strategic Substitutes are excluded ($p_{SS} = 0$), as their equilibria are typically unstable.

tical) distance from the cell to the hypotenuse of the triangle.

D Instructions and other screens

Preliminary instructions

At the start of the session, all participants received a printed copy of the instructions below, which were also read aloud by the experimenter.

Welcome to the CeDEx Lab!

Thank you very much for participating in this scientific experiment.

This is an experiment on decision-making. By carefully reading the instructions that will be shown on your computer, you can —depending on the decisions you make— earn an appreciable amount of money. It is therefore important that you pay close attention to the instructions.

We are interested in your individual choices. Therefore, communication is not allowed during the experiment. If you have a question, please just raise your hand.

In addition, there are some ground rules that need to be complied with to ensure fair conditions for all participants:

- **Focus on your own screen.** You should neither interfere with other participants nor look at their workstations (including their screens). Please do not lean backwards and always keep your chair close to your desk.
- **Be patient.** There will be some periods in the experiment in which you will be asked to wait for some time (for example, you might finish your tasks earlier than other people; or you might be asked to just plainly wait). In any case, please remain in your workstation, waiting in silence, without disturbing other participants.
- **No phones.** Please, put your phone(s) in silence mode and put them away (for instance, put them in your coat or bag): do not leave them on your desk. Please do it now. During this session, you should not check or use your phone at any time. If you check or use your phone, you will be asked to leave the session. This is to ensure fair play conditions for all participants.
- **Headphones at all times.** In this experiment, you are asked to wear the headphones provided next to your computer at all times. This is because sounds may be played to you as we will explain later. If you don't properly wear your headphones, you will be asked to leave the experiment.

If you do not comply with any of these ground rules, you will be asked to leave the experiment. Throughout the experiment, some of us, experimenters, will be walking around invigilating for these conditions to be met.

ou will soon be given instructions about today's experiment. All instructions will be provided on your computer screen.

Note that participation in this study is entirely voluntary. You can leave the experiment at any time you want. If you leave, we will take into account your progress up to that point to calculate your payment and will pay you accordingly.

In case you have any questions, feel free to ask the experimenter at any time, in a very quiet voice so that no one else is disturbed.

Now, please put on your headphones and then we will be ready to start the experiment.

After these instructions were read aloud, subsequent instructions were presented on participants' computer stations, as described in the next subsection.

Main instructions and decision screens

The following text reproduces the instructions displayed to participants via the computer stations. Each new screen in the instructions is marked below by a horizontal line. We provide clarifying notes in square brackets in italics.

General instructions for participants

Welcome and thank you for participating in this scientific experiment.

First, you will be asked for some contact information. Then, you will be given the instructions on how to earn money in this experiment.

Contact Information

In the following, we will ask for the email address associated with your PayPal account, and your university email address.

This information will be stored only for payment and research purposes.

Please, **be sure that your email addresses are typed correctly. Otherwise, we may not be able to make the payment.**

Please input the email address associated to your PayPal account. [Open text box]

Please type it again. [Open text box]

Now input your university email address. [Open text box]

Input it again. [Open text box]

Please input your **seat number** (this is the number written in the plastic card that you have drawn when you entered into the lab).

[Open text box]

Your consent to participate

Please be informed that the present study involves the possibility of listening to sounds that have been rated as aversive (i.e., unpleasant or disturbing) by participants in other studies. The safety of the participants is a priority for the CeDEx research centre. Therefore, technical measures have been put in place to ensure the hearing safety of the participants.

In order to participate in today's study, your explicit consent to participate is needed. Please note that participation in this study is entirely voluntary.

If you do not give your explicit consent, you will withdraw from this study. There will be no further consequences, and we will keep inviting you to other studies. We will give you £3 as a token of appreciation for coming today.

- ☐ "I do NOT GIVE my explicit consent to participate"
 - ☐ "I GIVE my explicit consent to participate"
-

In this experiment, you and other participants in the room will be part of the same group.

Groups are typically composed of around 30 participants.

Introduction

In today's experiment, you will be asked to do some tasks, which will be explained as you progress through the session. Depending on your performance (and your performance alone) in those tasks, your earnings may increase.

For your participation in today's session, you will receive £5, plus additional earnings based on your performance in the tasks.

Instructions: The Quiz (1/2)

In this experiment, you will be participating in a Quiz. The Quiz questions cover different topics (for instance, sports, geography, history, arts, music, etc.).

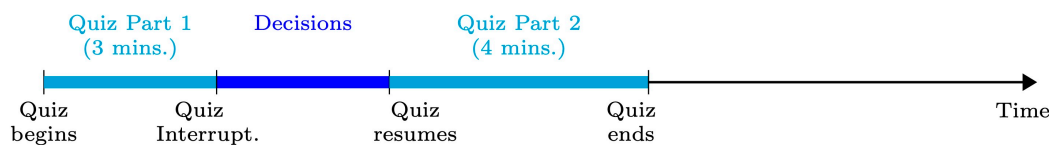
For each question, you will be provided with 4 possible answers, out of which exactly one will be correct.

In this Quiz task, **the more questions you answer correctly, the more you earn.**

The Quiz consists of **two parts**, namely Part 1 and Part 2.

- **Quiz Part 1:** You start with Part 1 of the quiz. You have 3 minutes to answer as many quiz questions as possible.
- **Quiz Interruption and Decisions:** After those 3 minutes, the Quiz will be briefly interrupted. During this interruption, you will be asked to make some decisions, as will be explained on the following screens.
- **Quiz Part 2:** After the interruption, you will answer Part 2 of the quiz, for 4 minutes.

The following figure illustrates the timeline:



The Quiz (2/2)

Below we explain how your earnings are calculated. The key feature is that **you should try to maximise the number of questions answered correctly**.

The number of correct answers in Parts 1 and 2 will be added up to calculate your earnings.

Your earnings increase in accordance with the level you reach. You start at Level 1. To reach Level 2, you need to answer 10 questions correctly (you do not need to correctly answer 10 questions in a row, what counts is the total number of correct answers). Then, to reach each additional level you need to answer correctly 20 extra questions per level. Thus, your earnings from the quiz are determined as follows:

- Level 1 (0-9 correct answers) = £5
- Level 2 (10-29 correct answers) = £6
- Level 3 (30-49 correct answers) = £7
- Level 4 (50-69 correct answers) = £9
- Level 5 (70-89 correct answers) = £13
- Level 6 (90+ correct answers) = £20

Your earnings from the quiz will be added to the £5 you will earn just for participating in this experiment.

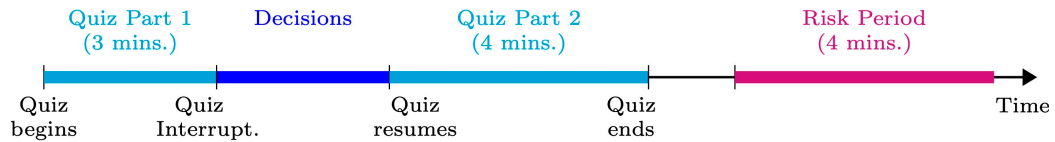
Example: suppose you answer correctly 25 questions in Part 1 and 35 in Part 2. Then the total number of correct answers is 60 ($=25+35$), so the level you reach is 4. Therefore, your earnings from the quiz would be £9. Adding up the £5 from your participation in this experiment, your total earnings would be £14 ($=£9 + £5$).

You can see that your earnings increase substantially, the higher the level that you reach.

The Risk Period

After the quiz finishes, a period called the “Risk Period” will start. The Risk Period lasts for 4 minutes.

During the Risk Period, a series of events may or may not happen as explained on the next screens.



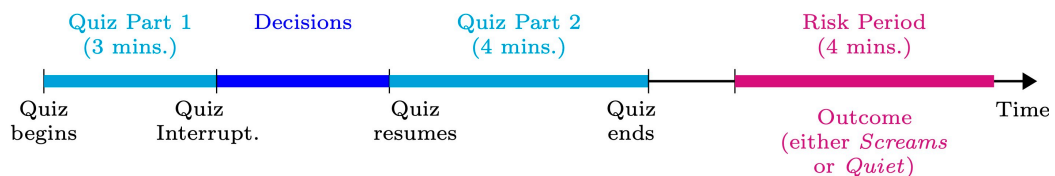
The Outcome

In the Risk Period, one of two outcomes will happen:

- **Screams** outcome: During the Risk Period, your group (including you) will hear a series of **unpleasant distress screams** from your headphones. These will happen at *unpredictable* moments during the Risk Period.

Previous research has shown that people consider similar screams as more aversive (disturbing) than the sound of nails sliding on a chalkboard.

- **Quiet** outcome: you and your group will not hear any screams during the whole experiment.



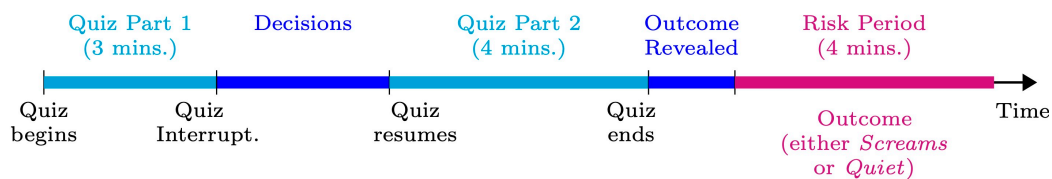
The Lottery

The Outcome (*Screams* or *Quiet*) in the Risk Period is the **same for everyone** in your group, and will be randomly determined by a computerised lottery.

Revelation of Outcome

The Outcome (i.e., *Screams* or *Quiet*) will be **revealed to everyone**, just a few screens **before the Risk Period**.

Therefore, everybody will already know the Outcome before the Risk Period starts.



Your Decision (1/3)

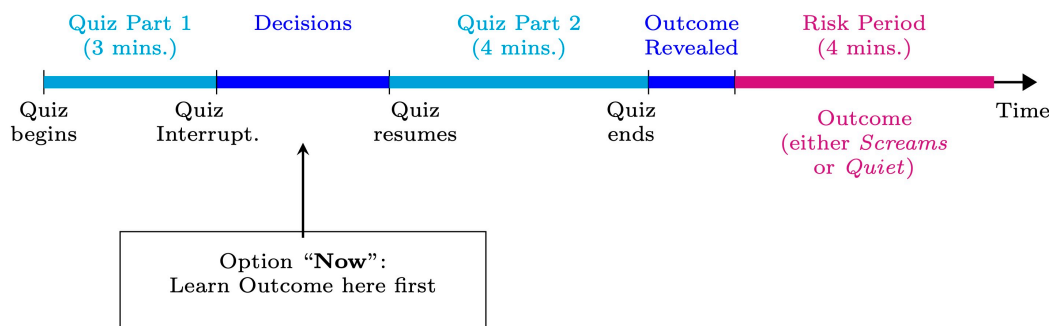
However, you can choose whether you want to know about the Outcome (*Screams* or *Quiet*) earlier or whether you prefer not to be aware of the Outcome until the beginning of the Risk Period.

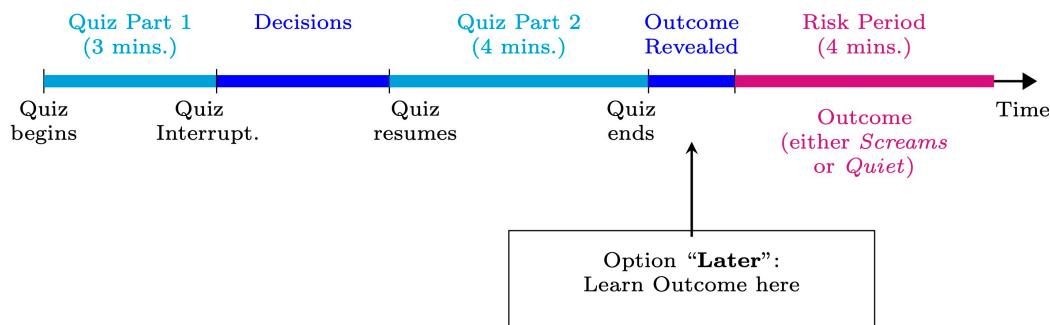
At the end of Part 1 of the quiz, you will choose one of two options:

- Option “**Now**” implies that you will learn the Outcome (*Screams* or *Quiet*) **immediately** (i.e., during the interruption of the quiz).
- Option “**Later**” implies that you will **not** learn about the Outcome (*Screams* or *Quiet*) immediately, and rather you will learn it just before the Risk Period.

This means that, during Part 2 of the quiz, you will not know whether in the forthcoming Risk Period you will hear the screams or not.

So the difference between “Now” and “Later” is **whether in Part 2 of the Quiz you are aware** of the Outcome or not. However, the Outcome is not affected by the option you choose.





The Volume of Screams

Although the Outcome is not affected by the option you choose, the **volume** of the screams (in case the Outcome is *Screams*) is affected by the choices that you and everyone else make:

- Each group member who chooses "**Later**" **increases the volume level by about 2 points out of 100.**
- Hence, if everybody chooses "**Later**", then the screams will be played at maximum volume (we call this level of volume "level 100").
- If nobody chooses "**Later**", then the screams will be played at volume "level 50".

This means that, if the Outcome is *Screams*, your choice between "Now" and "Later" would generate a difference in volume of about **2 points, out of 100.**

If instead the Outcome is *Quiet*, your choice has no implications beyond the timing in which you will find out the Outcome.

Check your Understanding

To check your understanding of the instructions, please imagine you are in each of the situations below and answer the following questions.

Situation 1a) Suppose that the Outcome is *Screams*. In addition, suppose that **everybody** in your group (including you) chooses "**Later**". What would be the volume of the screams?

- ☐ Level 50
- ☐ Level 52
- ☐ Level 98
- ☐ Level 100
- ☐ There would be no screams

Situation 1b) Suppose that the Outcome is *Screams*. In addition, suppose that **everybody else** in your group chooses “**Later**”, but **you** choose “**Now**”. What would roughly be the volume of the screams?

- ☐ Level 50
- ☐ Level 52
- ☐ Level 98
- ☐ Level 100
- ☐ There would be no screams

Situation 2a) Suppose that the Outcome is *Screams*. In addition, suppose that **everybody** in your group (including you) chooses “**Now**”. What would be the volume of the screams?

- ☐ Level 50
- ☐ Level 52
- ☐ Level 98
- ☐ Level 100
- ☐ There would be no screams

Situation 2b) Suppose that the Outcome is *Screams*. In addition, suppose that **everybody else** in your group chooses “**Now**”, but **you** choose “**Later**”. What would roughly be the volume of the screams?

- ☐ Level 50
- ☐ Level 52
- ☐ Level 98
- ☐ Level 100
- ☐ There would be no screams

Situation 3) Suppose that the Outcome is *Screams*. In that case:

- Your decision between “Now” and “Later” will...
 - ☐ Only affect the volume of the screams to be heard by you
 - ☐ Only affect the volume of the screams to be heard by participants other than you
 - ☐ Affect the volume of the screams to be heard by everyone (including you)
- The decisions between “Now” and “Later” **by participants other than you** will...
 - ☐ Only affect the volume of the screams to be heard by you
 - ☐ Only affect the volume of the screams to be heard by participants other than you
 - ☐ Affect the volume of the screams to be heard by everyone (including you)

Situation 4) Suppose that the Outcome is *Quiet*. In addition, suppose that **everybody** in your group (including you) chooses “Later”. What would roughly be the volume of the screams?

- ☐ Level 50
- ☐ Level 52
- ☐ Level 98
- ☐ Level 100
- ☐ There would be no screams at all

When you are ready, press the “Continue” button below.

(In case you are not taken to the next screen: some of your answers to the questions above might be wrong, please review them).

Instructions: Your Decision (2/3)

As mentioned, the difference between “Now” and “Later” is whether in Part 2 of the Quiz you are aware of the Outcome or not, and whether the volume of the screams (if the Outcome is Screams) changes by 2 points out of 100.

However, your choice between “Now” and “Later” does **not** change any of the following:

- the duration of the session;

- the sequence of the quiz;
- the amount of time you can spend on the quiz;
- when or whether you may potentially hear the screams.

Finally, note that for this choice between “Now” and “Later”, there is no right or wrong choice. You should simply decide based on your own preferences.

Instructions: Your Decision (3/3)

Recall that if you choose “Now”, then you will be shown a screen that informs you about the Outcome.

Upon being informed about the Outcome, you will be asked to select a button that correctly reflects it. You will need to do it to proceed to the next screens, which include Part 2 of the Quiz.

Differently, if you choose “Later”, then you will be shown a screen that will ask you to wait, and you will not learn the Outcome until Quiz Part 2 is finished, just before the Risk Period.

Control Questions

To briefly check that you understood the instructions correctly, please answer the following questions.

- If you choose option “Now”...
 - ☐ You will learn the Outcome immediately
 - ☐ You will not learn the Outcome immediately, but rather just before the Risk Period
 - ☐ You will never learn the Outcome
- If you choose option “Later”...
 - ☐ You will learn the Outcome immediately
 - ☐ You will not learn the Outcome immediately, but rather just before the Risk Period

- You will never learn the Outcome
- If the Outcome is *Screams*, the **timing** in which you will hear the screams...
 - will be earlier if you choose option “Now”
 - will be earlier if you choose option “Later”
 - will be the same independently of whether you choose “Now” or “Later”
- If the Outcome is *Screams*, the **volume** of the screams...
 - will be 2 points higher if you choose option “Later”
 - will be 2 points higher if you choose option “Now”
 - will be the same independently of whether you choose “Now” or “Later”
- Just before the Risk Period starts, you will be informed about the Outcome...
 - Only if you have chosen “Now”
 - Only if you have chosen “Later”
 - In any case, independently of whether you have chosen “Now” or “Later”
 - You will not be informed about the Outcome at that point

When you are ready, press the “Continue” button below.

(In case you are not taken to the next screen: some of your answers to the questions above might be wrong, please review them).

A Pilot Study

[Note: The text below was shown to participants in the Many Avoided condition. The version for the Few Avoided condition differed, with deviations indicated in square brackets.]

The information below gives you an idea of the choices of other people in a similar situation.

We ran a pilot study similar to this experiment, and found that, out of a set of 10 University of Nottingham students, **8 [2] chose the option “Later”**, preferring not to find out the Outcome until the end of Part 2. Therefore, in that group, **the overwhelming majority [minority] chose “Later”**.

If people in your group behave similarly to those in the pilot study, you can expect around 23 [6] people other than you to choose Later, and around 6 [23] to choose Now. In that case, the resulting volume level would roughly be 89 [60] if you choose Now, and 91 [62] if you choose Later.

Out of any 10 members of your group (*other than you*), how many do you believe will choose “Later”?

[Slider ranging from “0 out of 10 will choose “Later”” to “10 out of 10 will choose “Later””]

You are now ready to start the quiz

From the next screen, the quiz will start.

When you feel ready to start, press “Continue”, and both the quiz and the 3-minute timer will start at once.

Quiz

Please, answer the following questions to achieve a higher level and earn a higher payment.

[List of multiple choice quiz questions is displayed.]

Your quiz time has ended.

Please, press the Continue button.

The quiz has been interrupted. You will now be asked to make a series of decisions, including your preferred choice between knowing your lottery Outcome “Now” and “Later”.

What do you think the Outcome of your group's lottery will be? (*Screams* or *Quiet*)

[Radiobuttons (10 levels), with extremes labeled as "Definitely Quiet" and "Definitely Screams".]

[Note: this screen elicits the Kreps-Porteus questions (KP questions)]

Before choosing between "Now" and "Later", please answer the following questions about how happy you would be in different (hypothetical) situations.

- Imagine that you choose "Now", and then a moment later discover that the Outcome is Screams. How happy would you be in that moment *thinking about the Outcome ahead*?

[Slider ranging from -10="Very unhappy" to 10="Very happy".]

- Imagine that you choose "Now", and then a moment later discover that the Outcome is Quiet. How happy would you be in that moment *thinking about the Outcome ahead*?

[Slider ranging from -10="Very unhappy" to 10="Very happy".]

- Imagine that you choose "Later", and therefore you do not learn the Outcome until later. How happy would you be in that situation *thinking about the Outcome ahead*?

[Slider ranging from -10="Very unhappy" to 10="Very happy".]

Do you want to know the Outcome "Now" or "Later"?

Choose whether you want to know "Now" or "Later" whether the screams are upcoming.

- ☐ Later
- ☐ Now

How strong is your preference for your selected option, over the other?

[Radiobuttons (10), ranging from “I am **indifferent** between the two options” to “I have a **very strong** preference for the selected option”.]

A quick question

You have chosen the option [“Now”/“Later”]. Why did you prefer this option?

Please take a moment to answer this question.

[Open response textbox.]

Instructions: The Decisions Table

One participant will have the opportunity to **condition their decision on what the other participants chose** in the previous screen. This participant will be selected randomly by the computer.

Suppose you are that randomly selected participant. Then, the decision between “Now” and “Later” that you complete in the table on the following screen will be implemented for you.

See the image below for an example. In the table, you have the possible choices that other participants have made in the previous screen. Each row represents a different situation. **For each row, you can decide which option you would prefer.** Then, according to what the other players have *actually* chosen in the previous screen, the relevant row of the table will be determined, and the option selected in that row of the table will actually be implemented for you.

You have a positive chance of being selected as this player. Therefore, you must complete the table carefully according to your preferences.

(Please scroll down to see the ‘Continue’ button)

Out of 10 other participants, those who chose <i>Later</i> are...	In that situation, you prefer to know the Outcome...	Volume of screams if you choose Now	Volume of screams if you choose Later
0	Now Later	50	52
2	Now Later	60	62
4	Now Later	69	71
6	Now Later	79	81
8	Now Later	89	91
10	Now Later	98	100

Please, select your preferred option for each of the following situations.

Out of 10 other participants, those who chose <i>Later</i> are...	In that situation, you prefer to know the Outcome...	Volume of screams if you choose Now	Volume of screams if you choose Later
0	Now Later	50	52
2	Now Later	60	62
4	Now Later	69	71
6	Now Later	79	81
8	Now Later	89	91
10	Now Later	98	100

You will now be asked a series of questions. To answer them, please imagine you are in each of the situations presented.

Questions (1/5)

[Note: This and following screens elicit the Information Preferences Scale (IPS; [Ho et al., 2021](#))]

Please, imagine you are in each of the following situations, and answer the questions:

1. As part of a medical checkup, your doctor asks you a series of questions. The answers to these questions can be used to estimate your life expectancy (the age you are predicted to live to). Do you want to know how long you can expect to live?
 - ☐ Definitely don't want to know
 - ☐ Probably don't want to know
 - ☐ Probably want to know
 - ☐ Definitely want to know
2. You provide some genetic material to a testing service to learn more about your ancestors. You are then told that the same test can, at no additional cost, tell you whether you have an elevated risk of developing Alzheimer's (a brain disorder that slowly destroys memory and thinking skills and, eventually, the ability to carry out the simplest tasks). Do you want to know whether you have a high risk of developing Alzheimer's?
 - ☐ Definitely don't want to know
 - ☐ Probably don't want to know
 - ☐ Probably want to know
 - ☐ Definitely want to know
3. At your medical checkup, you are given the option to see the results of a diagnostic test, which can identify the extent to which your body has suffered long-term effects from stress. Do you want to know how much lasting damage your body has suffered from stress?
 - ☐ Definitely don't want to know
 - ☐ Probably don't want to know
 - ☐ Probably want to know
 - ☐ Definitely want to know

Questions (2/5)

4. Ten years ago, you had the opportunity to invest in two retirement funds: Fund A and Fund B. For the past 10 years, you have invested all your retirement savings in Fund A. Do you want to know the balance you would have if you had invested in Fund B instead?

- ☐ Definitely don't want to know
 - ☐ Probably don't want to know
 - ☐ Probably want to know
 - ☐ Definitely want to know
5. You decide to go to the theater for your birthday and give your close friend (or partner) your credit card so they can purchase tickets for the two of you, which they do. You aren't sure but suspect that the tickets may have been expensive. Do you want to know how much the tickets cost?
- ☐ Definitely don't want to know
 - ☐ Probably don't want to know
 - ☐ Probably want to know
 - ☐ Definitely want to know
6. You bought an electronic device at a store at what seemed like a reasonable, though not particularly low, price. A month has passed, and the item is no longer returnable. You see the same device displayed in another store with a sign announcing "SALE." Do you want to know the price you could have bought it for?
- ☐ Definitely don't want to know
 - ☐ Probably don't want to know
 - ☐ Probably want to know
 - ☐ Definitely want to know

Questions (3/5)

7. You gave a close friend one of your favorite books for her birthday. Visiting her apartment a couple of months later, you notice the book on her shelf. She never said anything about it; do you want to know if she liked the book?
- ☐ Definitely don't want to know
 - ☐ Probably don't want to know
 - ☐ Probably want to know
 - ☐ Definitely want to know
8. Someone has described you as quirky, which could be interpreted in a positive or negative sense. Do you want to know which interpretation he intended?

- ☐ Definitely don't want to know
 - ☐ Probably don't want to know
 - ☐ Probably want to know
 - ☐ Definitely want to know
9. You gave a toast at your best friend's wedding. Your best friend says you did a good job, but you aren't sure if he or she meant it. Later, you overhear people discussing the toasts. Do you want to know what people really thought of your toast?
- ☐ Definitely don't want to know
 - ☐ Probably don't want to know
 - ☐ Probably want to know
 - ☐ Definitely want to know
10. As part of a fundraising event, you agree to post a picture of yourself and have people guess your age (the closer they get, the more they win). At the end of the event, you have the option to see people's guesses. Do you want to learn how old people guessed that you are?
- ☐ Definitely don't want to know
 - ☐ Probably don't want to know
 - ☐ Probably want to know
 - ☐ Definitely want to know
11. You have just participated in a psychological study in which all of the participants rate others' attractiveness. The experimenter gives you an option to see the results for how people rated you. Do you want to know how attractive other people think you are?
- ☐ Definitely don't want to know
 - ☐ Probably don't want to know
 - ☐ Probably want to know
 - ☐ Definitely want to know

Questions (4/5)

12. Some people seek out information even when it might be painful. Others avoid getting information that they suspect might be painful, even if it could be useful. How would you describe yourself?

- ☐ Definitely don't want to know
 - ☐ Probably don't want to know
 - ☐ Probably want to know
 - ☐ Definitely want to know
13. If people know bad things about my life that I don't know, I would prefer not to be told.
- ☐ Definitely don't want to know
 - ☐ Probably don't want to know
 - ☐ Probably want to know
 - ☐ Definitely want to know

Questions (5/5)

The questions below ask about what you would decide in the workplace. If you have previously worked in a company, imagine yourself in it and answer the questions. If you had not worked for a company, read the following questions as imagining you are working in a company.

14. Employers commonly encourage workers to complete health risk assessments that gauge their health characteristics and risk. These are usually used to estimate workplace-wide risks the company may face. If you had the opportunity, would you like to get your individual risk assessment?
- ☐ Definitely don't want to know
 - ☐ Probably don't want to know
 - ☐ Probably want to know
 - ☐ Definitely want to know
15. Most people spend some time at work on activities they are not proud of (e.g., browsing social media). Suppose you could compose a list of these activities and have your computer track how much time you spend doing them. The information would not be shared with your employer. Would you like to know what fraction of each hour at work you spend on these activities?
- ☐ Definitely don't want to know
 - ☐ Probably don't want to know

- ☐ Probably want to know
 - ☐ Definitely want to know
16. As part of a unit-wide evaluation, you and your coworkers are asked how easy it is to get along with each other and what your strengths and weaknesses are. Would you like to know how your coworkers rated you?
- ☐ Definitely don't want to know
 - ☐ Probably don't want to know
 - ☐ Probably want to know
 - ☐ Definitely want to know
17. The rise of artificial intelligence is likely to lead to job losses in a wide range of occupations, affecting workers in industries from long-distance trucking to healthcare. Would you like to know whether your job is at risk owing to automation in the next 10 years?
- ☐ Definitely don't want to know
 - ☐ Probably don't want to know
 - ☐ Probably want to know
 - ☐ Definitely want to know
18. One option in your employer's retirement plan allows you to compare your investment return with that of your coworkers. Would you like to know how your investment return compares with the average and highest investment returns earned by employees at your firm?
- ☐ Definitely don't want to know
 - ☐ Probably don't want to know
 - ☐ Probably want to know
 - ☐ Definitely want to know

Please answer the following questions about how happy you would be in different (hypothetical) situations.

- Imagine that the Outcome is *Screams*, and that the **volume of the screams** will be **level 50**. How happy would you thinking about the Outcome ahead?
[Slider from -10="Very unhappy" to 10="Very happy"]

- Imagine that the Outcome is *Screams*, and that the **volume of the screams** will be **level 100**. How happy would you thinking about the Outcome ahead?
[Slider from -10=“Very unhappy” to 10=“Very happy”]
-

Thank you for answering the questions.

In the next screen, you will get to know the Outcome if you selected option Now, or remain without knowing it, if you selected option Later. If you were selected as the random player, then the option you selected in the relevant row of the table will be implemented.

*[Note: this screen was displayed to participants whose implemented option was **Now**]*

Based on your previous decisions, the choice implemented for you is “Now”.

The Outcome is **Screams** [Quiet].

Therefore, you **will hear the screams** [not hear any screams] during the Risk Period.

Please, indicate below which is the Outcome.

- ☐ Screams
 - ☐ Quiet
-

*[Note: this screen was displayed to participants whose implemented option was **Later**]*

Based on your previous decisions, the choice implemented for you is “Later”.

You may or may not hear the screams during the Risk Period. You will be informed about the Outcome later, just before the Risk Period starts.

Press Continue to proceed to Quiz Part 2.

Quiz

Please, answer the following questions to achieve a higher level and earn a higher payment.

[The same list of multiple choice quiz questions is displayed.]

The Quiz has ended.

You will be asked a series of questions, and then move to the Risk Period.

Questions (1/5)

[Note: This and following screens elicit social and economic preferences from the Global Preferences Survey (GPS; [Falk et al., 2018](#))]

We now ask for your **willingness to act** in a certain way in four different areas.

Please indicate your answer on a scale from 0 to 10.

How willing are you to give up something that is beneficial for you today in order to benefit more from that in the future? [Slider: 0=“Completely unwilling to do so”, 10=“Very willing to do so”.]

In general, how willing or unwilling are you to take risks? [Slider: 0=“Completely unwilling to take risks”, 10=“Very willing to take risks”.]

Questions (2/5)

1. When someone does me a favor I am willing to return it.

[Slider: 0=“Does not describe me at all”, 10=“Describes me perfectly”.]

2. Please think about what you would do in the following situation:

You are in an area you are not familiar with, and you realize you lost your way. You ask a stranger for directions. The stranger offers to take you to your destination. Helping you costs the stranger about £20 in total. However, the stranger says he or she does not want any money from you. You have six presents with you. The cheapest present costs £5, the most expensive one costs £30. Do you give one of the presents to the stranger as a "thank-you" gift? If so, which present do you give to the stranger?

- ☐ No present
- ☐ Present worth £5
- ☐ Present worth £10
- ☐ Present worth £15
- ☐ Present worth £20
- ☐ Present worth £25
- ☐ Present worth £30

Questions (3/5)

If I am treated very unjustly, I will take revenge at the first occasion, even if there is a cost to do so. [Slider: 0="Does not describe me at all", 10="Describes me perfectly".]

How willing are you to punish someone who treats others unfairly, even if there may be costs for you? [Slider: 0="Completely unwilling to do so", 10="Very willing to do so".]

Questions (4/5)

Imagine the following situation: Today you unexpectedly received £1,000. How much of this amount would you donate to a good cause? [Open text box; inputs restricted to numbers between 0 and 1000]

How willing are you to give to good causes without expecting anything in return? [Slider: 0="Completely unwilling to do so", 10="Very willing to do so".]

Questions (5/5)

"I assume that people have only the best intentions." [Slider: 0="Does not describe me at all", 10="Describes me perfectly".]

Instructions: the Risk Period (1/2)

Thanks for kindly answering the questions.

You are now about to enter the Risk Period.

Before the Risk Period starts, two things will happen:

1. In the next screen, you will be provided with a sample scream, to get an idea of the kind of screams that may happen during the Risk Period.
2. After that, everybody will be informed about: a) which is the Outcome (*Screams* or *Quiet*); and b) the volume of the screams as the result of the group's choices, if the Outcome is *Screams*.

After that, the Risk Period will start. It will last 4 minutes, and you will be provided with the answers to the quiz to review if you want to.

Pre-experiencing a sample scream (1/2)

To give you an idea of what the screams sound like, on the next screen, you will hear a sample scream, similar to the screams you would hear if your lottery outcome were *Screams*.

On the next screen, you will hear only 1 scream, whereas during the Risk Period you may hear more than 1 scream if your lottery Outcome is *Screams*, and 0 if your lottery outcome is *Quiet*.

To ensure your safety, the volumes of the screams have been calibrated to comply with the safe hearing levels recommended in the UK.

Please, keep your headphones on at all times. You are also reminded you may leave the session at any time you would like.

Please, try to remain silent throughout.

Now, if you wish to proceed, press Continue.

Pre-experiencing a sample scream (2/2)

To hear the sample scream, press the “Play” button. From the moment you press the “Play” button below, the sample scream will be played at a random moment within 30 seconds.

You will be able to hear the sample scream only once. In other words, the button will not do anything from the second time you press it.

Please, try to remain silent throughout.

After the 30 seconds, the Continue button will appear. You will be able to proceed by pressing it.

[Play button]

The Outcome, revealed

[If the lottery Outcome was Screams, the following text was displayed:]

Your group’s lottery Outcome is **Screams**.

Therefore, your group **will hear the screams** during the Risk Period.

The volume level of the screams is [**volume level**]. This is the result of the choices of all participants.

[If the lottery Outcome was Quiet, the following text was displayed:]

Your lottery outcome is **Quiet**.

Therefore, you will **not hear any screams** during the Risk Period.

Risk Period

The Risk Period has started. Please, remain in silence until it ends.

In the meantime, you can read the list of quiz questions and answers below.

At the end of the Risk Period, a “Continue” button will appear, at the bottom. (You may need to scroll down)

Quiz Answers

Below is the list of quiz questions. The correct answers to these are underlined.

[List of all quiz questions and their answers is displayed]

Thank you for your patience. The Risk Period has ended.

You will now be asked a few final questions.

Please, press the Continue button to proceed.

Final Questions (1/4)

Please, kindly answer the following questions.

- How easy to understand were the instructions? [Radiobuttons (5): “Very easy” to “Very hard”]
 - How hard to answer were the quiz questions? [Radiobuttons (5): “Very easy” to “Very hard”]
 - Do you have any experience at work, even casually? [“Yes” / “No”]
-

Final Questions (2/4)

- All in all, how do you feel about this study? [Open text box]

- Is there anything else you would want to tell us? [Open text box]
-

Final Questions (4/4)

- What is your age? [Open text box, restricted to numbers]
 - What is your gender? [Female/Male/Other]
-

You have answered [Q] questions correctly. Therefore, you have reached performance level [P]. In consequence, you have earned £[e] from the quiz.

Adding the £5 for your participation, your **total earnings** from the experiment are £[e + 5].

You will be contacted about your payment in the next couple of days.

Thanks for your participation

We appreciate that you took the time to come and participate in today's experiment. Thanks a lot, and hope to see you around soon!

Your session is ending now. You can now leave the lab. Please, do it in silence, so other participants are not disturbed.